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**Witte et al.**

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(54) **SURGICAL INSTRUMENTS WITH  
ACTUATABLE TAILPIECE**

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(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 435 days.

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(51) **Int. Cl.**  
**A61B 34/00** (2016.01)  
**A61B 17/00** (2006.01)  
(Continued)

(52) **U.S. Cl.**  
CPC ..... **A61B 34/71** (2016.02); **A61B 17/00234** (2013.01); **A61B 34/37** (2016.02);  
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(58) **Field of Classification Search**  
None

See application file for complete search history.

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*Primary Examiner* — James M Kish

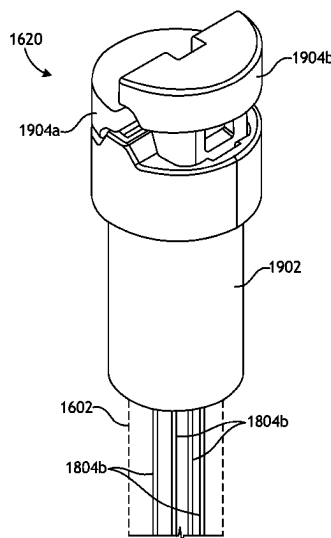
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(57) **ABSTRACT**

A robotic surgical tool includes a handle providing a plurality of drive inputs, an elongate shaft including a plurality of sliding rack gears movably nested within a corresponding plurality of longitudinal channels defined along the shaft, and an end effector arranged at a distal end of the shaft and an articulable wrist interposing the end effector and the distal end. An actuation system housed in the handle actuates the drive inputs to move the sliding rack gears and thereby causes operation of at least one of the end effector and the wrist. A tailpiece is arranged at a proximal end of the shaft and is operably coupled to the sliding rack gears, the tailpiece includes a housing and a manual actuation device mounted to the housing and manually actuatable to act on the sliding rack gears and thereby operate at least one of the end effector and the wrist.

**20 Claims, 37 Drawing Sheets**



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- (51) **Int. Cl.**  
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*A61B 34/37* (2016.01)  
*B25J 9/10* (2006.01)  
*A61B 17/29* (2006.01)
- (52) **U.S. Cl.**  
CPC ..... *B25J 9/102* (2013.01); *A61B 2017/00477*  
(2013.01); *A61B 2017/2927* (2013.01); *A61B*  
*2034/305* (2016.02)
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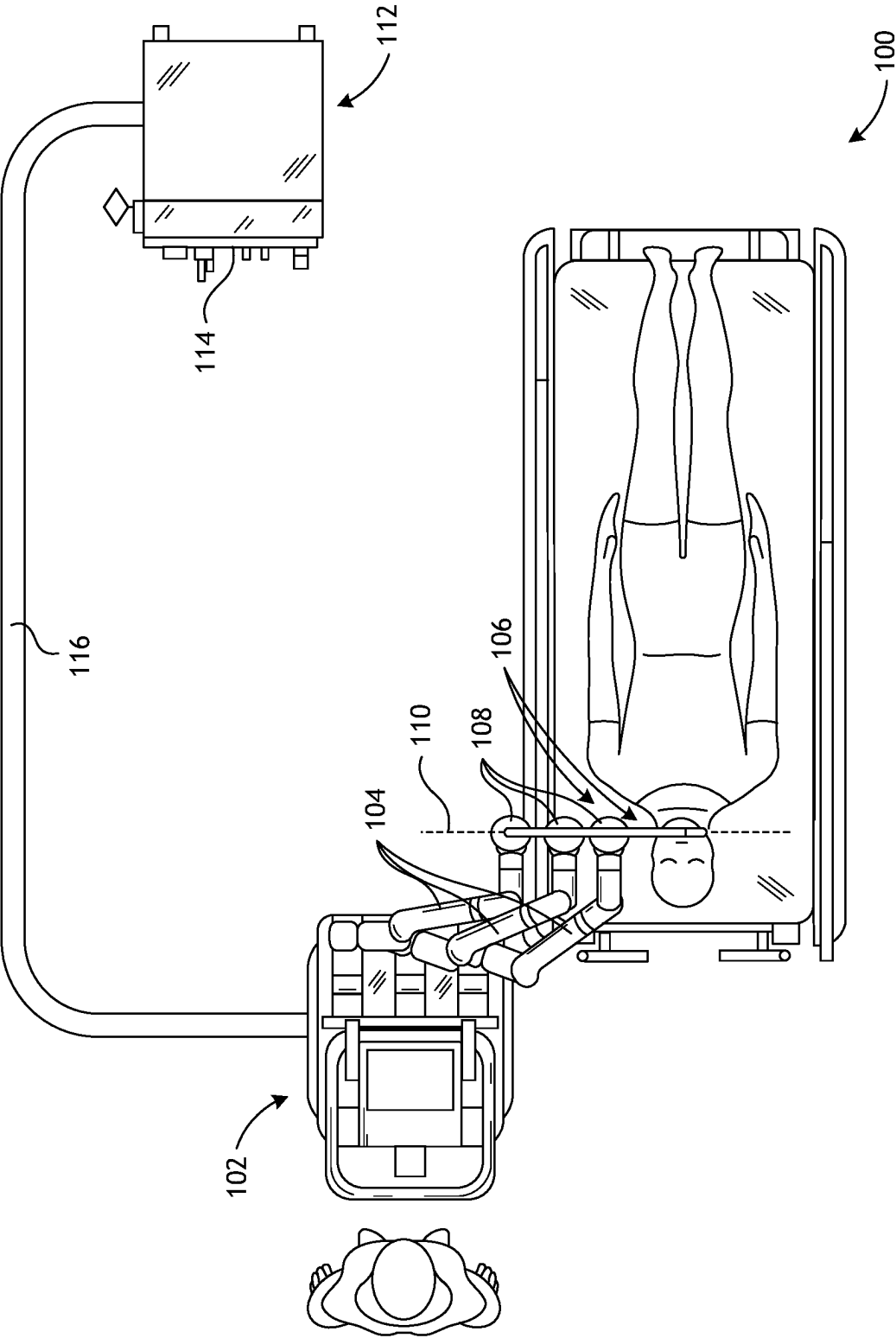
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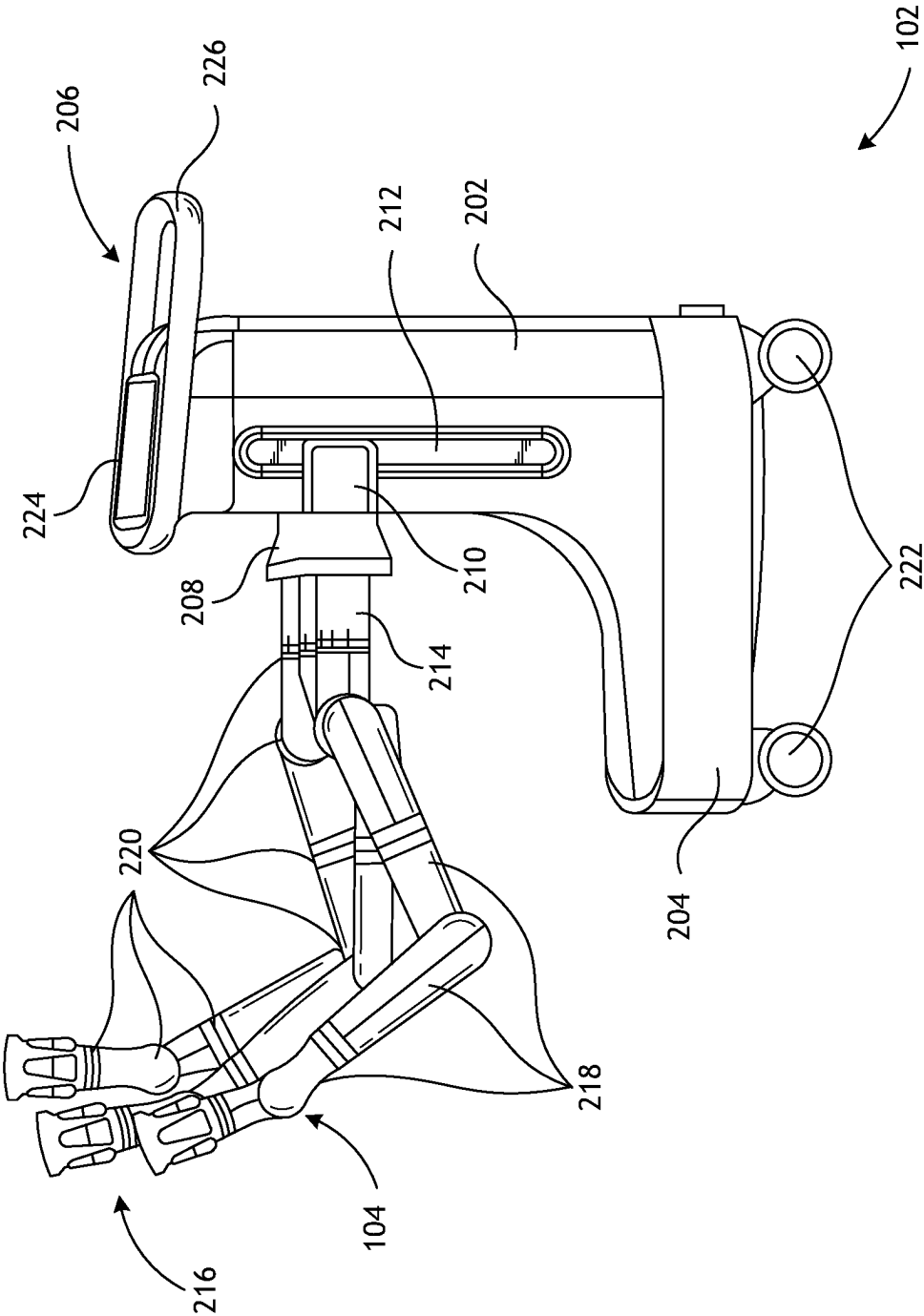


FIG. 2

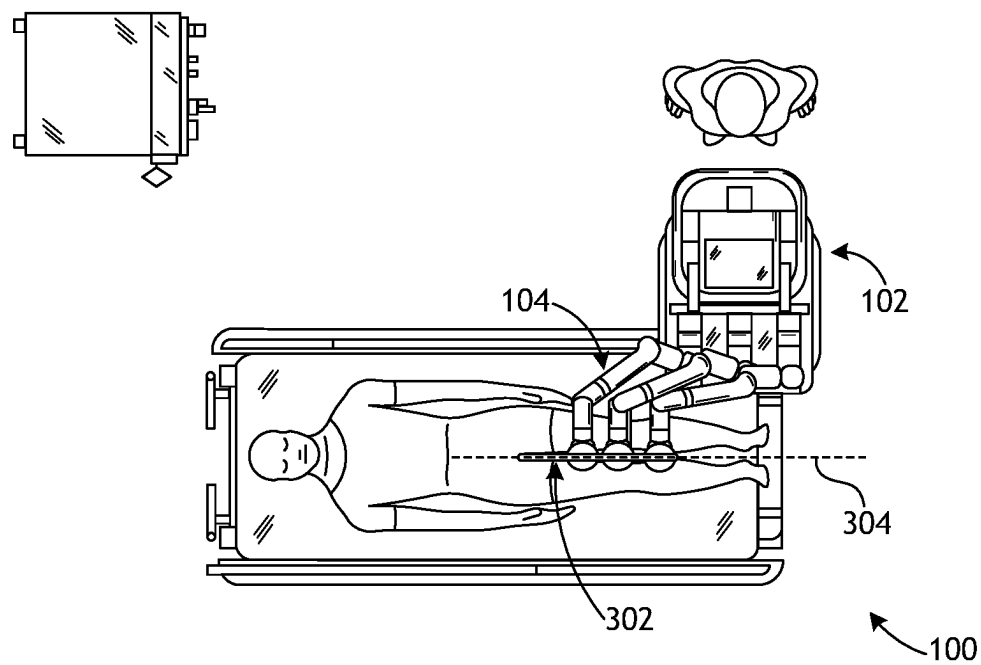


FIG. 3A

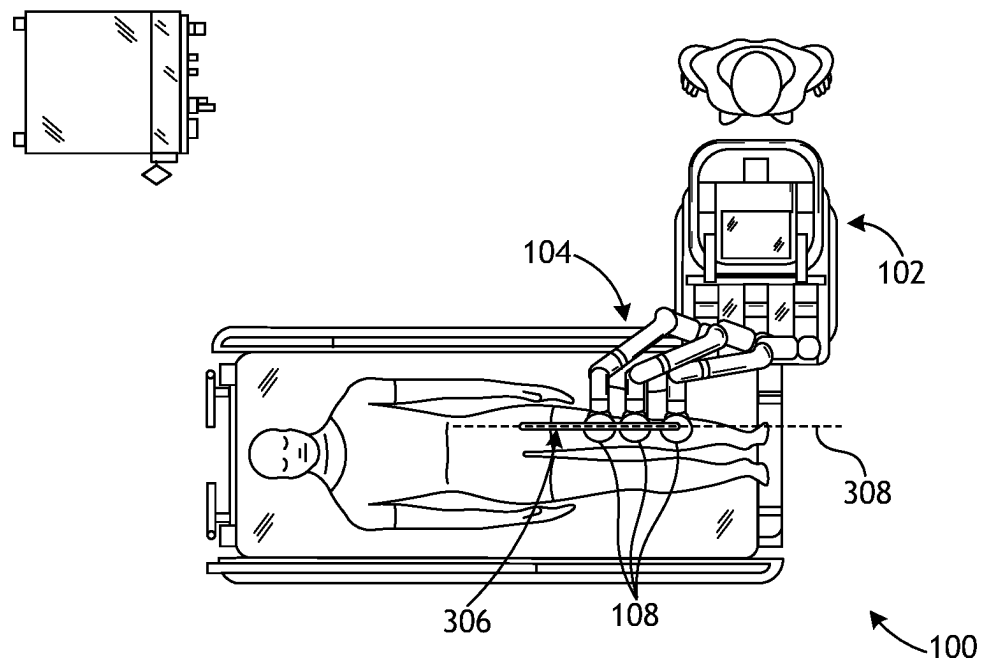


FIG. 3B

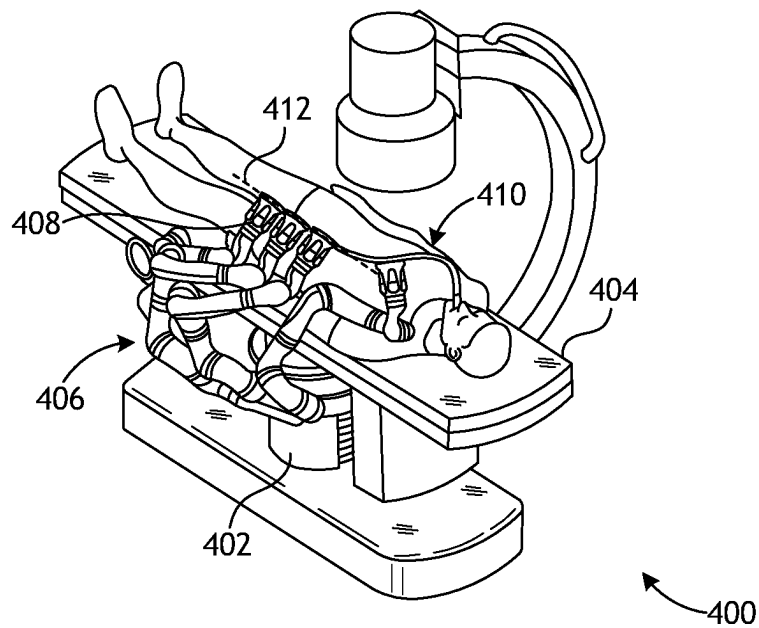


FIG. 4

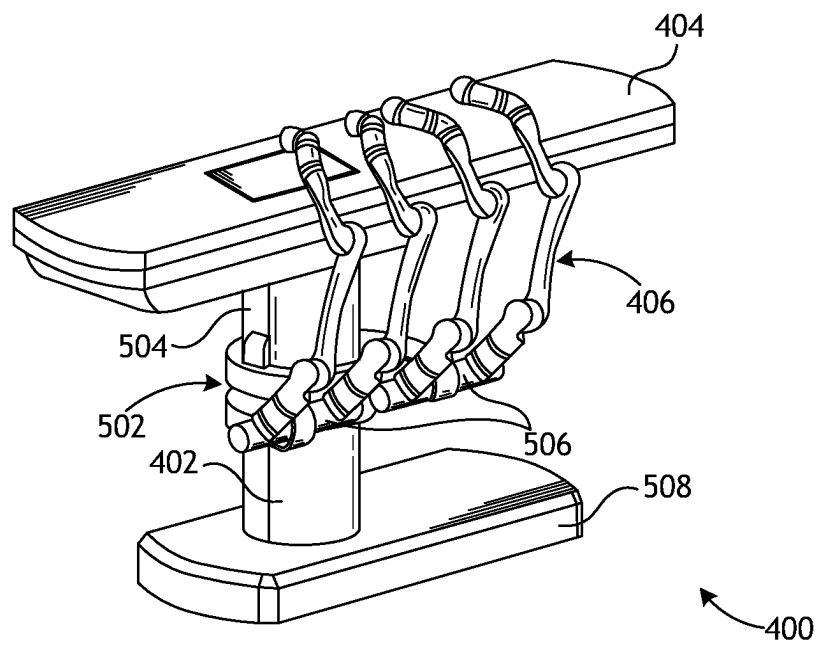


FIG. 5

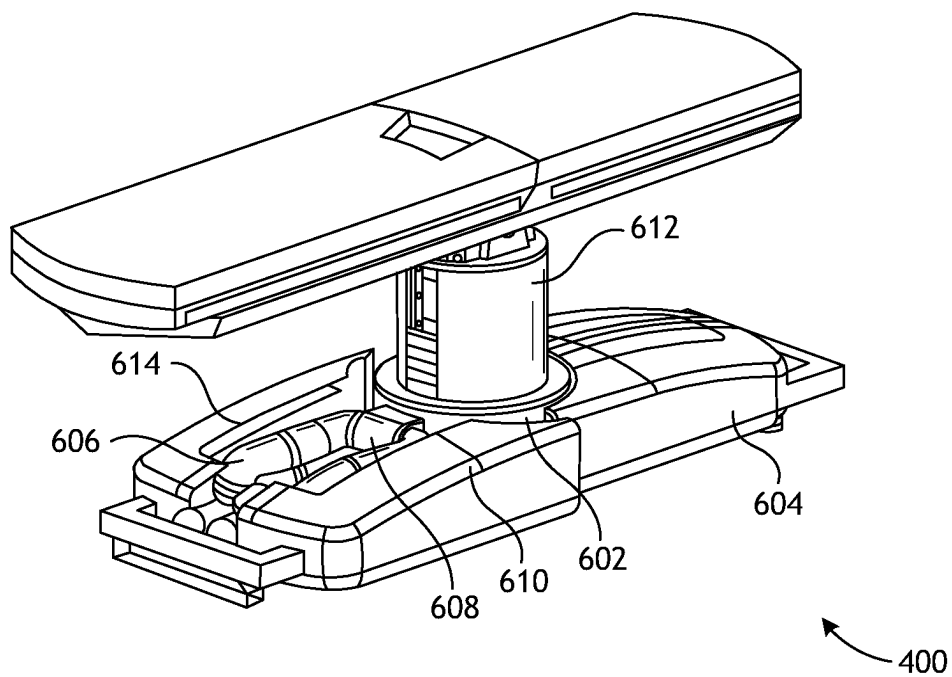


FIG. 6

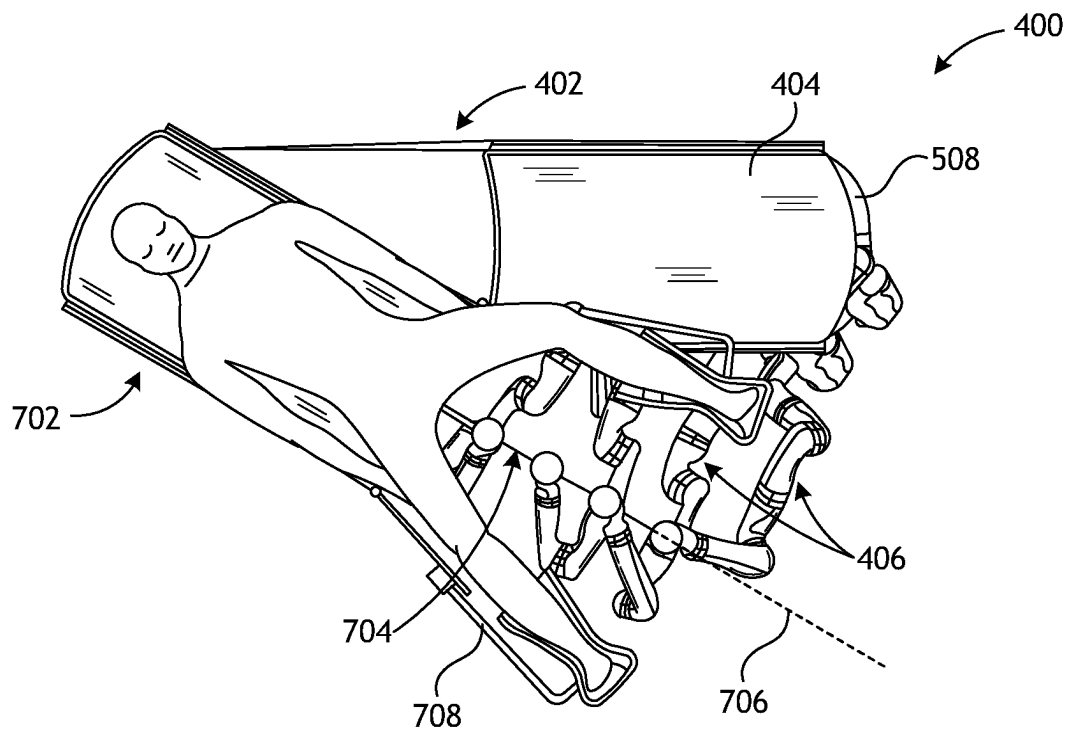


FIG. 7A

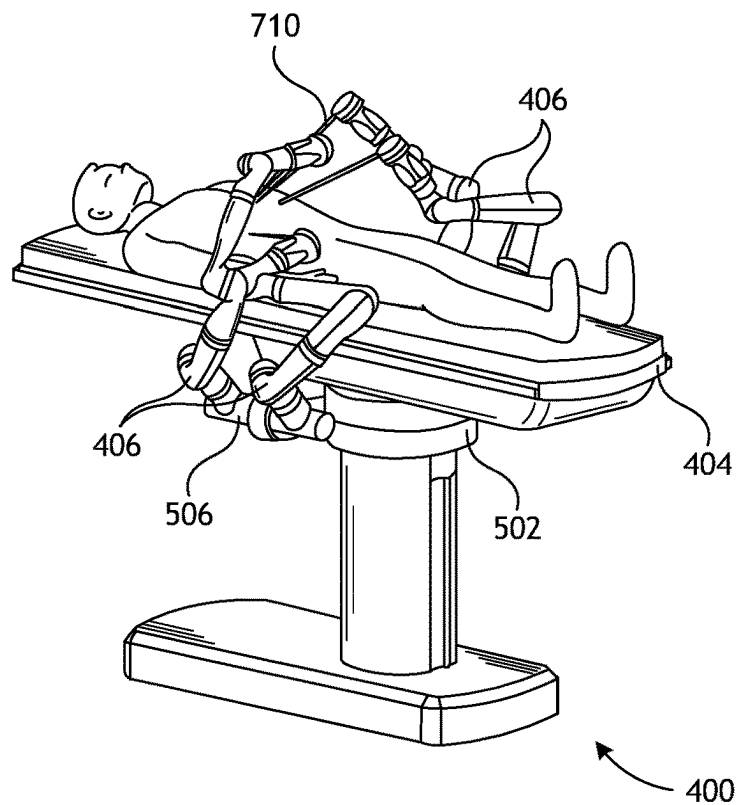


FIG. 7B

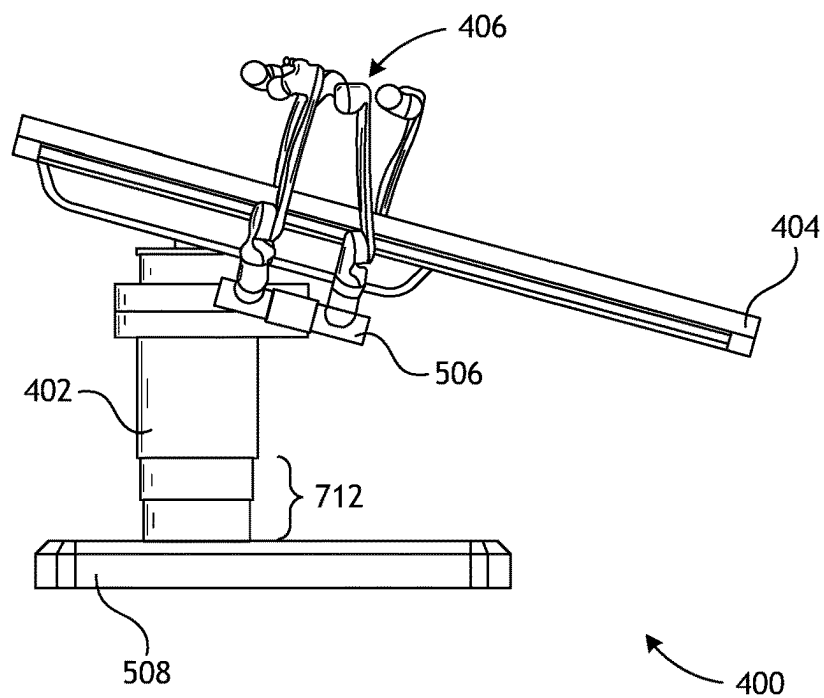


FIG. 7C

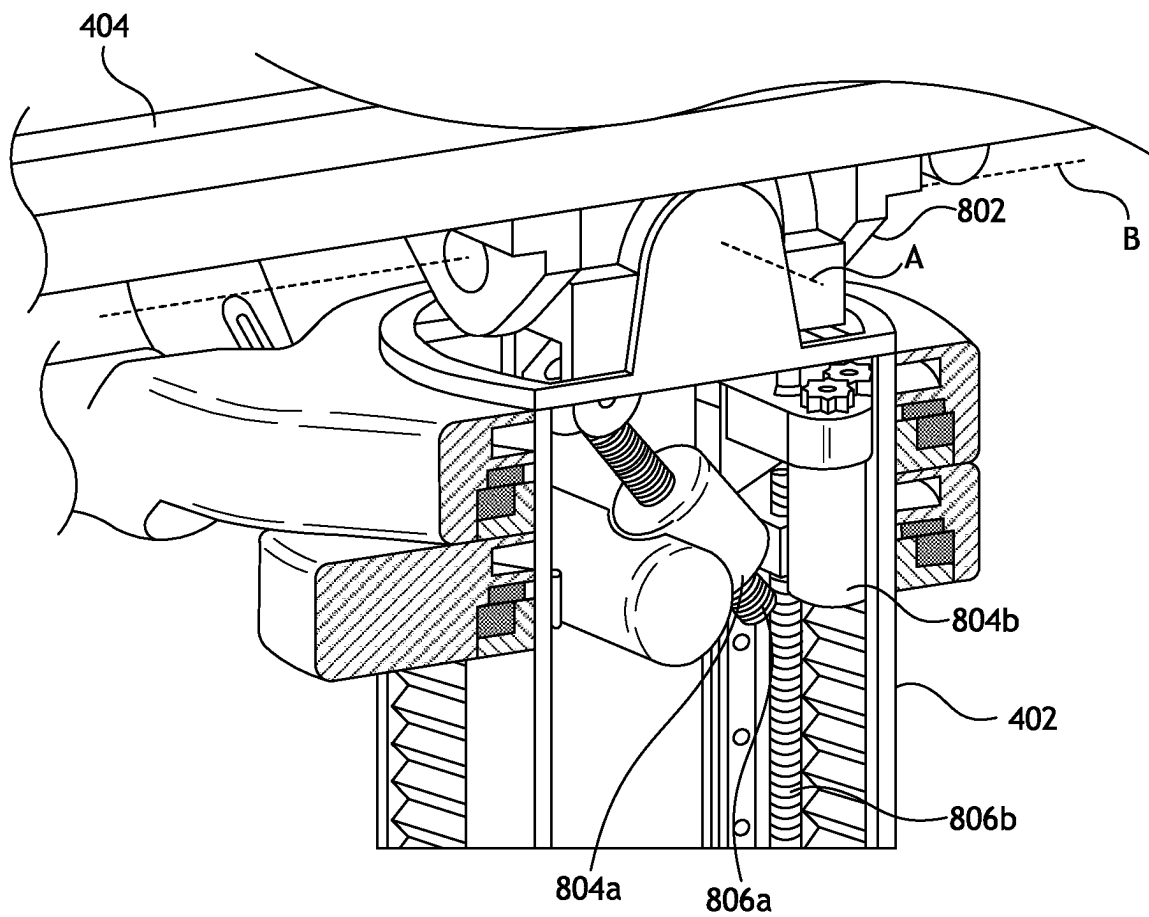


FIG. 8

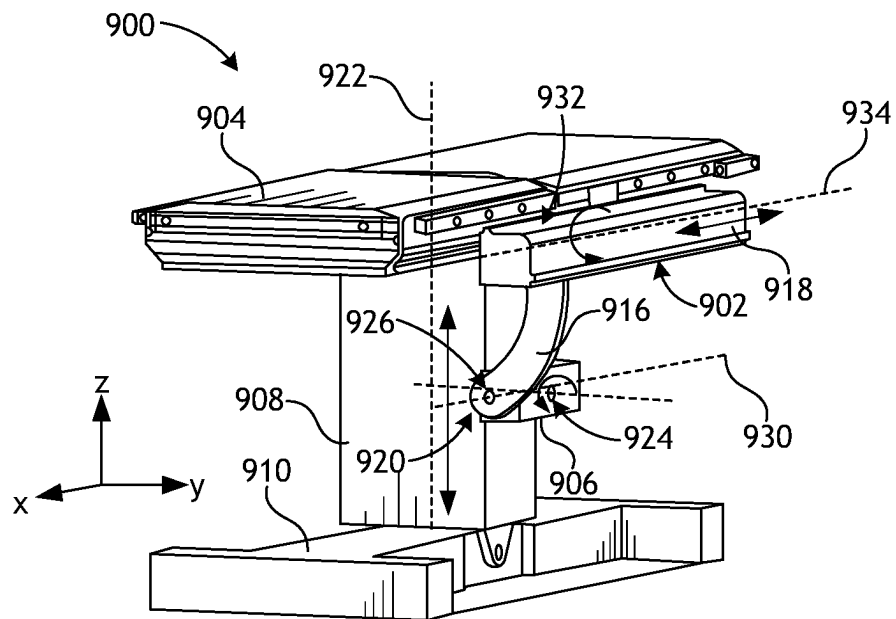


FIG. 9A

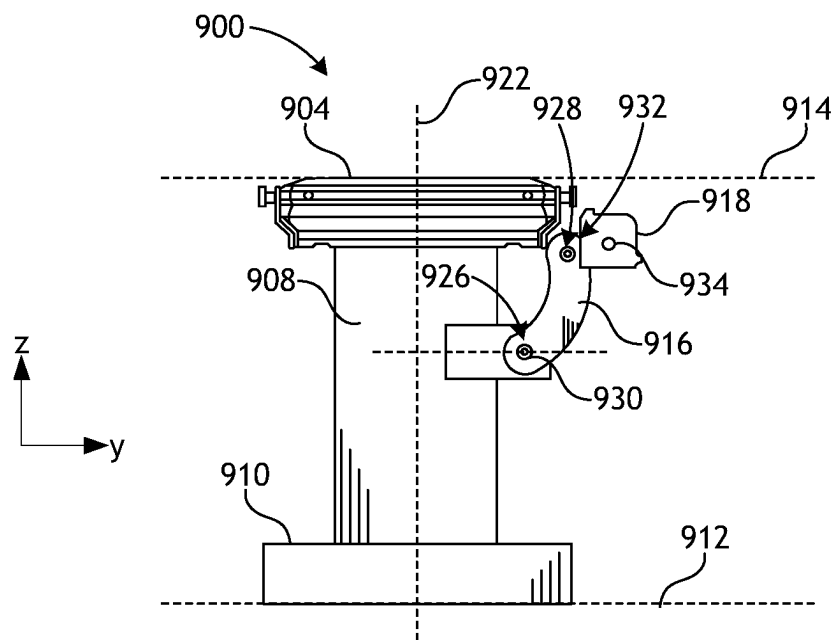
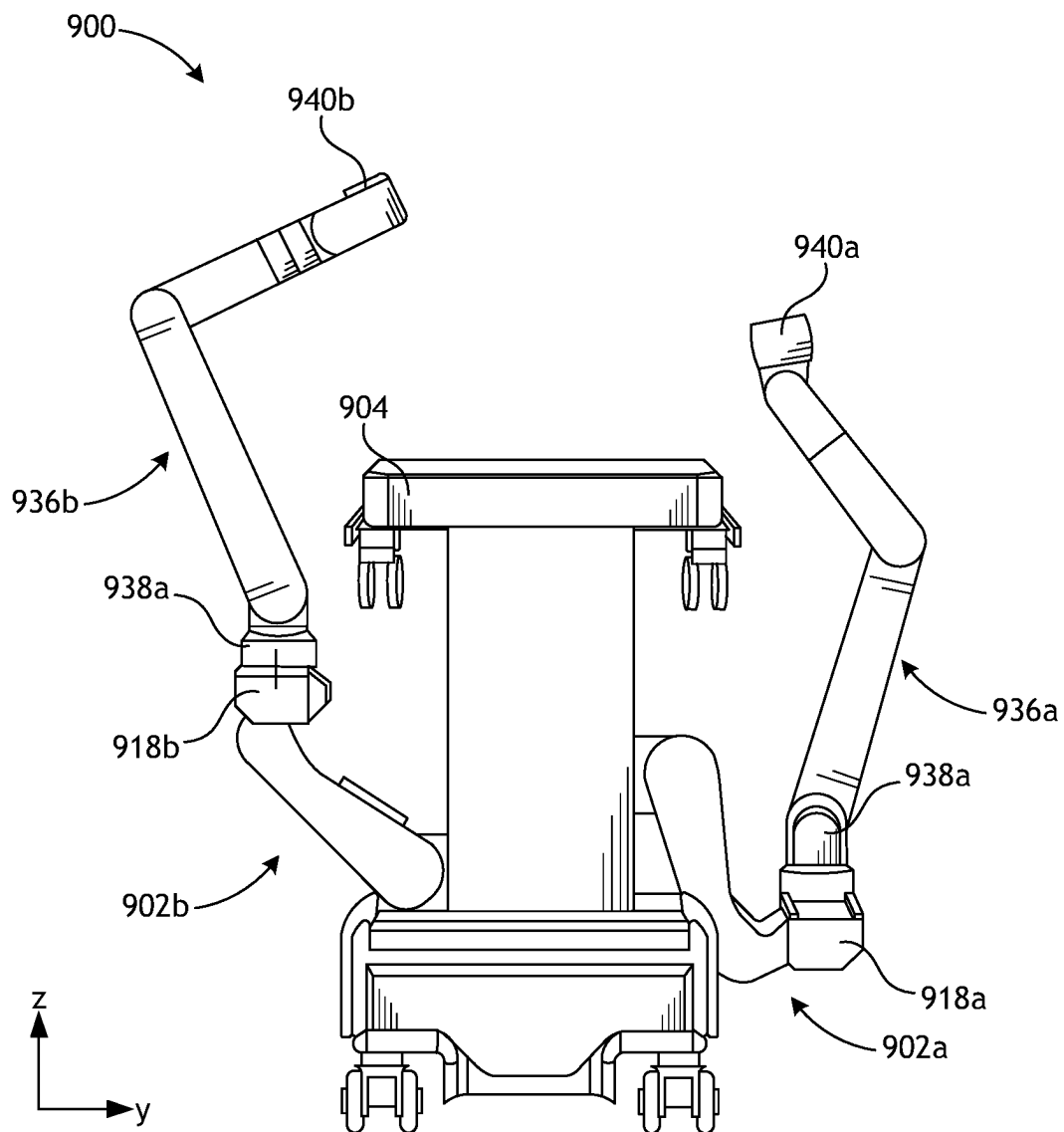


FIG. 9B

*FIG. 9C*

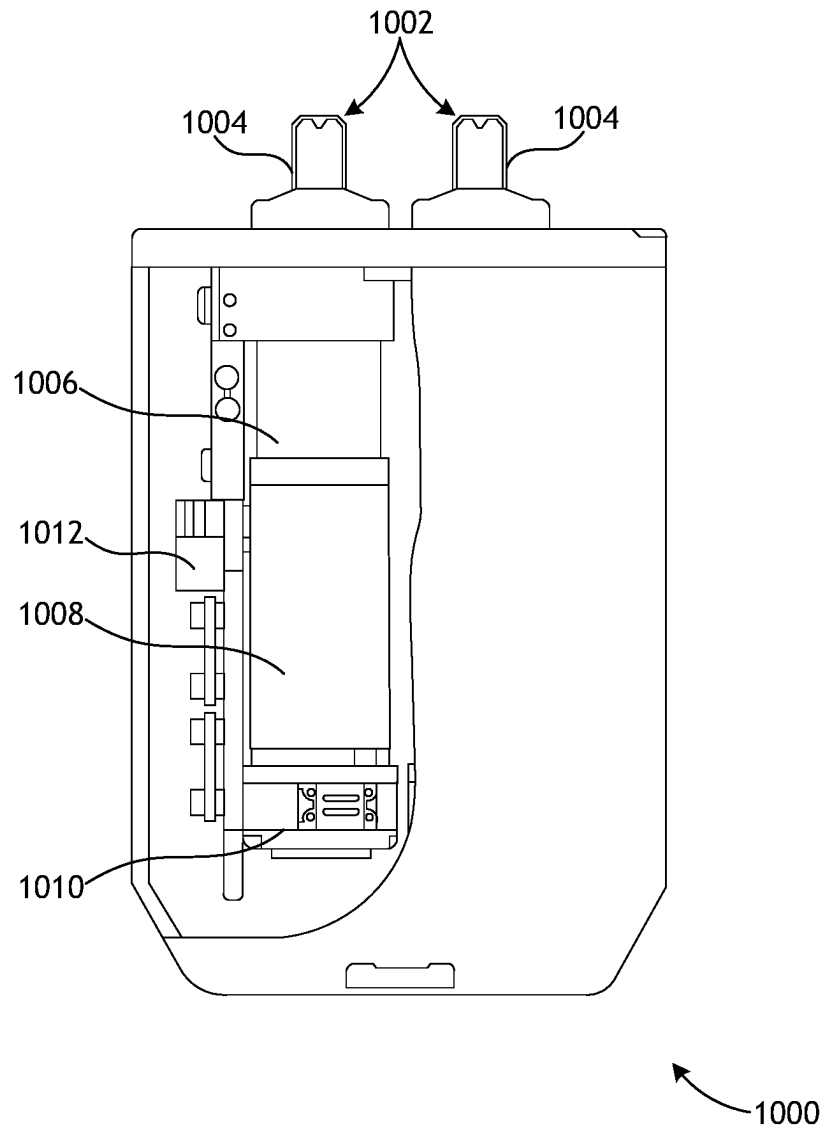


FIG. 10

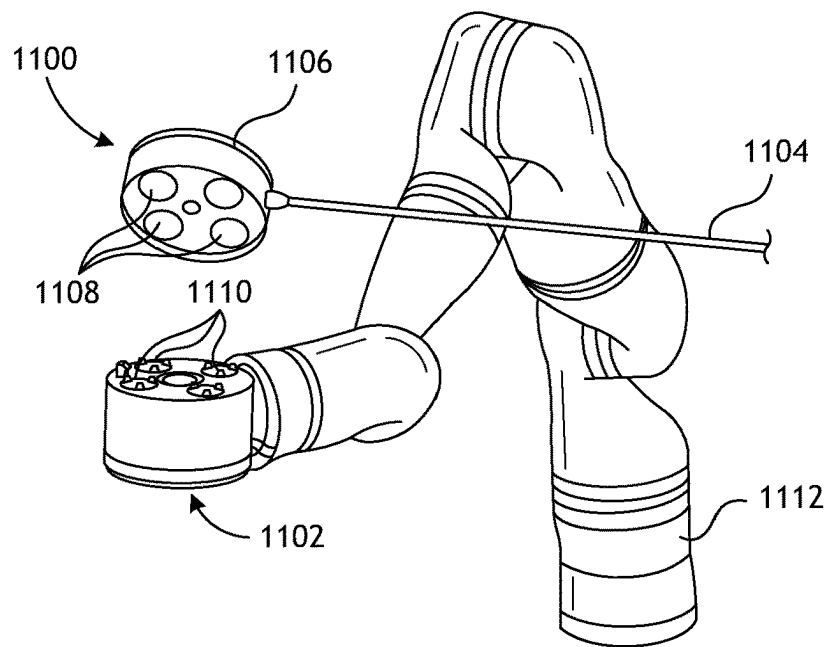


FIG. 11

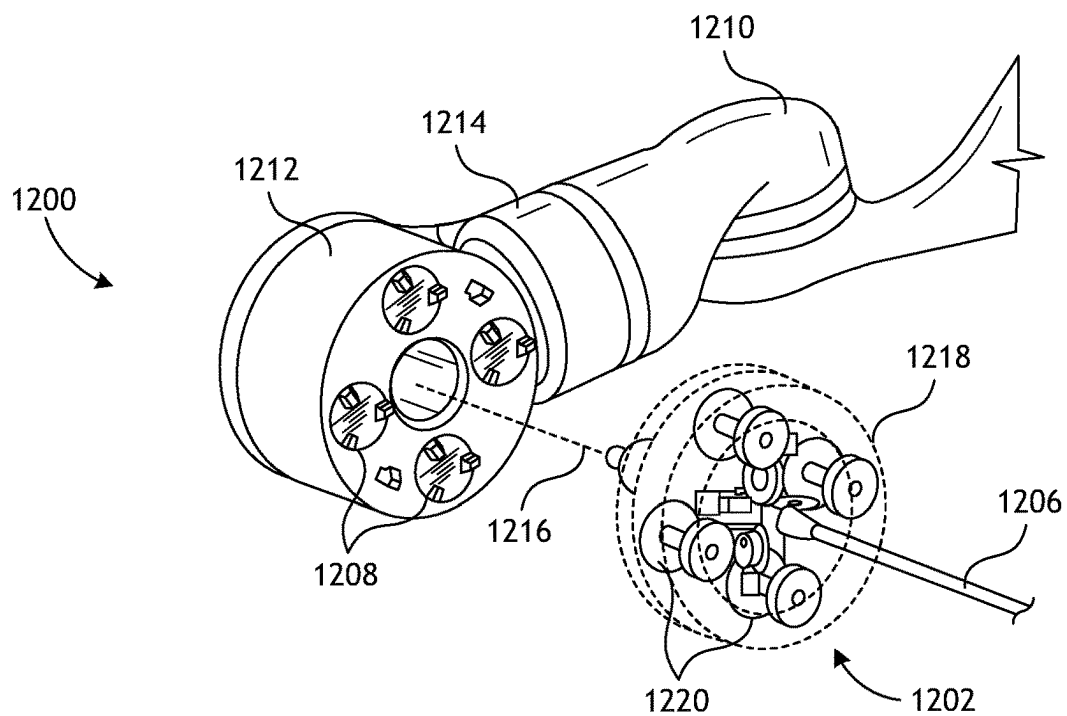


FIG. 12

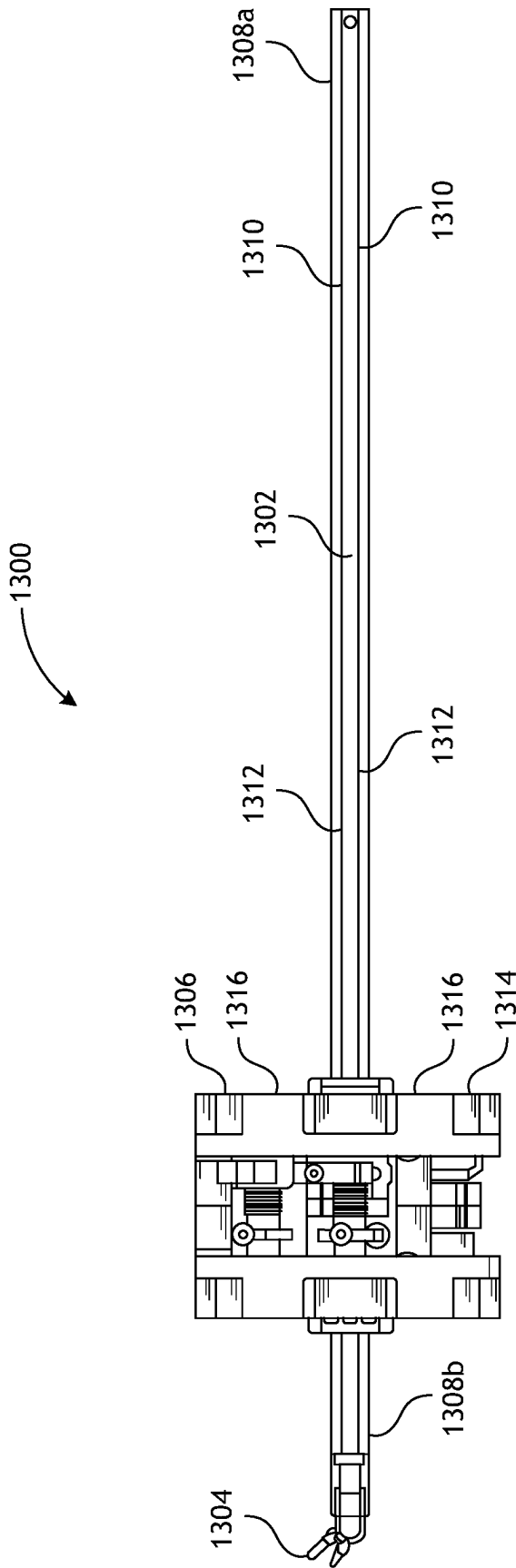


FIG. 13

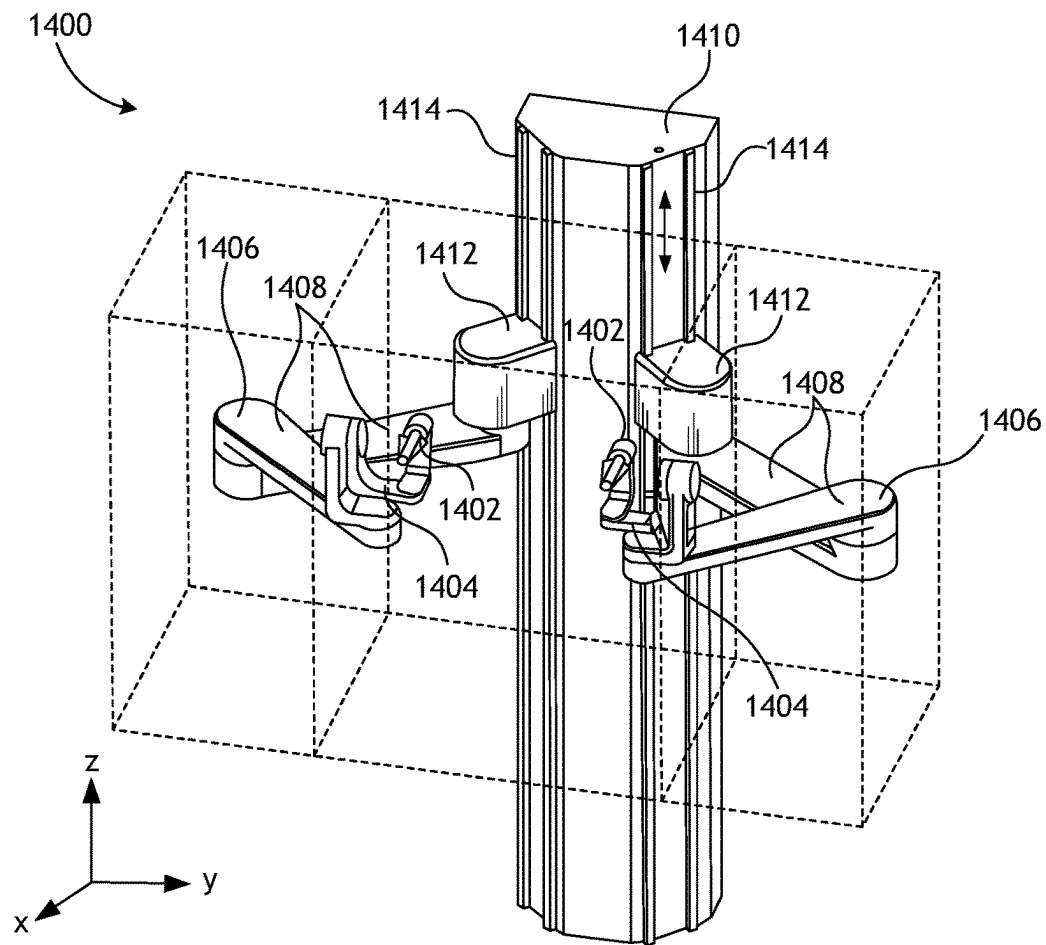


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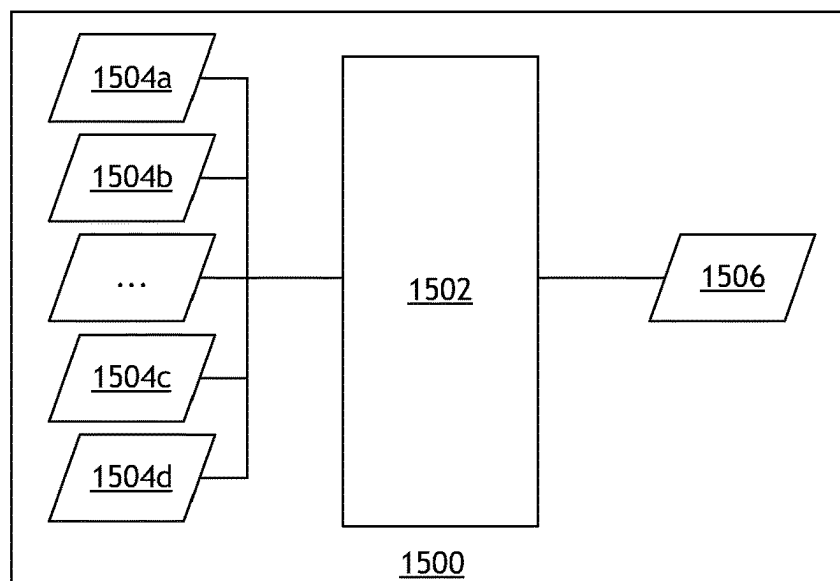


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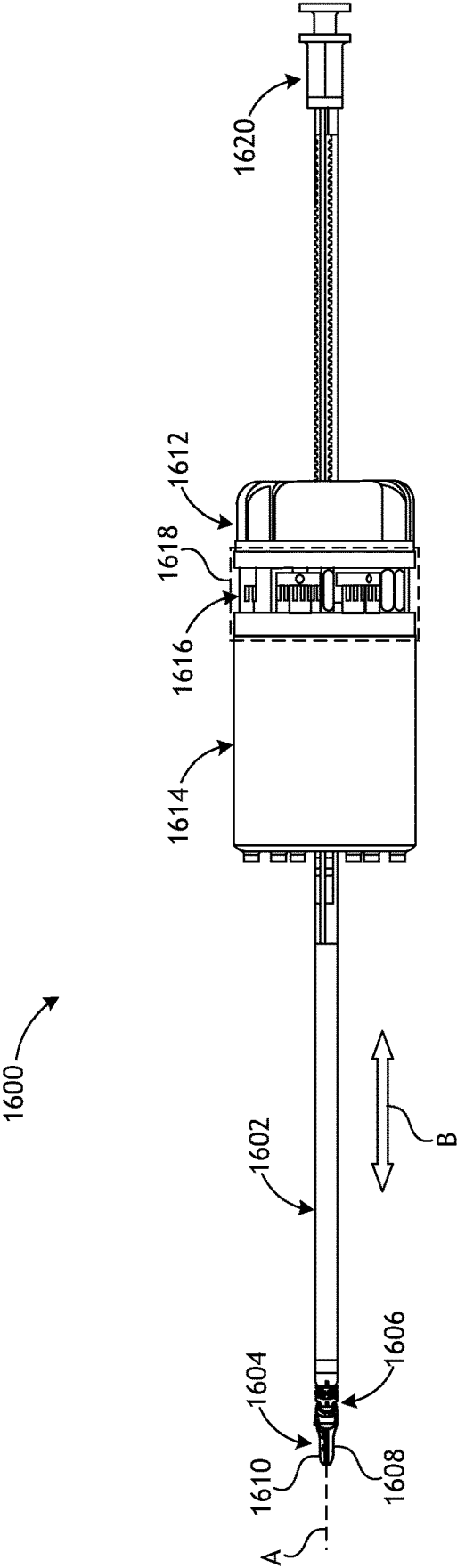
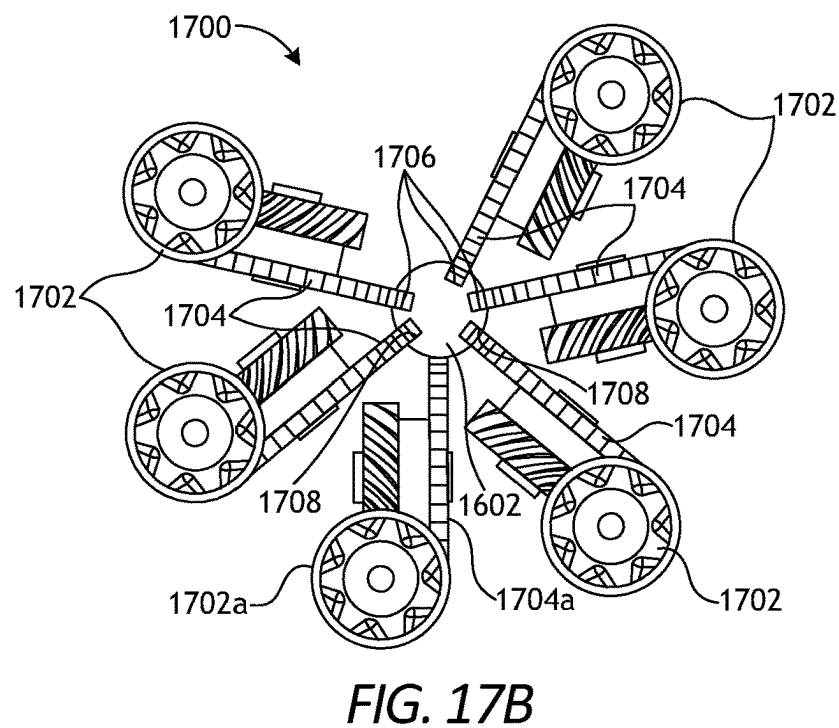
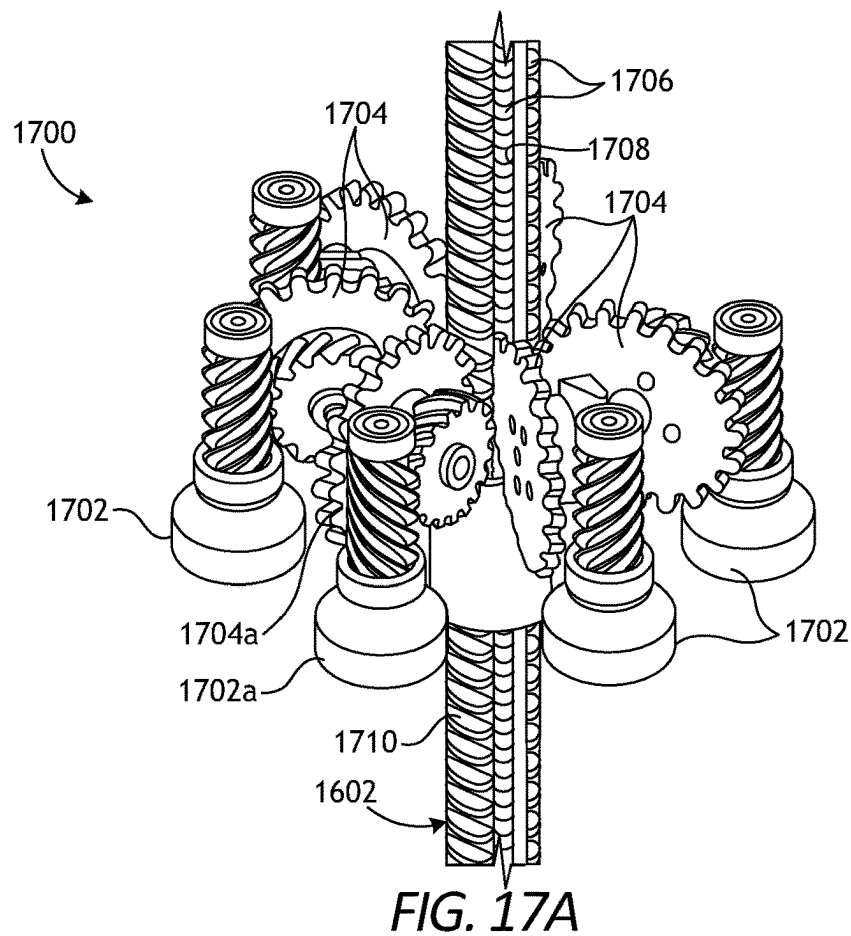


FIG. 16



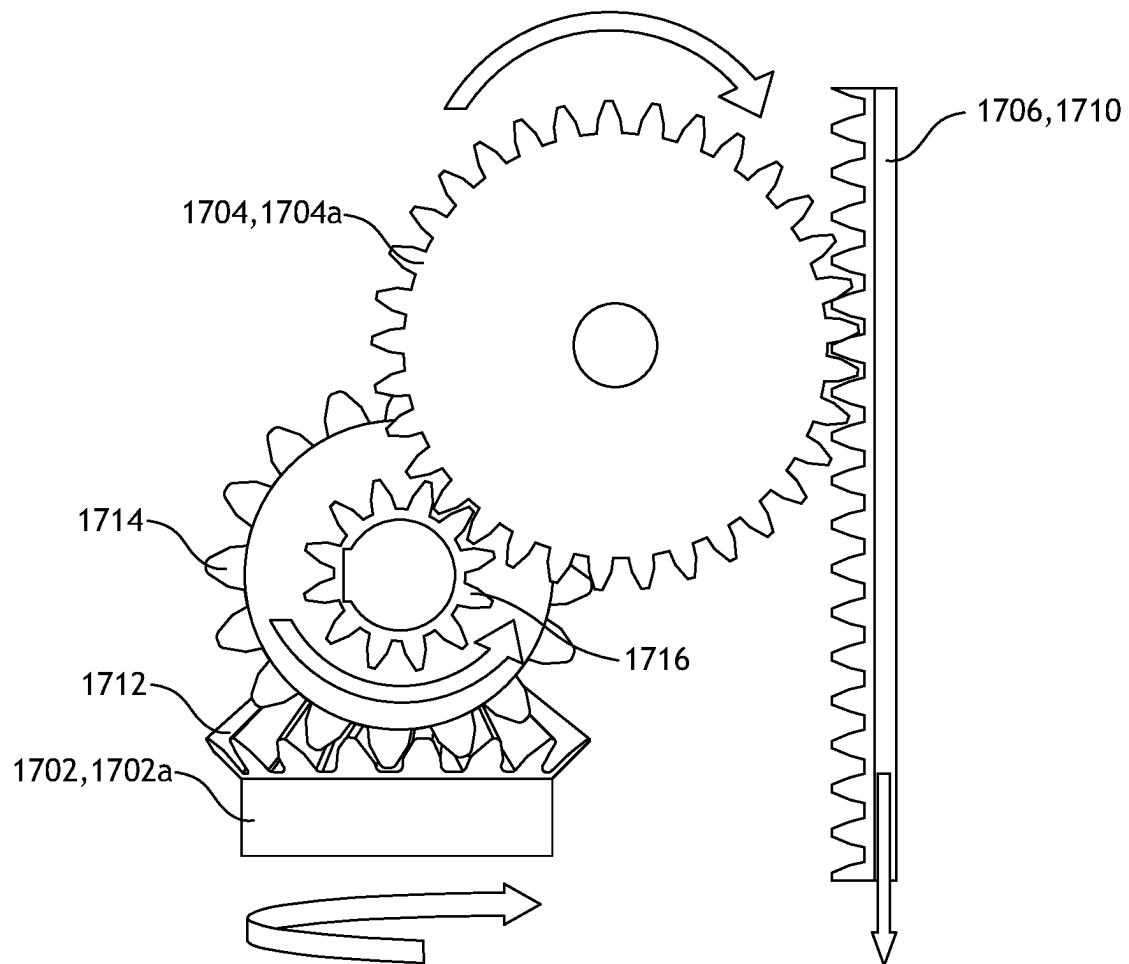


FIG. 17C

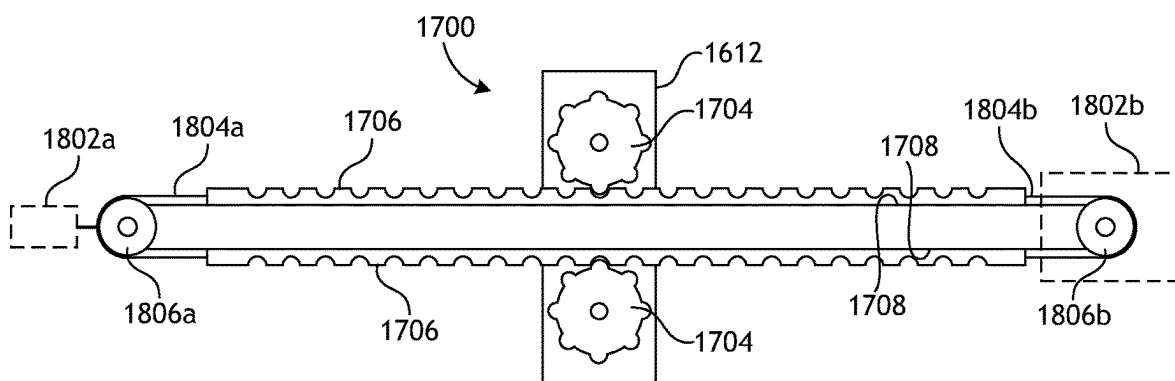


FIG. 18A

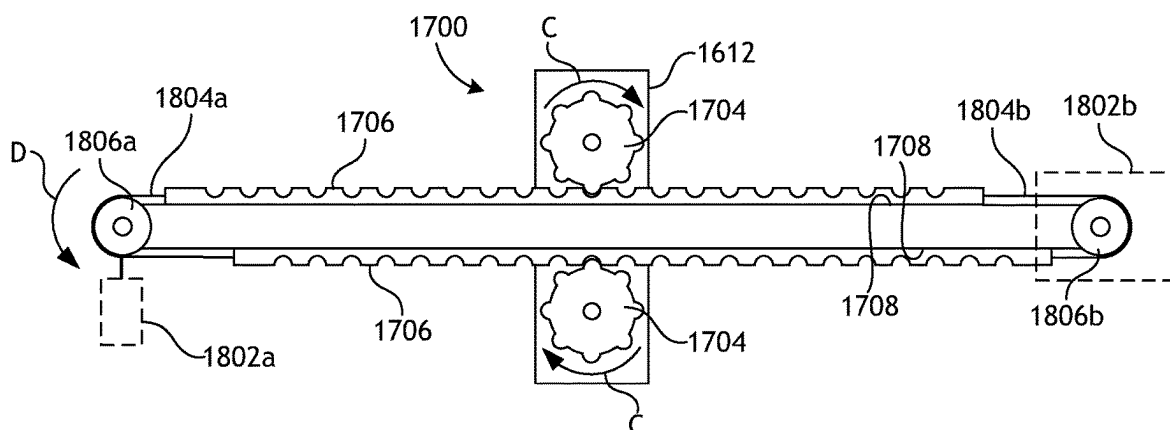


FIG. 18B

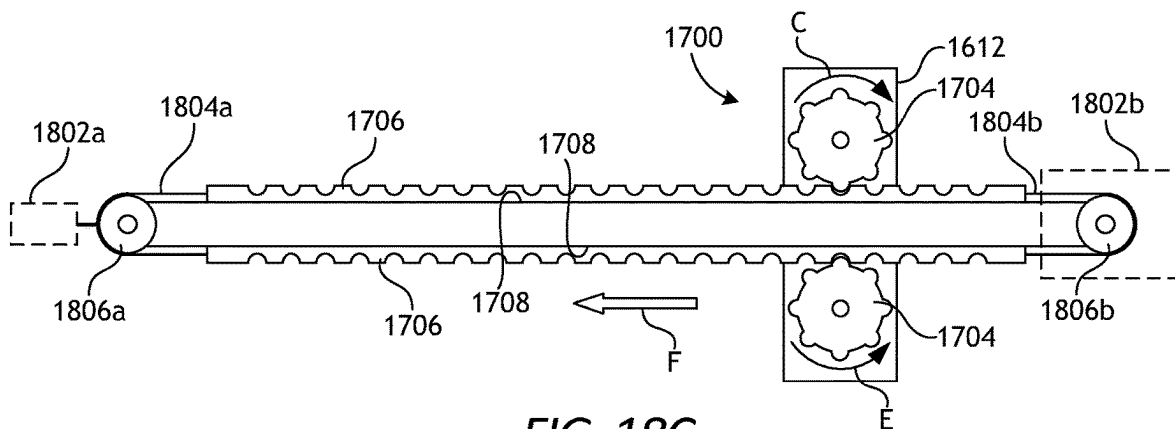


FIG. 18C

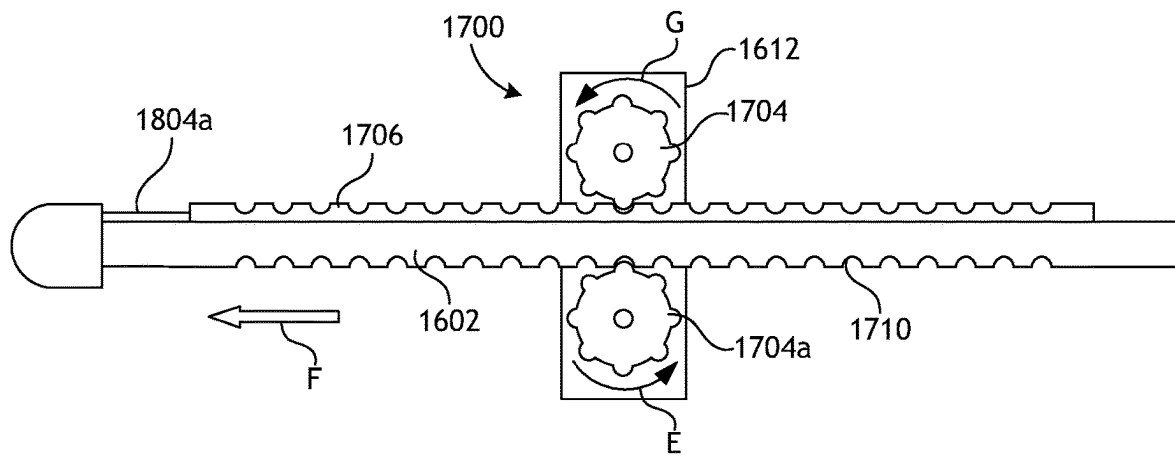


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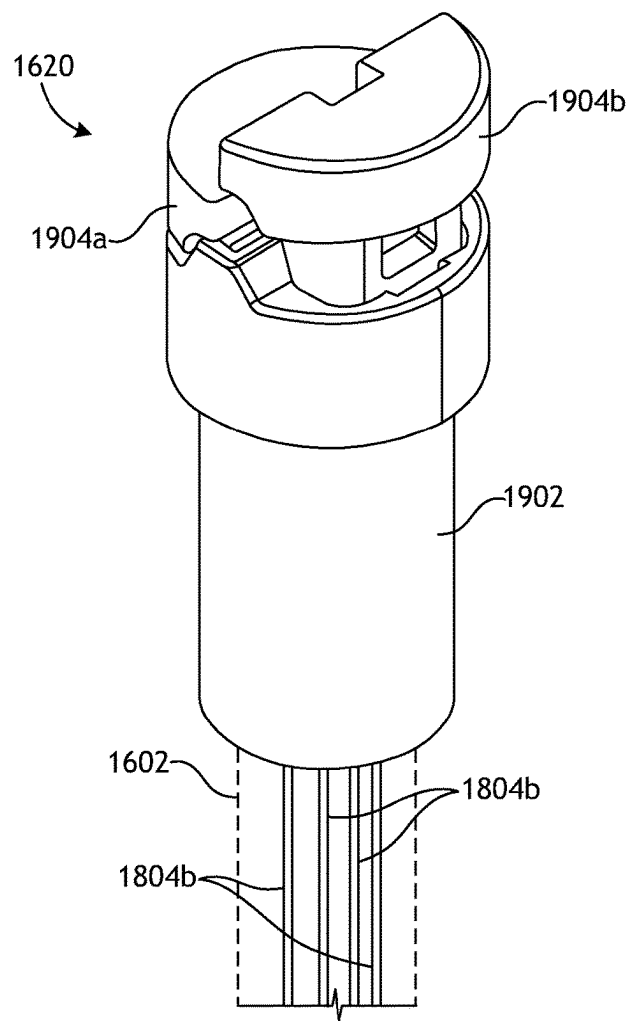


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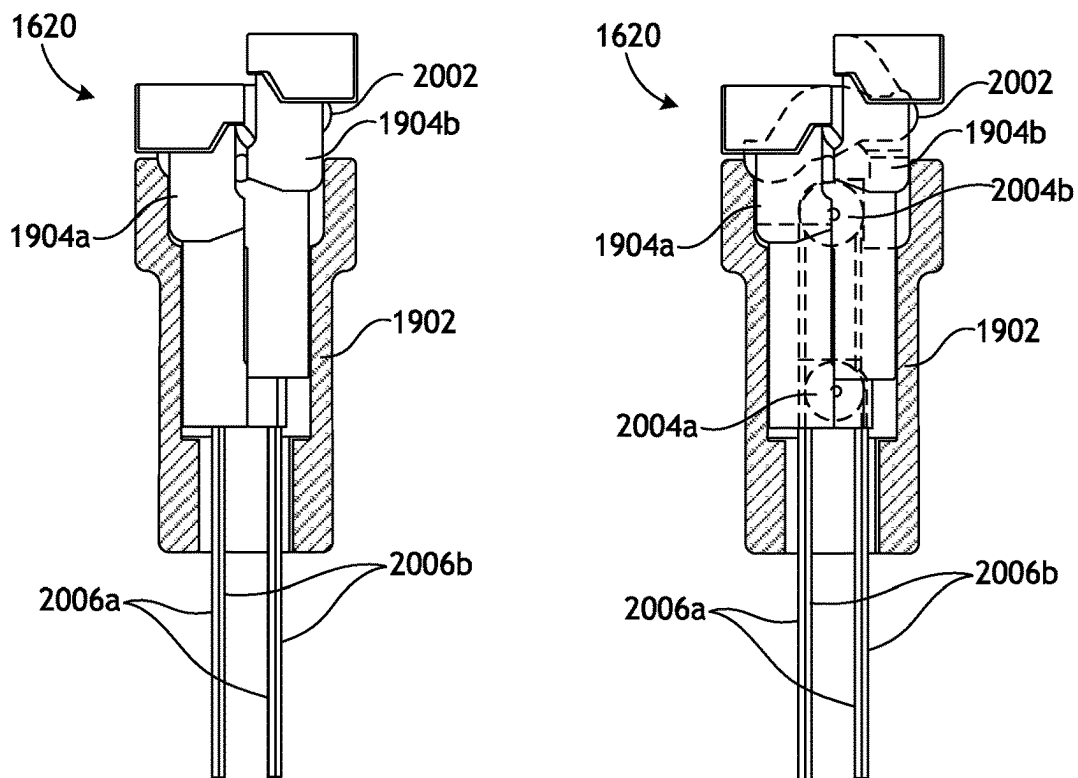


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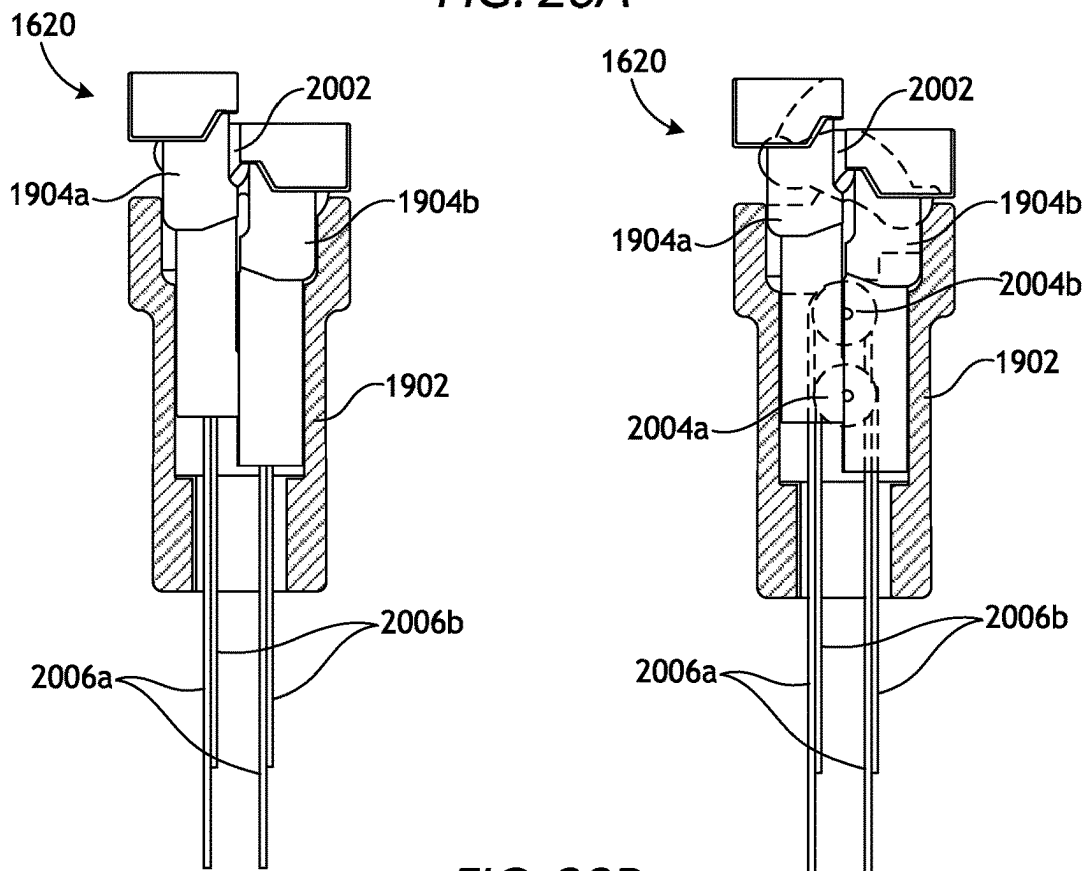


FIG. 20B

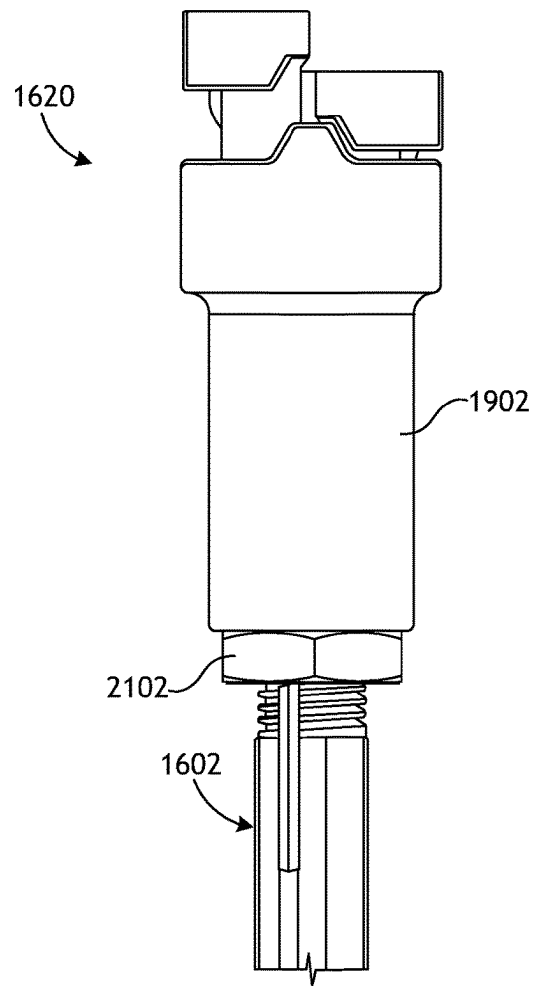


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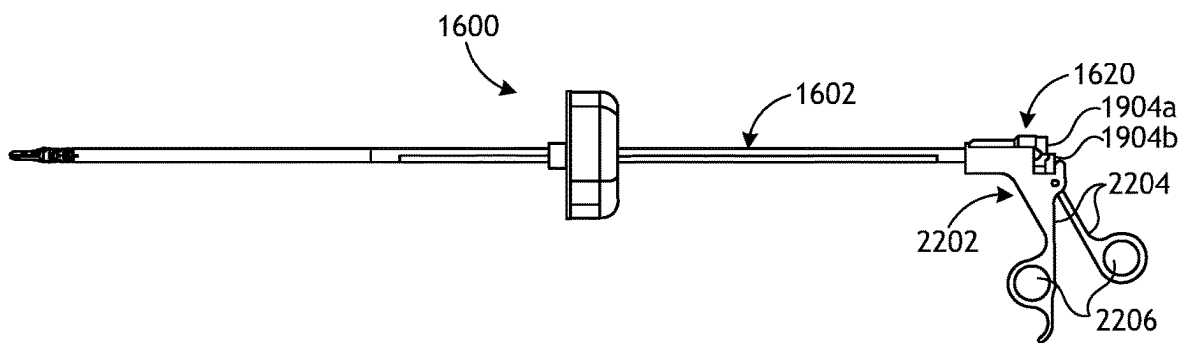


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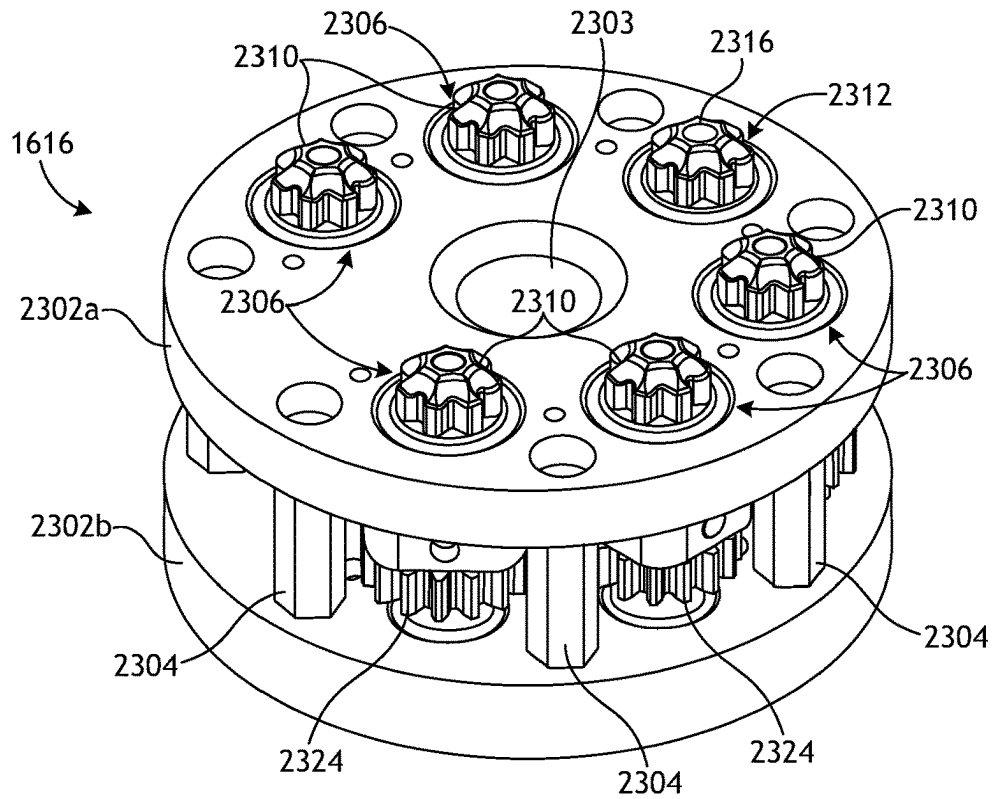


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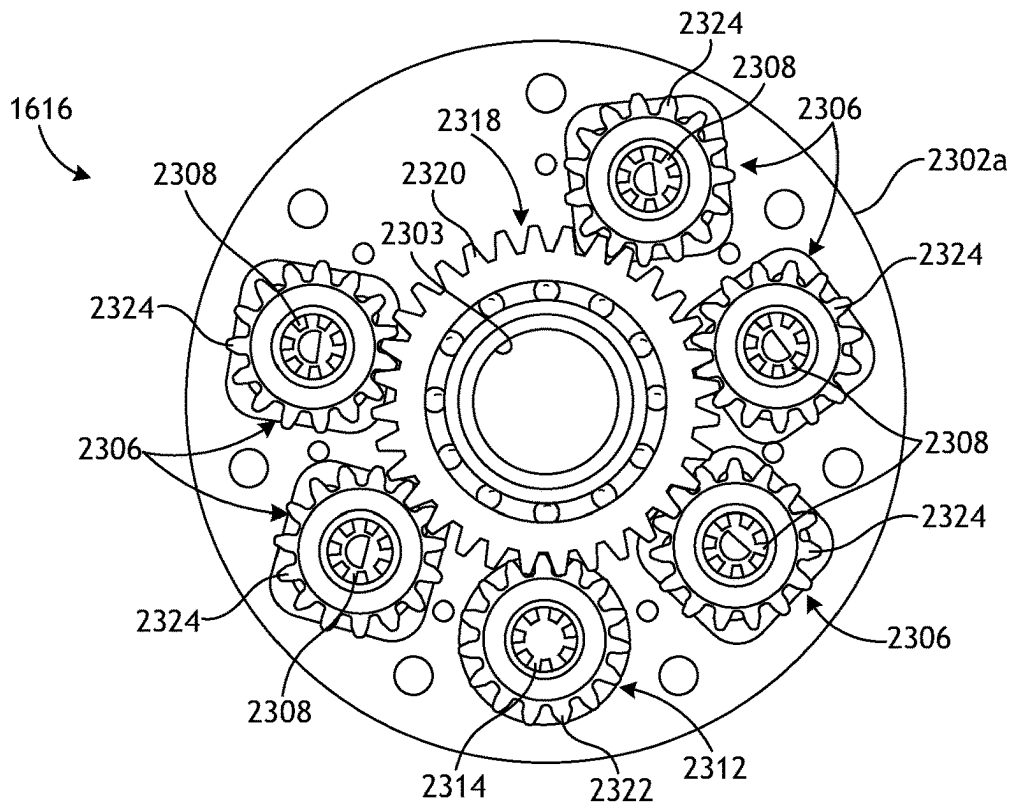


FIG. 23B

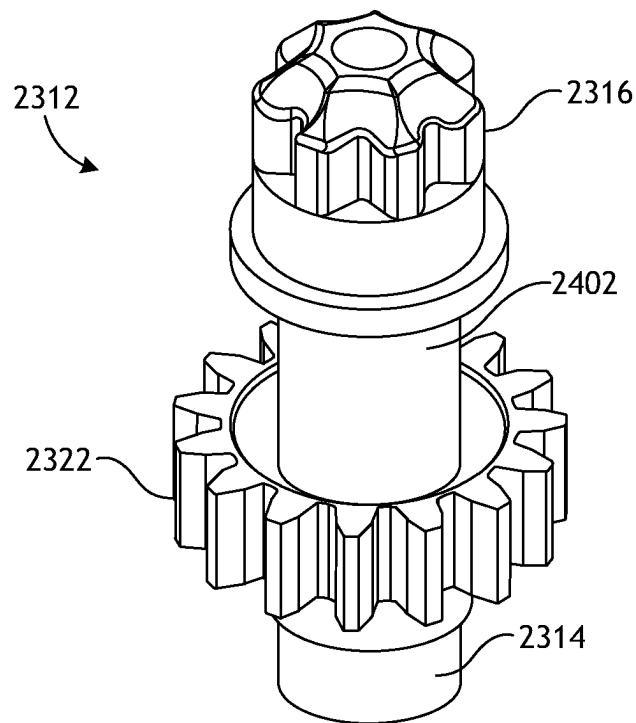


FIG. 24

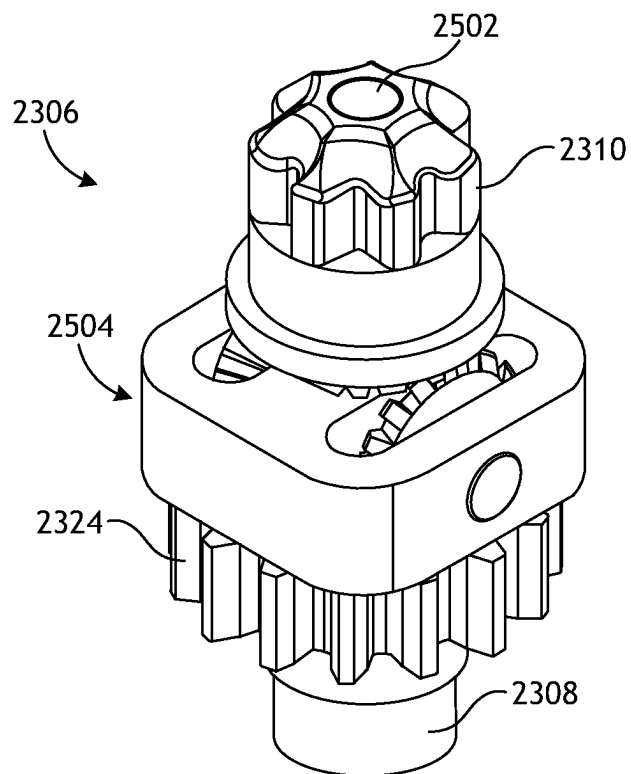


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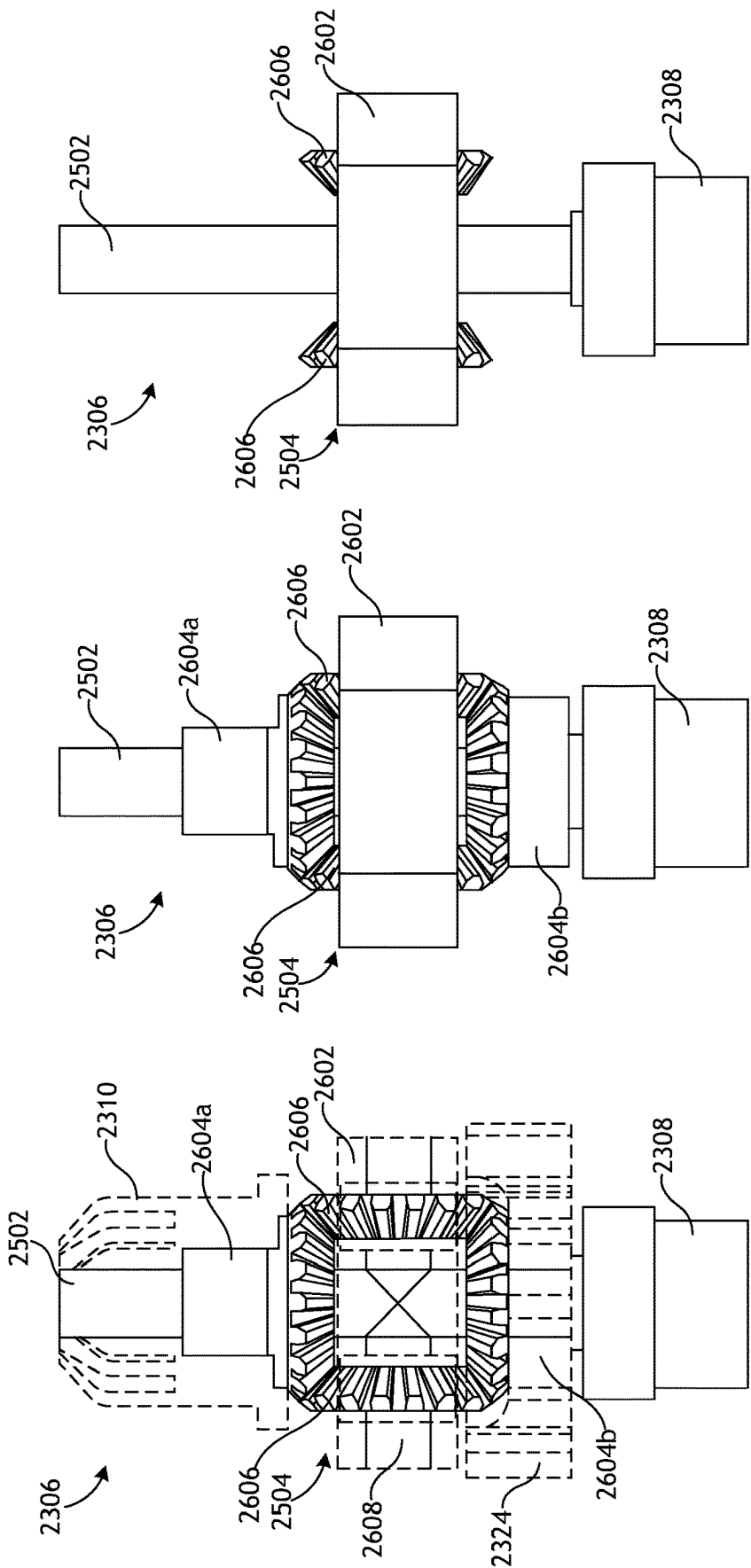


FIG. 26C

FIG. 26B

FIG. 26A

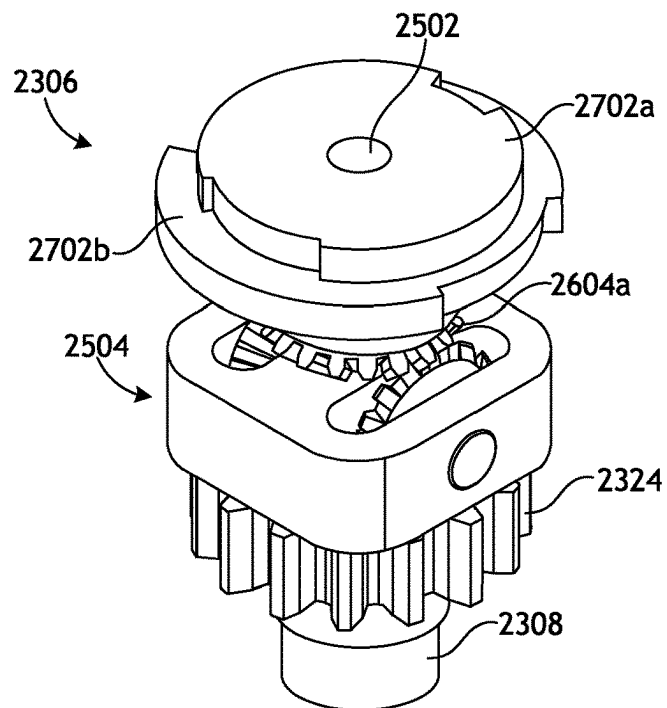


FIG. 27A

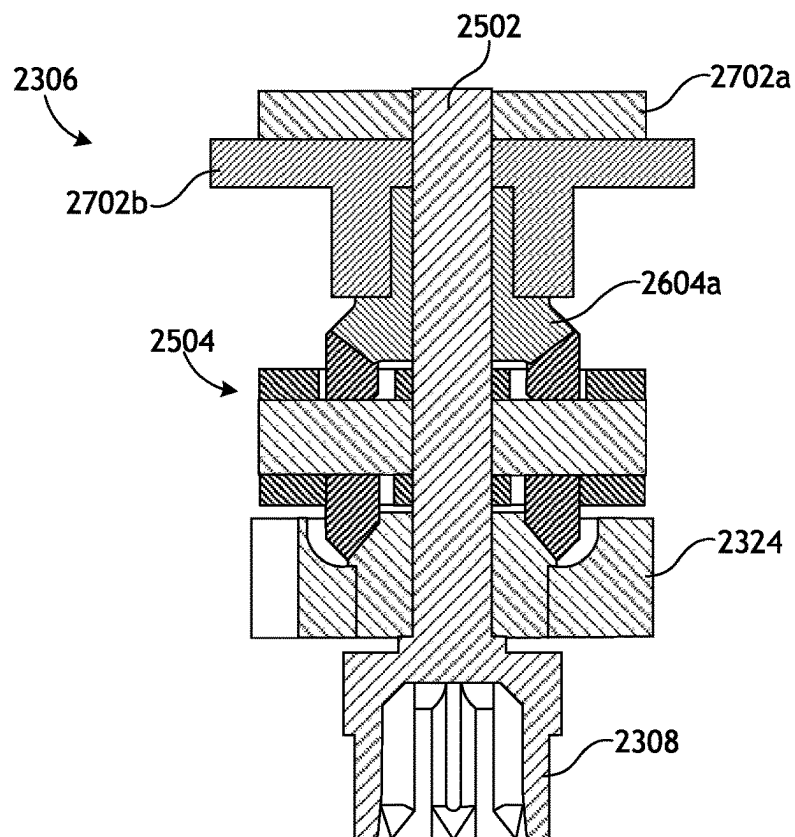


FIG. 27B

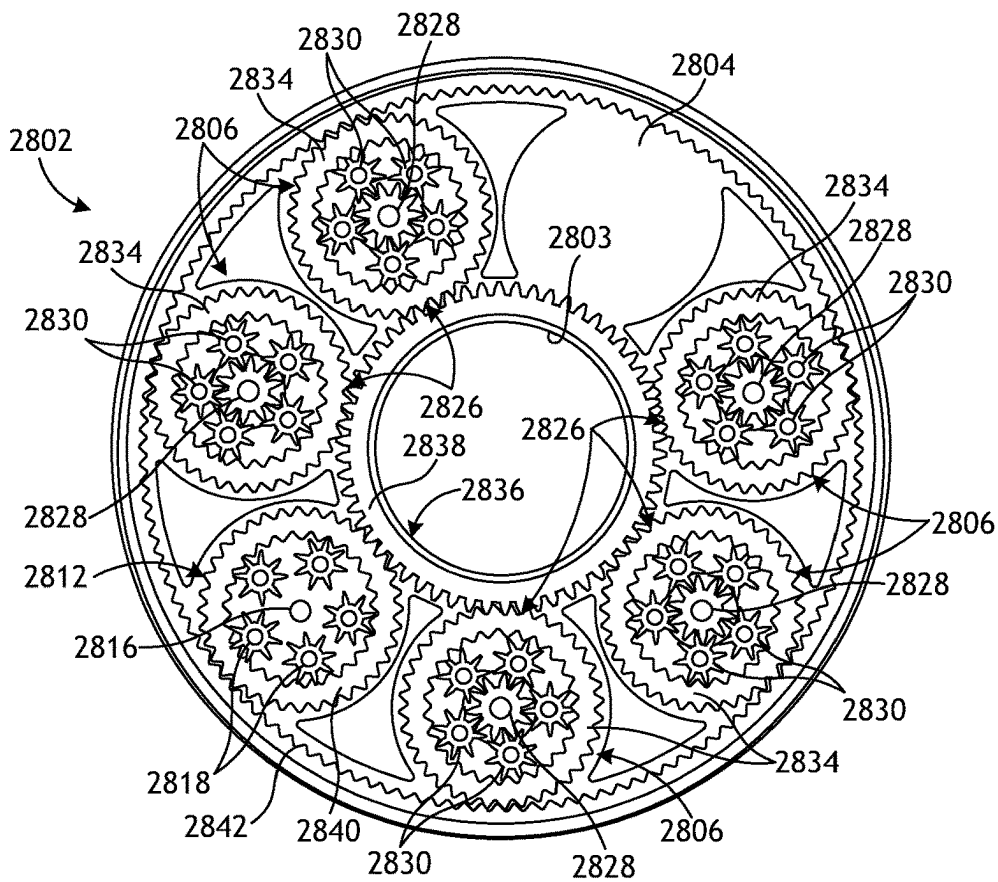


FIG. 28

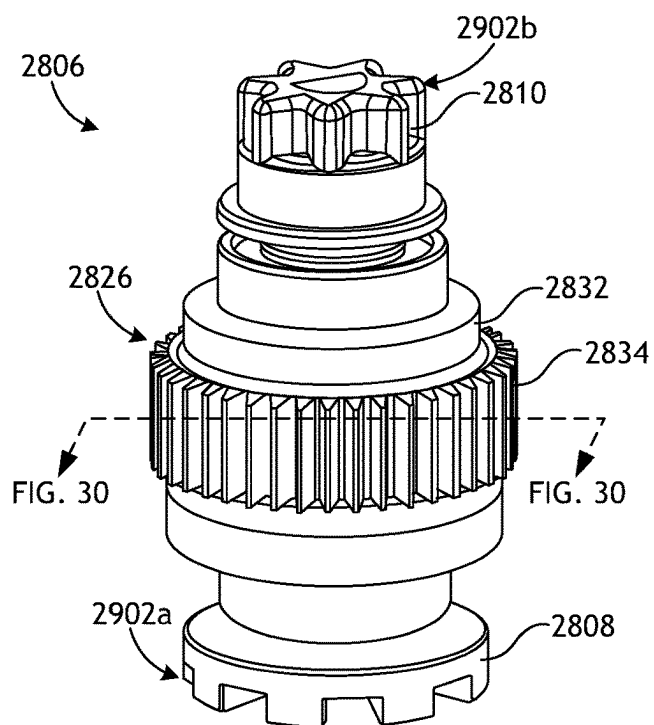


FIG. 29

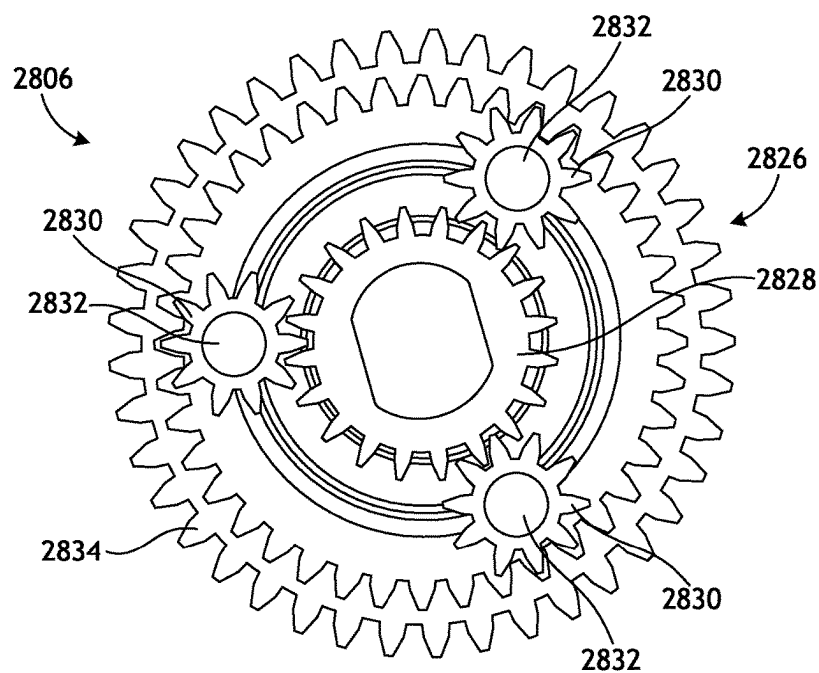


FIG. 30

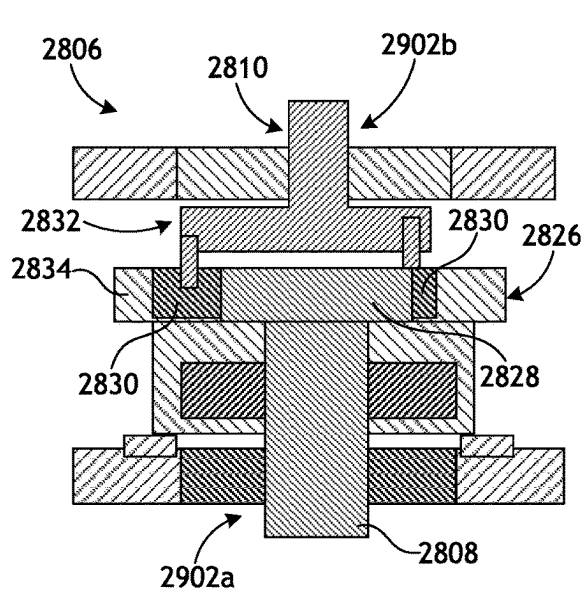


FIG. 31

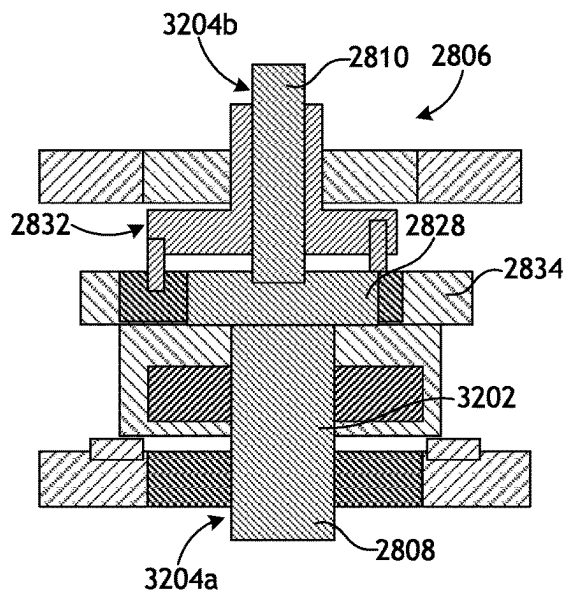


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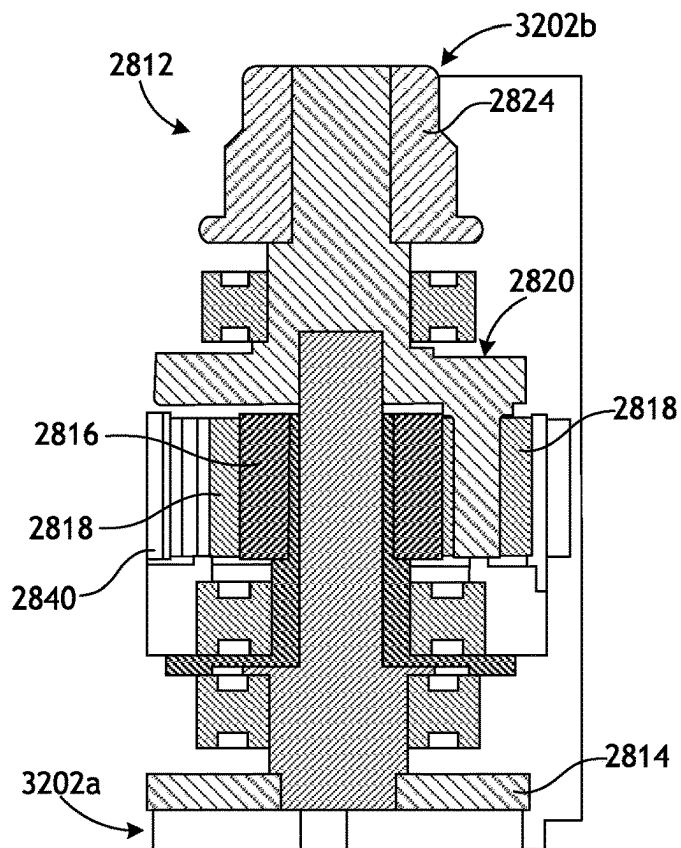


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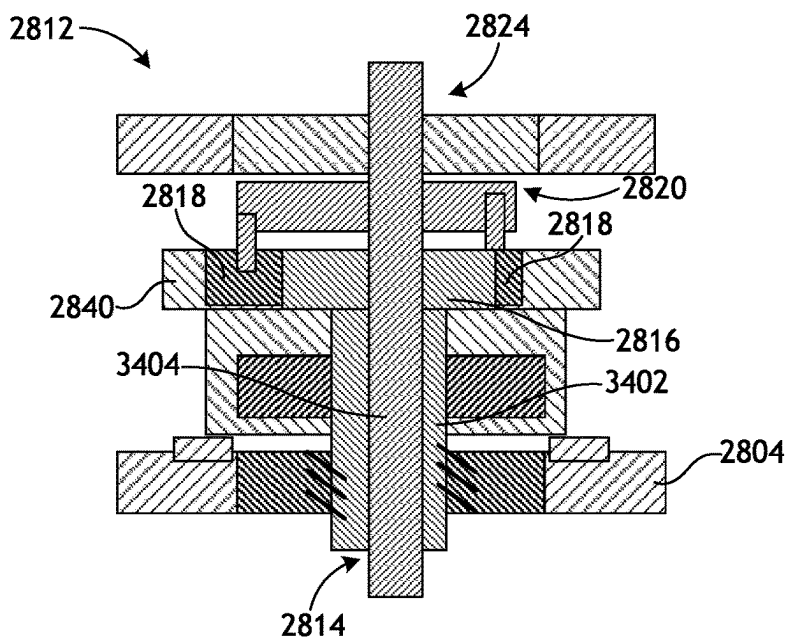


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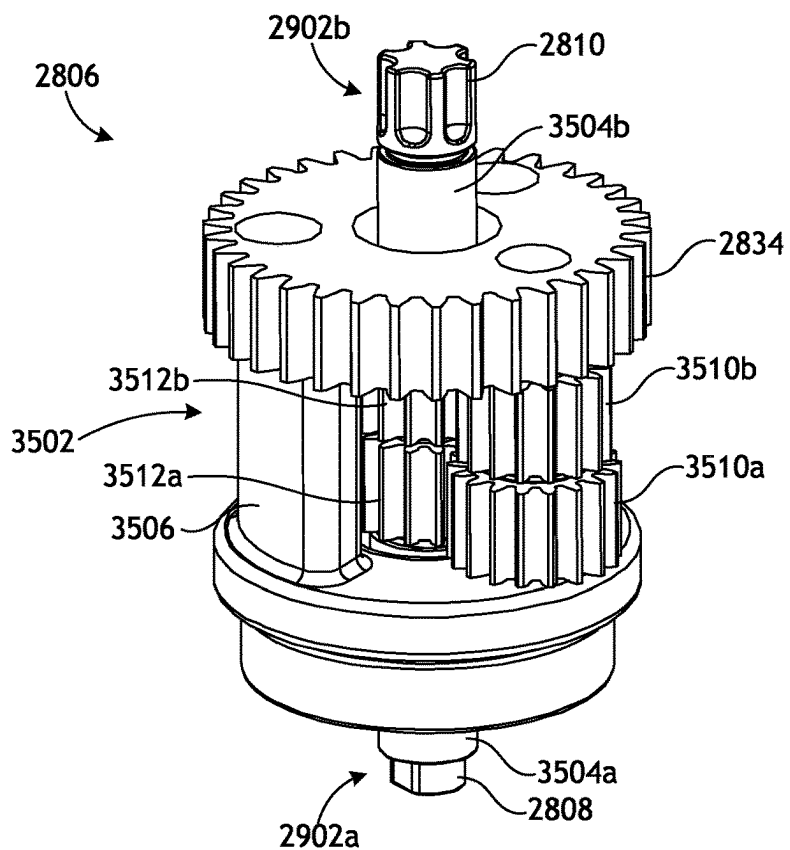


FIG. 35A

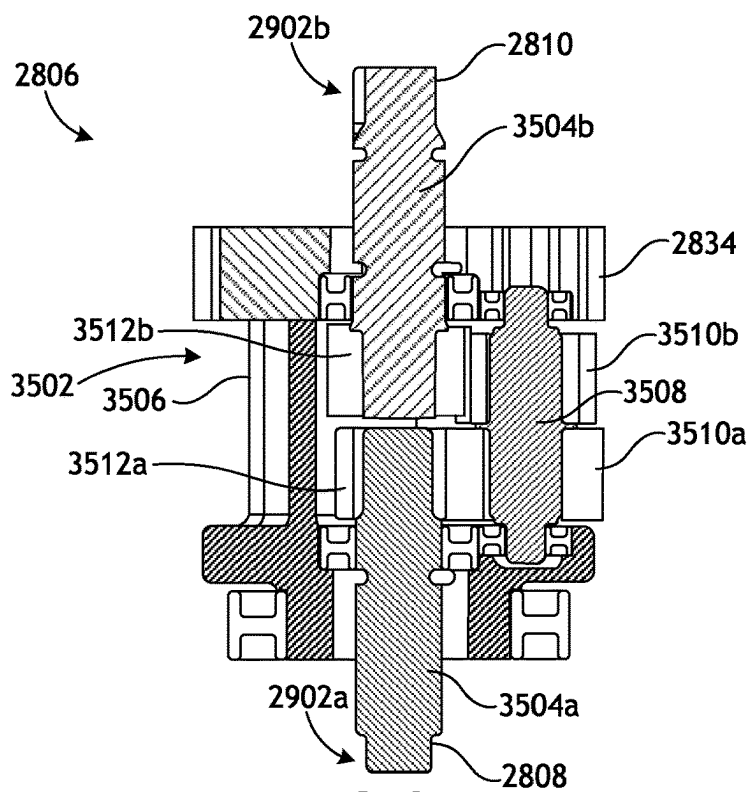


FIG. 35B

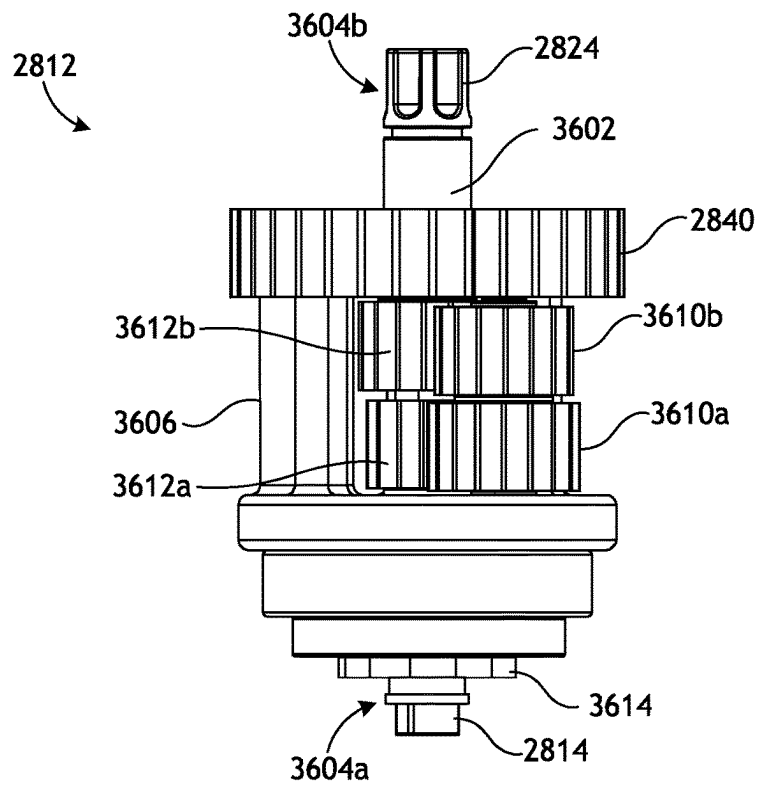


FIG. 36A

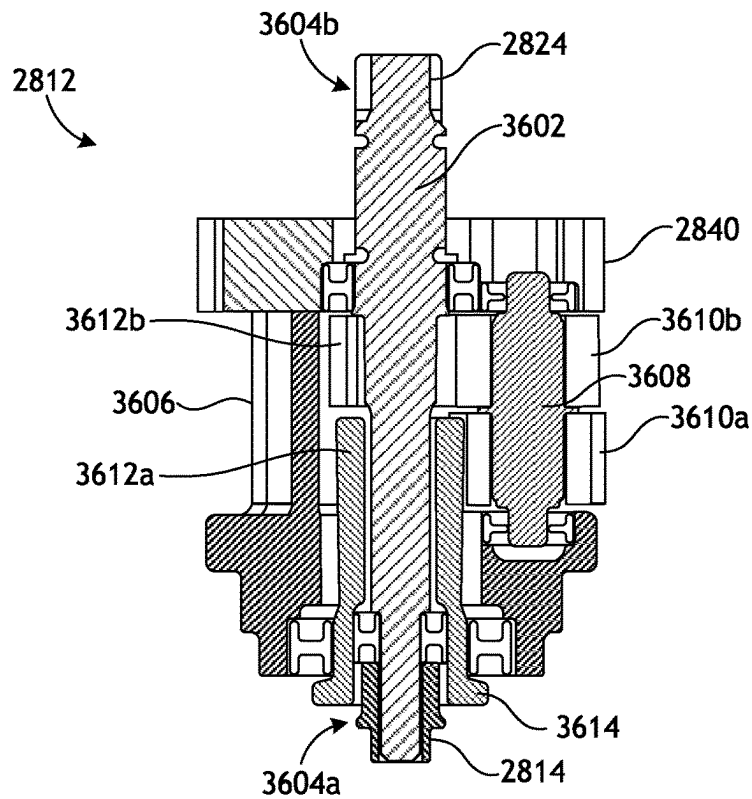


FIG. 36B

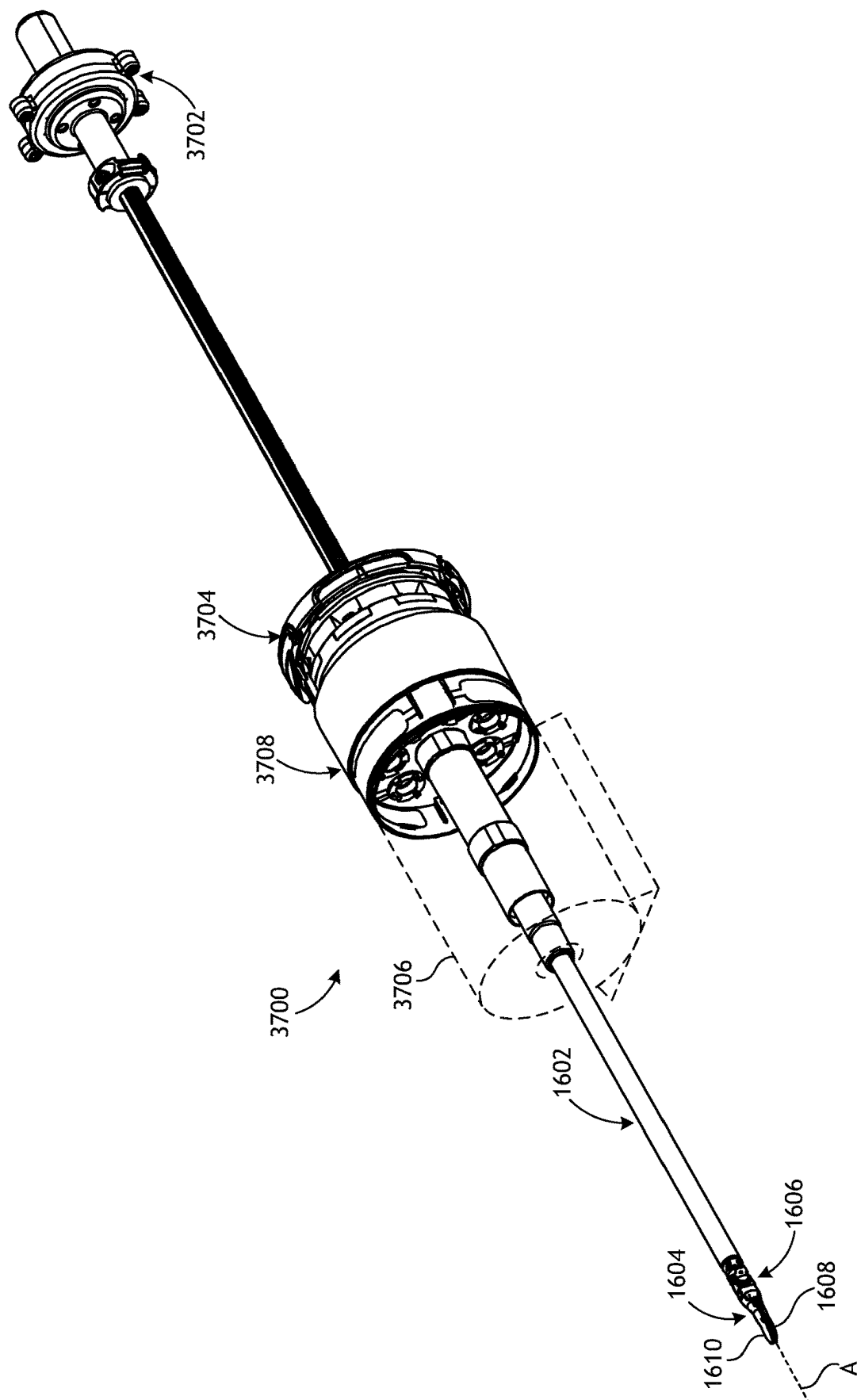
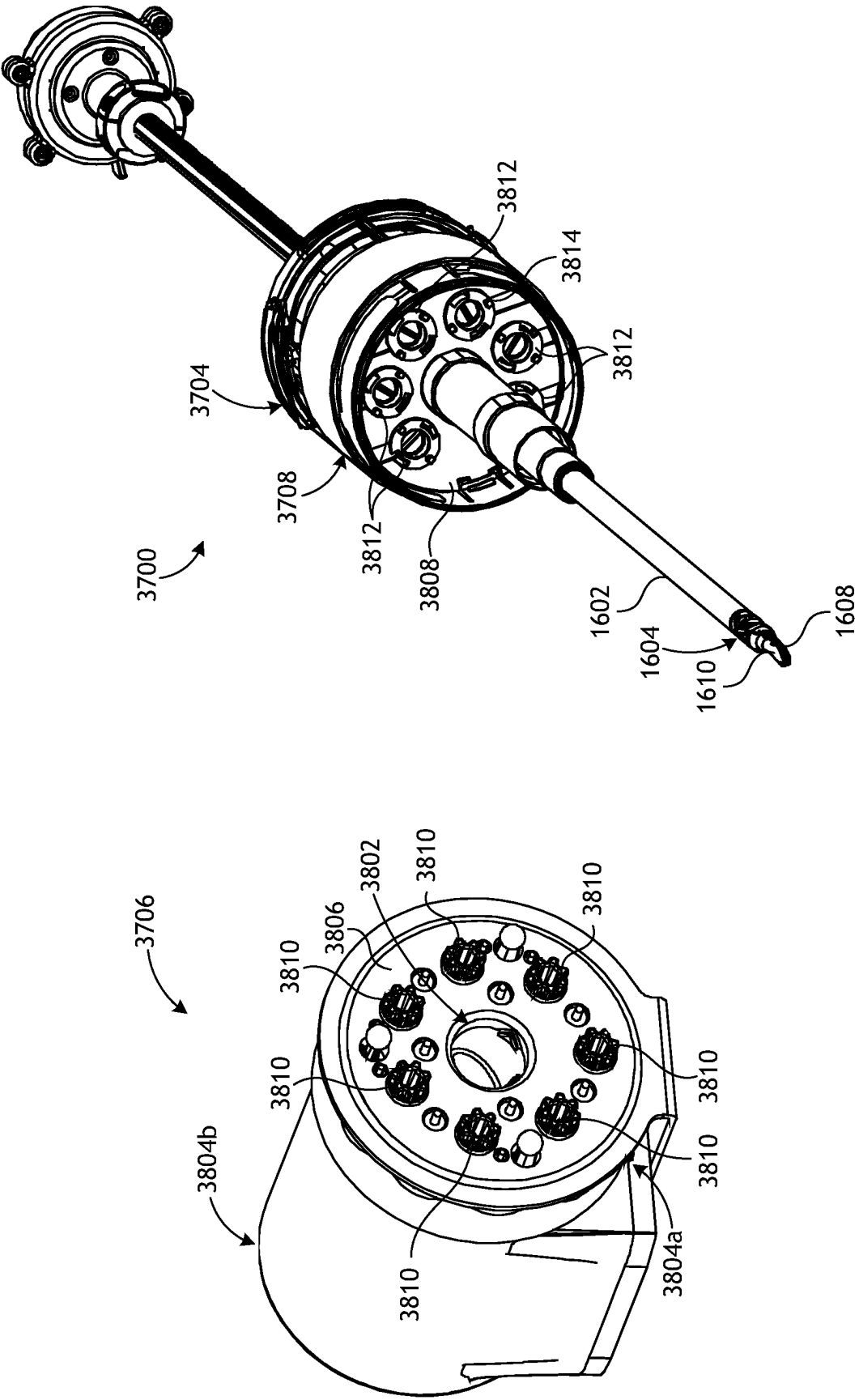


FIG. 37



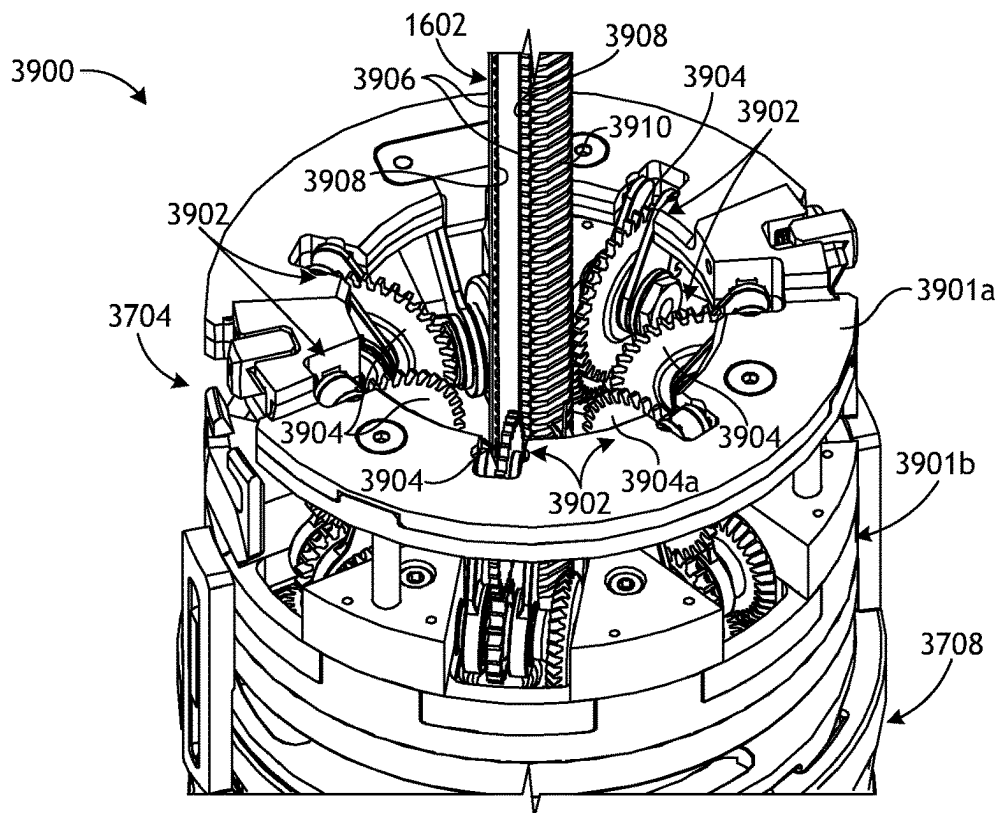
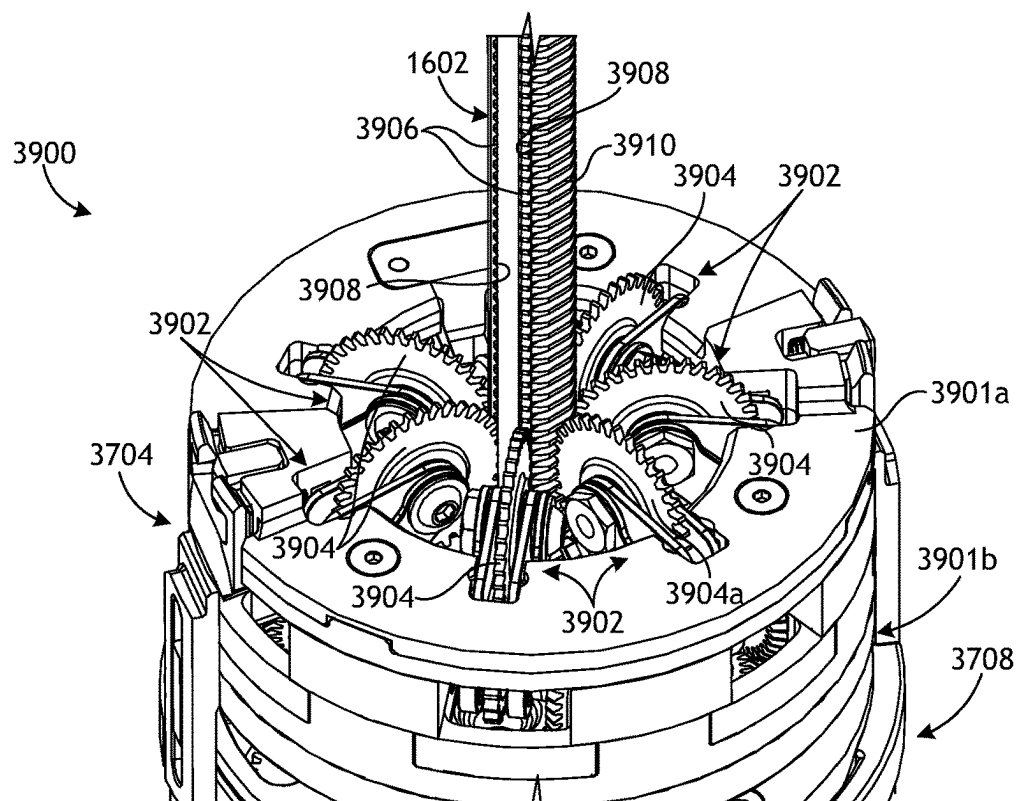


FIG. 39A



**FIG. 39B**

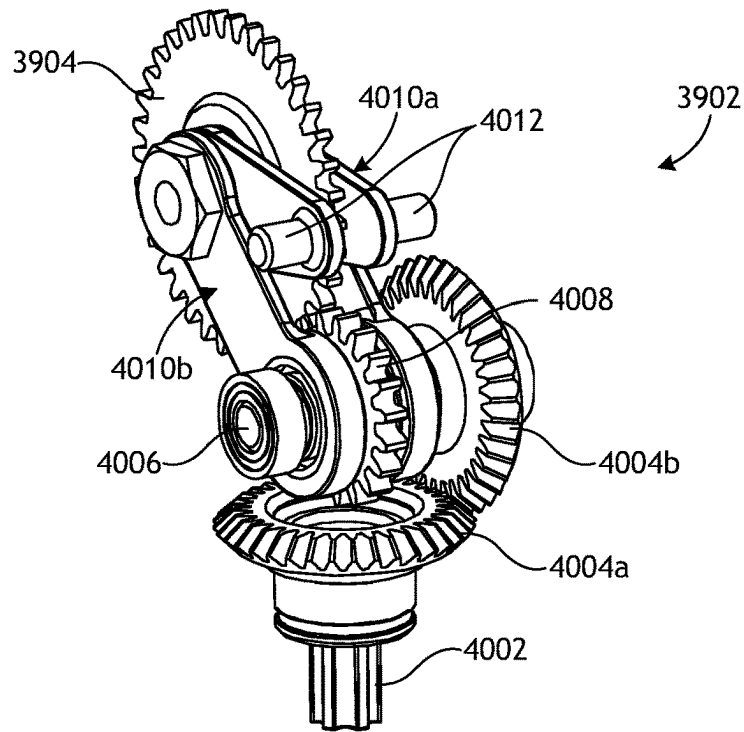


FIG. 40

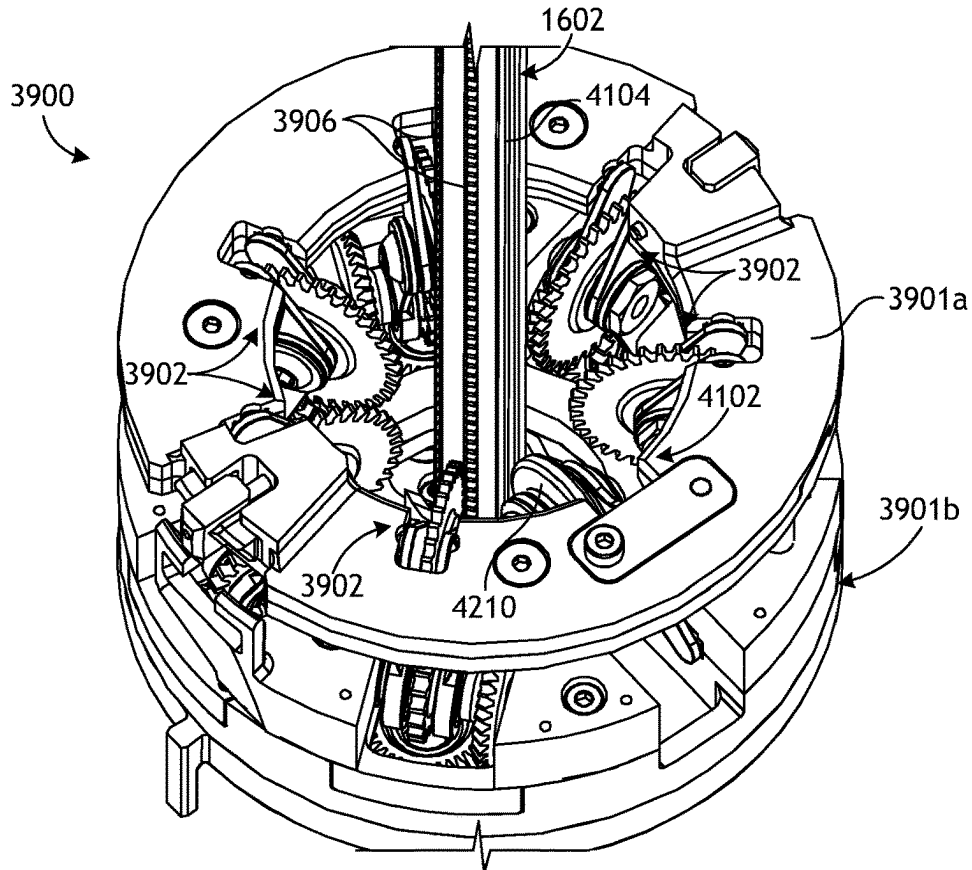


FIG. 41

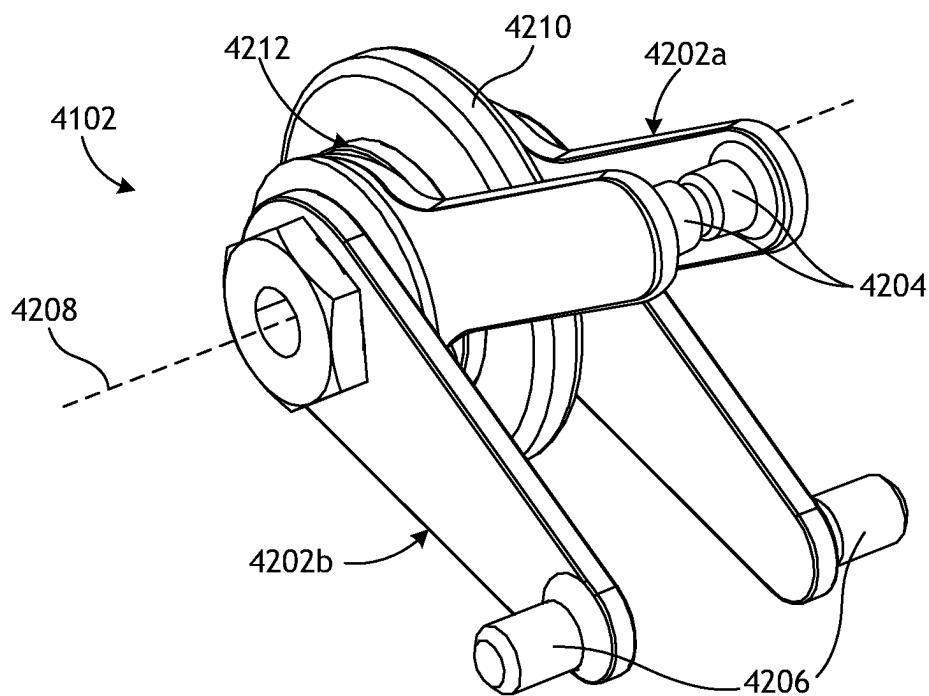


FIG. 42

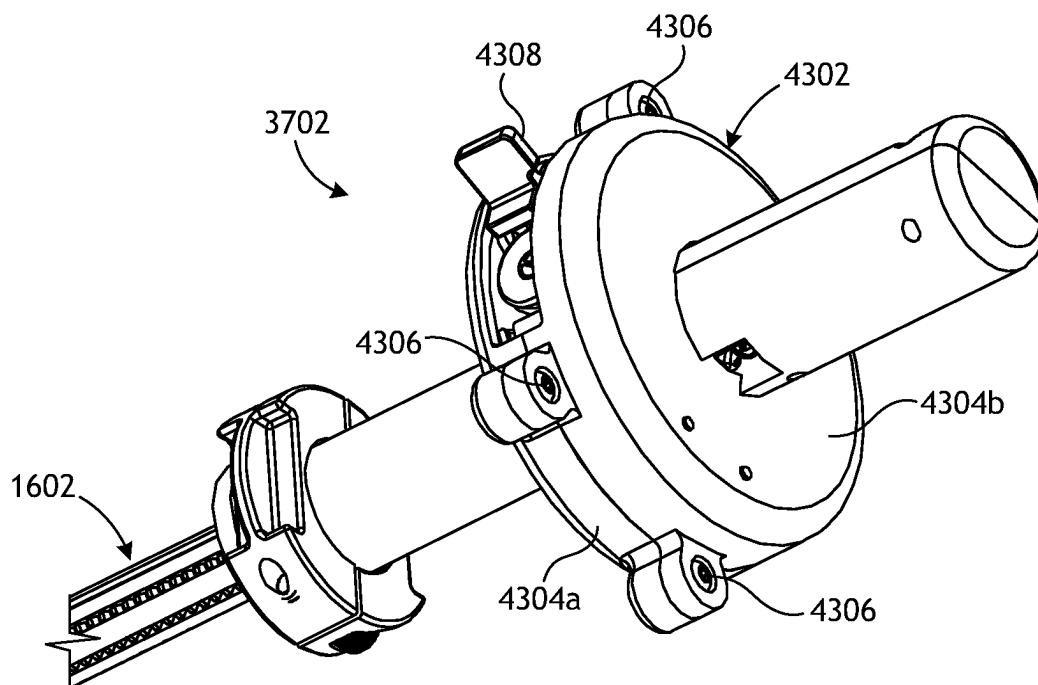


FIG. 43

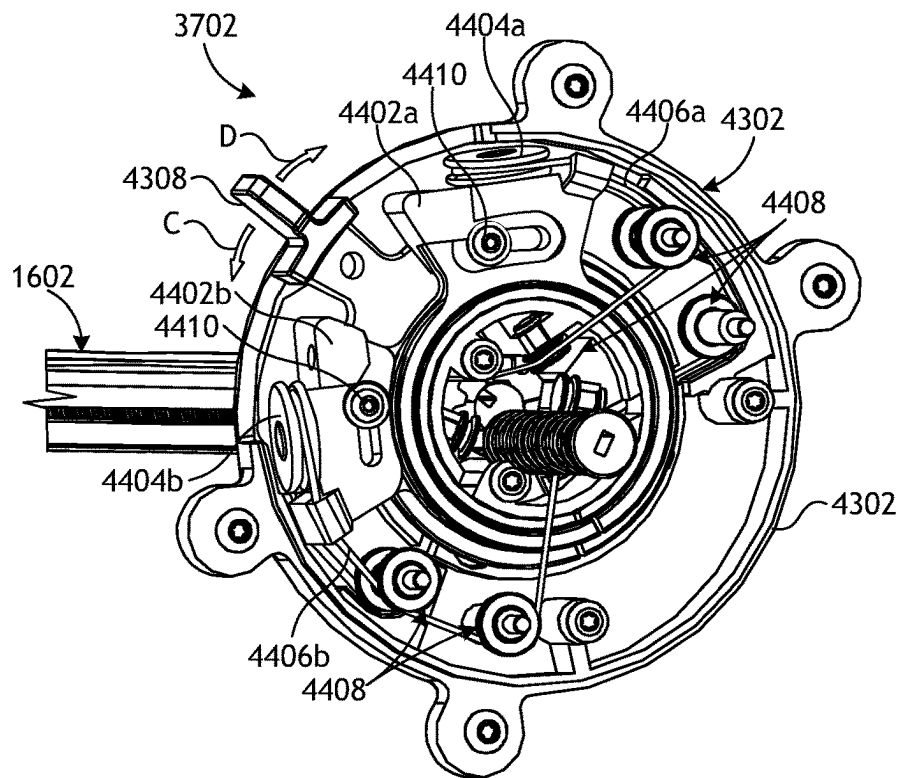


FIG. 44

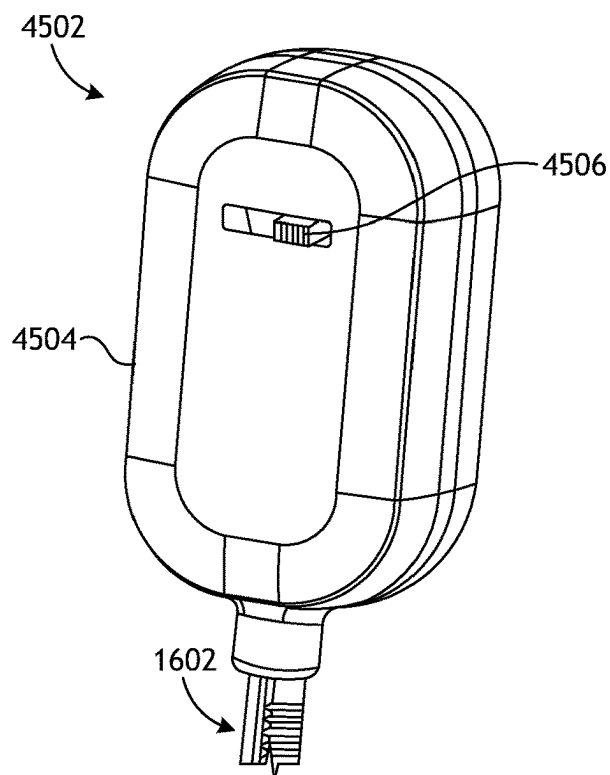


FIG. 45

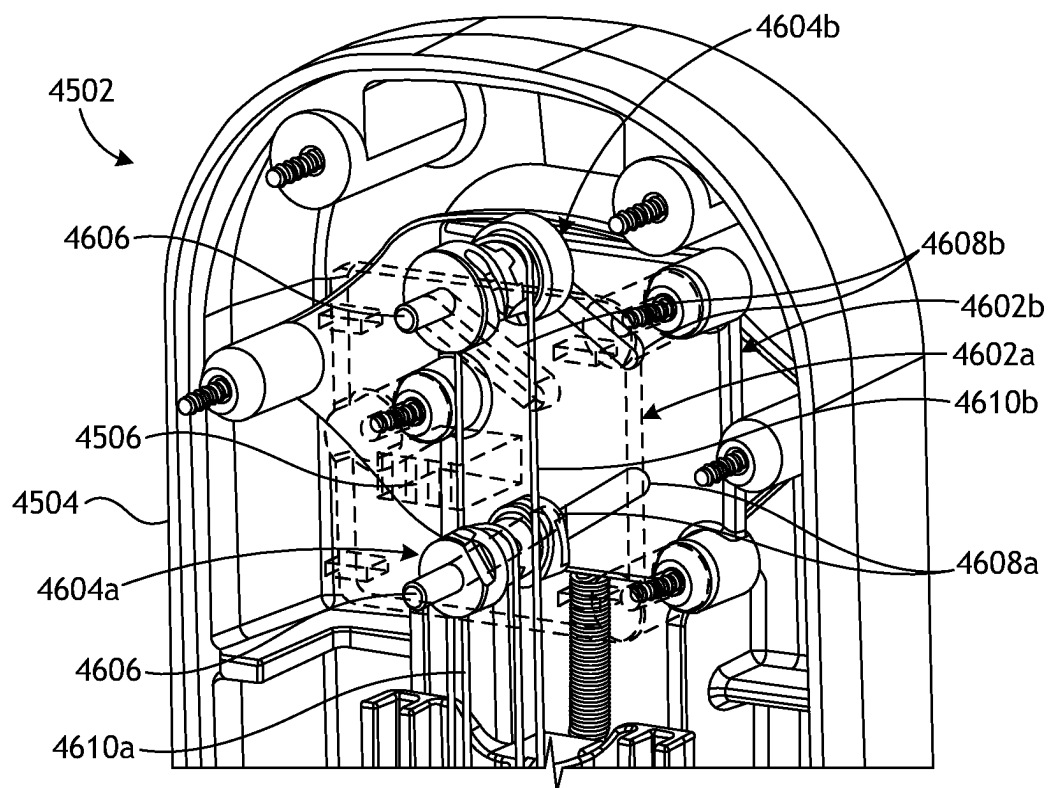


FIG. 46A

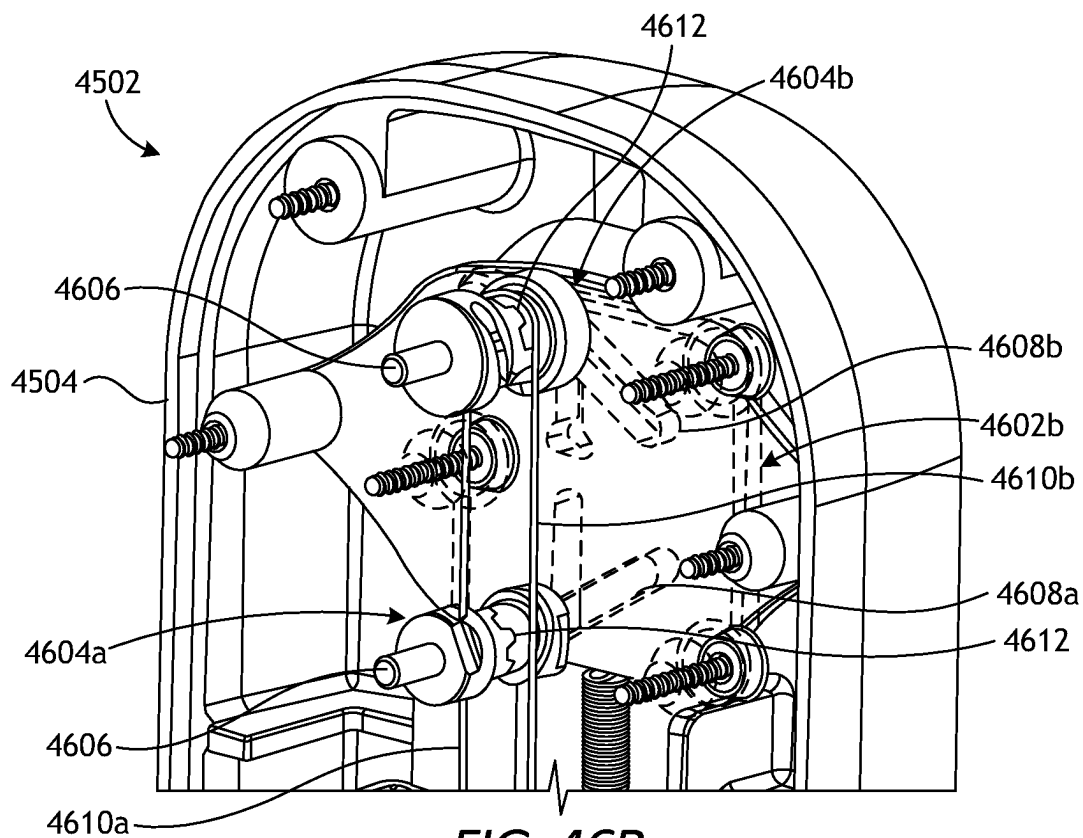


FIG. 46B

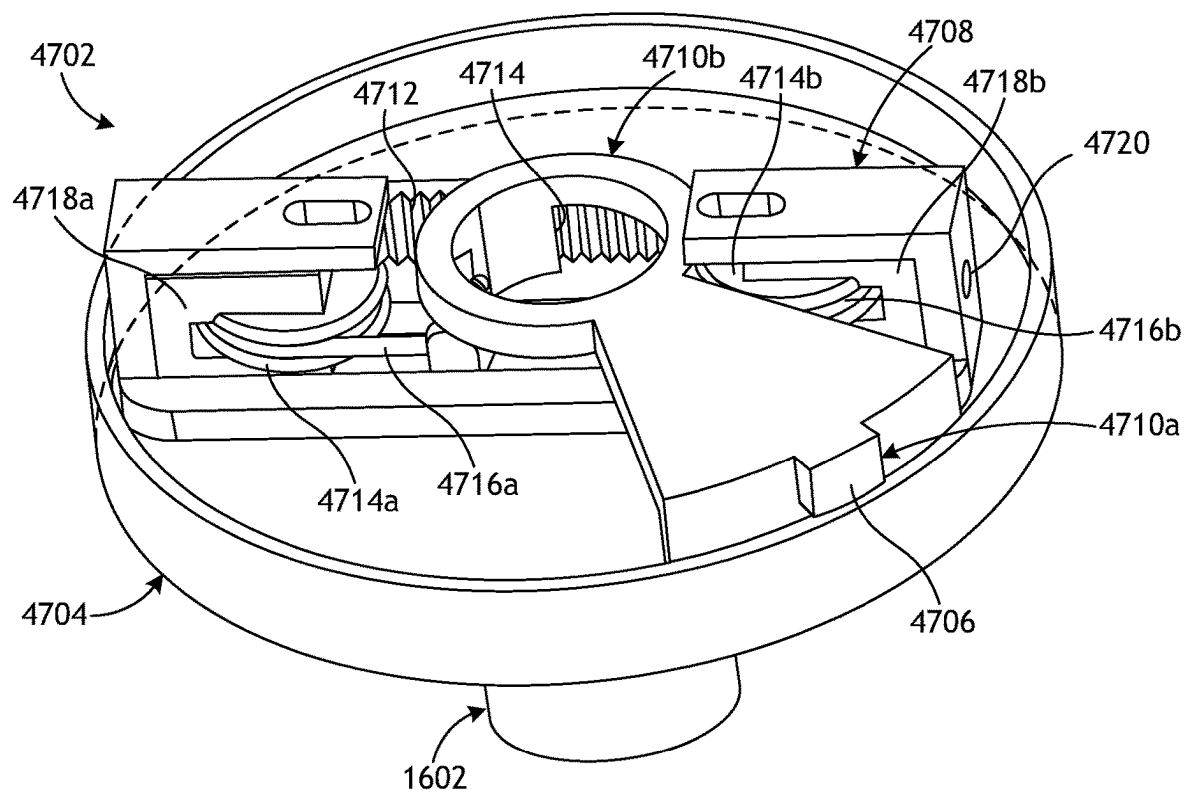


FIG. 47

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## SURGICAL INSTRUMENTS WITH ACTUATABLE TAILPIECE

### TECHNICAL FIELD

The systems and methods disclosed herein are directed to robotic surgical tools and, more particularly to, actuatable tailpiece assemblies for surgical tools that allow a user to manually manipulate operation of the surgical tool.

### BACKGROUND

Minimally invasive surgical (MIS) instruments are often preferred over traditional open surgical devices due to the reduced post-operative recovery time and minimal scarring. The most common MIS procedure may be endoscopy, and the most common form of endoscopy is laparoscopy, in which one or more small incisions are formed in the abdomen of a patient and a trocar is inserted through the incision to form a pathway that provides access to the abdominal cavity. The cannula and sealing system of the trocar are used to introduce various instruments and tools into the abdominal cavity, as well as to provide insufflation to elevate the abdominal wall above the organs. The instruments can be used to engage and/or treat tissue in a number of ways to achieve a diagnostic or therapeutic effect.

Various robotic systems have recently been developed to assist in MIS procedures. Robotic systems can allow for more instinctive hand movements by maintaining natural eye-hand axis. Robotic systems can also allow for more degrees of freedom in movement by including an articulable “wrist” joint that creates a more natural hand-like articulation. In such systems, an end effector positioned at the distal end of the instrument can be articulated (moved) using a cable driven motion system having one or more drive cables (or other elongate members) that extend through the wrist joint. A user (e.g., a surgeon) is able to remotely operate the end effector by grasping and manipulating in space one or more controllers that communicate with a tool driver coupled to the surgical instrument. User inputs are processed by a computer system incorporated into the robotic surgical system, and the tool driver responds by actuating the cable driven motion system and thereby actively controlling the tension balance in the drive cables. Moving the drive cables articulates the end effector to desired angular positions and configurations.

Improvements to robotically-enabled medical systems will provide physicians with the ability to perform endoscopic and laparoscopic procedures more effectively and with improved ease.

### BRIEF DESCRIPTION OF THE DRAWINGS

The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements.

FIG. 1 illustrates an embodiment of a cart-based robotic system arranged for diagnostic and/or therapeutic bronchoscopy procedure(s).

FIG. 2 depicts further aspects of the robotic system of FIG. 1.

FIG. 3A illustrates an embodiment of the robotic system of FIG. 1 arranged for ureteroscopy.

FIG. 3B illustrates an embodiment of the robotic system of FIG. 1 arranged for a vascular procedure.

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FIG. 4 illustrates an embodiment of a table-based robotic system arranged for a bronchoscopy procedure.

FIG. 5 provides an alternative view of the robotic system of FIG. 4.

FIG. 6 illustrates an example system configured to stow robotic arm(s).

FIG. 7A illustrates an embodiment of a table-based robotic system configured for a ureteroscopy procedure.

FIG. 7B illustrates an embodiment of a table-based robotic system configured for a laparoscopic procedure.

FIG. 7C illustrates an embodiment of the table-based robotic system of FIGS. 4-7B with pitch or tilt adjustment.

FIG. 8 provides a detailed illustration of the interface between the table and the column of the table-based robotic system of FIGS. 4-7.

FIG. 9A illustrates an alternative embodiment of a table-based robotic system.

FIG. 9B illustrates an end view of the table-based robotic system of FIG. 9A.

FIG. 9C illustrates an end view of a table-based robotic system with robotic arms attached thereto.

FIG. 10 illustrates an exemplary instrument driver.

FIG. 11 illustrates an exemplary medical instrument with a paired instrument driver.

FIG. 12 illustrates an alternative design for an instrument driver and instrument where the axes of the drive units are parallel to the axis of the elongated shaft of the instrument.

FIG. 13 illustrates an instrument having an instrument-based insertion architecture.

FIG. 14 illustrates an exemplary controller.

FIG. 15 depicts a block diagram illustrating a localization system that estimates a location of one or more elements of the robotic systems of FIGS. 1-7C, such as the location of the instrument of FIGS. 11-13, in accordance to an example embodiment.

FIG. 16 is a side view of an example surgical tool that may incorporate some or all of the principles of the present disclosure.

FIGS. 17A and 17B are isometric and bottom views, respectively, of an example actuation system, according to one or more embodiments.

FIG. 17C is another example of a gear train that may be used to transfer rotational torque from the drive inputs to the rack gears.

FIGS. 18A-18D are schematic diagrams of example operation of the actuation system of FIGS. 17A-17B, according to one or more embodiments.

FIG. 19 is an isometric view of one example of the tailpiece of FIG. 16, according to one or more embodiments.

FIGS. 20A and 20B show example operation of the tailpiece (pantograph) of FIG. 19, according to one or more embodiments.

FIG. 21 is an enlarged side view of the tailpiece (pantograph) of FIG. 19, according to one or more embodiments.

FIG. 22 is a side view of the surgical tool of FIG. 16, according to one or more additional embodiments.

FIGS. 23A and 23B are isometric and bottom views, respectively, of one example of the decoupler of FIG. 16, according to one or more embodiments.

FIG. 24 is an isometric view of one example of the insertion assembly of FIGS. 23A-23B, according to one or more embodiments.

FIG. 25 is an isometric view of an example differential assembly of FIGS. 23A-23B, according to one or more embodiments.

FIGS. 26A-26C are side views of the differential assembly of FIG. 25 in various stages of deconstruction, according to one or more embodiments.

FIGS. 27A and 27B are isometric and cross-sectional side views, respectively, of another example differential assembly of FIG. 25, according to one or more additional embodiments.

FIG. 28 is a top view of another example of the decoupler of FIG. 16, according to one or more additional embodiments.

FIG. 29 is an isometric view of an example differential assembly.

FIG. 30 is a cross-sectional top view of the differential assembly of FIG. 29 taken along the lines shown in FIG. 29.

FIG. 31 is a schematic cross-sectional side view of the differential assembly of FIG. 29, according to one or more embodiments.

FIG. 32 is a schematic, cross-sectional side view of yet another example of the differential assembly of FIG. 28, according to one or more additional embodiments.

FIG. 33 is a cross-sectional side view of one example of the insertion assembly of FIG. 28, according to one or more embodiments.

FIG. 34 is a schematic, cross-sectional side view of another example of the insertion assembly of FIG. 28, according to one or more additional embodiments.

FIGS. 35A and 35B are isometric and cross-sectional side views of another example of the differential assembly of FIG. 28, according to one or more additional embodiments.

FIGS. 36A and 36B are isometric and cross-sectional side views, respectively, of another example of the insertion assembly of FIG. 28, according to one or more additional embodiments.

FIG. 37 is an isometric view of another example surgical tool that may incorporate some or all of the principles of the present disclosure.

FIG. 38 depicts separated isometric end views of the instrument driver and the surgical tool of FIG. 37.

FIGS. 39A and 39B are isometric open (disengaged) and closed (engaged) views, respectively, of another example actuation system, according to one or more additional embodiments.

FIG. 40 is an isometric view of an example spur linkage subassembly, according to one or more embodiments.

FIG. 41 is another isometric view of the actuation system of FIGS. 39A-39B, according to one or more additional embodiments.

FIG. 42 is an enlarged, isometric view of one example of the clocking wheel linkage subassembly, according to one or more embodiments.

FIG. 43 is an enlarged isometric view of the tailpiece of FIG. 37, according to one or more embodiments.

FIG. 44 is an isometric view of the interior of the tailpiece of FIG. 43, according to one or more embodiments.

FIG. 45 is an isometric view of another embodiment of the tailpiece of FIG. 37, according to one or more additional embodiments.

FIGS. 46A and 46B are isometric views of the interior of the tailpiece of FIG. 45, according to one or more embodiments.

FIG. 47 is an enlarged isometric view of another example tailpiece, according to one or more additional embodiments.

## DETAILED DESCRIPTION

### 1. Overview.

Aspects of the present disclosure may be integrated into a robotically-enabled medical system capable of performing a variety of medical procedures, including both minimally invasive (e.g., laparoscopy) and non-invasive (e.g., endoscopy) procedures. Among endoscopy procedures, the system may be capable of performing bronchoscopy, ureteroscopy, gastroscopy, etc.

In addition to performing the breadth of procedures, the system may provide additional benefits, such as enhanced imaging and guidance, to assist the physician. Additionally, the system may provide the physician with the ability to perform the procedure from an ergonomic position without the need for awkward arm motions and positions. Still further, the system may provide the physician with the ability to perform the procedure with improved ease of use such that one or more of the instruments of the system can be controlled by a single user.

Various embodiments will be described below in conjunction with the drawings for purposes of illustration. It should be appreciated that many other implementations of the disclosed concepts are possible, and various advantages can be achieved with the disclosed implementations. Headings are included herein for reference and to aid in locating various sections. These headings are not intended to limit the scope of the concepts described with respect thereto, as such concepts may have applicability throughout the entire specification.

#### A. Robotic System—Cart.

The robotically-enabled medical system may be configured in a variety of ways depending on the particular procedure. FIG. 1 illustrates an embodiment of a cart-based robotically-enabled system 100 arranged for a diagnostic and/or therapeutic bronchoscopy procedure. For a bronchoscopy procedure, the robotic system 100 may include a cart 102 having one or more robotic arms 104 (three shown) to deliver a medical instrument (alternately referred to as a “surgical tool”), such as a steerable endoscope 106 (e.g., a procedure-specific bronchoscope for bronchoscopy), to a natural orifice access point (i.e., the mouth of the patient) to deliver diagnostic and/or therapeutic tools. As shown, the cart 102 may be positioned proximate to the patient's upper torso in order to provide access to the access point. Similarly, the robotic arms 104 may be actuated to position the bronchoscope relative to the access point. The arrangement in FIG. 1 may also be utilized when performing a gastrointestinal (GI) procedure with a gastroscope, a specialized endoscope for GI procedures.

Once the cart 102 is properly positioned adjacent the patient, the robotic arms 104 are operated to insert the steerable endoscope 106 into the patient robotically, manually, or a combination thereof. The steerable endoscope 106 may comprise at least two telescoping parts, such as an inner leader portion and an outer sheath portion, where each portion is coupled to a separate instrument driver of a set of instrument drivers 108. As illustrated, each instrument driver 108 is coupled to the distal end of a corresponding one of the robotic arms 104. This linear arrangement of the instrument drivers 108, which facilitates coaxially aligning the leader portion with the sheath portion, creates a “virtual rail” 110 that may be repositioned in space by manipulating the robotic arms 104 into different angles and/or positions. Translation of the instrument drivers 108 along the virtual rail 110 telescopes the inner leader portion relative to the

outer sheath portion, thus effectively advancing or retracting the endoscope **106** relative to the patient.

As illustrated, the virtual rail **110** (and other virtual rails described herein) is depicted in the drawings using dashed lines, thus not constituting any physical structure of the system **100**. The angle of the virtual rail **110** may be adjusted, translated, and pivoted based on clinical application or physician preference. For example, in bronchoscopy, the angle and position of the virtual rail **110** as shown represents a compromise between providing physician access to the endoscope **106** while minimizing friction that results from bending the endoscope **106** into the patient's mouth.

After insertion into the patient's mouth, the endoscope **106** may be directed down the patient's trachea and lungs using precise commands from the robotic system **100** until reaching a target destination or operative site. In order to enhance navigation through the patient's lung network and/or reach the desired target, the endoscope **106** may be manipulated to telescopically extend the inner leader portion from the outer sheath portion to obtain enhanced articulation and greater bend radius. The use of separate instrument drivers **108** also allows the leader portion and sheath portion to be driven independent of each other.

For example, the endoscope **106** may be directed to deliver a biopsy needle to a target, such as, for example, a lesion or nodule within the lungs of a patient. The needle may be deployed down a working channel that runs the length of the endoscope **106** to obtain a tissue sample to be analyzed by a pathologist. Depending on the pathology results, additional tools may be deployed down the working channel of the endoscope for additional biopsies. After identifying a tissue sample to be malignant, the endoscope **106** may endoscopically deliver tools to resect the potentially cancerous tissue. In some instances, diagnostic and therapeutic treatments can be delivered in separate procedures. In those circumstances, the endoscope **106** may also be used to deliver a fiducial marker to "mark" the location of a target nodule as well. In other instances, diagnostic and therapeutic treatments may be delivered during the same procedure.

The system **100** may also include a movable tower **112**, which may be connected via support cables to the cart **102** to provide support for controls, electronics, fluidics, optics, sensors, and/or power to the cart **102**. Placing such functionality in the tower **112** allows for a smaller form factor cart **102** that may be more easily adjusted and/or repositioned by an operating physician and his/her staff. Additionally, the division of functionality between the cart/table and the support tower **112** reduces operating room clutter and facilitates improving clinical workflow. While the cart **102** may be positioned close to the patient, the tower **112** may alternatively be stowed in a remote location to stay out of the way during a procedure.

In support of the robotic systems described above, the tower **112** may include component(s) of a computer-based control system that stores computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, etc. The execution of those instructions, whether the execution occurs in the tower **112** or the cart **102**, may control the entire system or sub-system(s) thereof. For example, when executed by a processor of the computer system, the instructions may cause the components of the robotics system to actuate the relevant carriages and arm mounts, actuate the robotics arms, and control the medical instruments. For example, in response to receiving the

control signal, motors in the joints of the robotic arms **104** may position the arms into a certain posture or angular orientation.

The tower **112** may also include one or more of a pump, flow meter, valve control, and/or fluid access in order to provide controlled irrigation and aspiration capabilities to the system **100** that may be deployed through the endoscope **106**. These components may also be controlled using the computer system of the tower **112**. In some embodiments, irrigation and aspiration capabilities may be delivered directly to the endoscope **106** through separate cable(s).

The tower **112** may include a voltage and surge protector designed to provide filtered and protected electrical power to the cart **102**, thereby avoiding placement of a power transformer and other auxiliary power components in the cart **102**, resulting in a smaller, more moveable cart **102**.

The tower **112** may also include support equipment for sensors deployed throughout the robotic system **100**. For example, the tower **112** may include opto-electronics equipment for detecting, receiving, and processing data received from optical sensors or cameras throughout the robotic system **100**. In combination with the control system, such opto-electronics equipment may be used to generate real-time images for display in any number of consoles deployed throughout the system, including in the tower **112**. Similarly, the tower **112** may also include an electronic subsystem for receiving and processing signals received from deployed electromagnetic (EM) sensors. The tower **112** may also be used to house and position an EM field generator for detection by EM sensors in or on the medical instrument.

The tower **112** may also include a console **114** in addition to other consoles available in the rest of the system, e.g., a console mounted to the cart **102**. The console **114** may include a user interface and a display screen (e.g., a touchscreen) for the physician operator. Consoles in the system **100** are generally designed to provide both robotic controls as well as pre-operative and real-time information of the procedure, such as navigational and localization information of the endoscope **106**. When the console **114** is not the only console available to the physician, it may be used by a second operator, such as a nurse, to monitor the health or vitals of the patient and the operation of system, as well as provide procedure-specific data, such as navigational and localization information. In other embodiments, the console **114** may be housed in a body separate from the tower **112**.

The tower **112** may be coupled to the cart **102** and endoscope **106** through one or more cables **116** connections. In some embodiments, support functionality from the tower **112** may be provided through a single cable **116** extending to the cart **102**, thus simplifying and de-cluttering the operating room. In other embodiments, specific functionality may be coupled in separate cabling and connections. For example, while power may be provided through a single power cable to the cart **102**, support for controls, optics, fluidics, and/or navigation may be provided through one or more separate cables.

FIG. 2 provides a detailed illustration of an embodiment of the cart **102** from the cart-based robotically-enabled system **100** of FIG. 1. The cart **102** generally includes an elongated support structure **202** (also referred to as a "column"), a cart base **204**, and a console **206** at the top of the column **202**. The column **202** may include one or more carriages, such as a carriage **208** (alternatively "arm support") for supporting the deployment of the robotic arms **104**. The carriage **208** may include individually configurable arm mounts that rotate along a perpendicular axis to adjust the base **214** of the robotic arms **104** for better positioning

relative to the patient. The carriage **208** also includes a carriage interface **210** that allows the carriage **208** to vertically translate along the column **202**.

The carriage interface **210** is connected to the column **202** through slots, such as slot **212**, that are positioned on opposite sides of the column **202** to guide the vertical translation of the carriage **208**. The slot **212** contains a vertical translation interface to position and hold the carriage **208** at various vertical heights relative to the cart base **204**. Vertical translation of the carriage **208** allows the cart **102** to adjust the reach of the robotic arms **104** to meet a variety of table heights, patient sizes, and physician preferences. Similarly, the individually configurable arm mounts on the carriage **208** allow a base **214** of the robotic arms **104** to be angled in a variety of configurations.

In some embodiments, the slot **212** may be supplemented with slot covers (not shown) that are flush and parallel to the slot surface to prevent dirt and fluid ingress into the internal chambers of the column **202** and the vertical translation interface as the carriage **208** vertically translates. The slot covers may be deployed through pairs of spring spools positioned near the vertical top and bottom of the slot **212**. The covers are coiled within the spools until deployed to extend and retract from their coiled state as the carriage **208** vertically translates up and down. The spring-loading of the spools provides force to retract the cover into a spool when carriage **208** translates towards the spool, while also maintaining a tight seal when the carriage **208** translates away from the spool. The covers may be connected to the carriage **208** using, for example, brackets in the carriage interface **210** to ensure proper extension and retraction of the cover as the carriage **208** translates.

The column **202** may internally comprise mechanisms, such as gears and motors, which are designed to use a vertically aligned lead screw to translate the carriage **208** in a mechanized fashion in response to control signals generated in response to user inputs, e.g., inputs from the console **206**.

The robotic arms **104** may generally comprise robotic arm bases **214** and end effectors **216** (three shown), separated by a series of linkages **218** connected by a corresponding series of joints **220**, each joint **220** including an independent actuator, and each actuator including an independently controllable motor. Each independently controllable joint **220** represents an independent degree of freedom available to the corresponding robotic arm **104**. In the illustrated embodiment, each arm **104** has seven joints **220**, thus providing seven degrees of freedom. A multitude of joints **220** result in a multitude of degrees of freedom, allowing for “redundant” degrees of freedom. Redundant degrees of freedom allow the robotic arms **104** to position their respective end effectors **216** at a specific position, orientation, and trajectory in space using different linkage positions and joint angles. This allows for the system **100** to position and direct a medical instrument from a desired point in space while allowing the physician to move the arm joints **220** into a clinically advantageous position away from the patient to create greater access, while avoiding arm collisions.

The cart base **204** balances the weight of the column **202**, the carriage **208**, and the arms **104** over the floor. Accordingly, the cart base **204** houses heavier components, such as electronics, motors, power supply, as well as components that either enable movement and/or immobilize the cart. For example, the cart base **204** includes rolling casters **222** that allow for the cart to easily move around the room prior to a procedure. After reaching an appropriate position, the cast-

ers **222** may be immobilized using wheel locks to hold the cart **102** in place during the procedure.

Positioned at the vertical end of the column **202**, the console **206** allows for both a user interface for receiving user input and a display screen (or a dual-purpose device such as, for example, a touchscreen **224**) to provide the physician user with both pre-operative and intra-operative data. Potential pre-operative data on the touchscreen **224** may include pre-operative plans, navigation and mapping data derived from pre-operative computerized tomography (CT) scans, and/or notes from pre-operative patient interviews. Intra-operative data on the touchscreen **224** may include optical information provided from the tool, sensor and coordinate information from sensors, as well as vital patient statistics, such as respiration, heart rate, and/or pulse. The console **206** may be positioned and tilted to allow a physician to access the console from the side of the column **202** opposite carriage **208**. From this position, the physician may view the console **206**, the robotic arms **104**, and the patient while operating the console **206** from behind the cart **102**. As shown, the console **206** also includes a handle **226** to assist with maneuvering and stabilizing cart **102**.

FIG. 3A illustrates an embodiment of the system **100** of FIG. 1 arranged for ureteroscopy. In a ureteroscopic procedure, the cart **102** may be positioned to deliver a ureteroscope **302**, a procedure-specific endoscope designed to traverse a patient's urethra and ureter, to the lower abdominal area of the patient. In ureteroscopy, it may be desirable for the ureteroscope **302** to be directly aligned with the patient's urethra to reduce friction and forces on the sensitive anatomy. As shown, the cart **102** may be aligned at the foot of the table to allow the robotic arms **104** to position the ureteroscope **302** for direct linear access to the patient's urethra. From the foot of the table, the robotic arms **104** may insert the ureteroscope **302** along a virtual rail **304** directly into the patient's lower abdomen through the urethra.

After insertion into the urethra, using similar control techniques as in bronchoscopy, the ureteroscope **302** may be navigated into the bladder, ureters, and/or kidneys for diagnostic and/or therapeutic applications. For example, the ureteroscope **302** may be directed into the ureter and kidneys to break up kidney stone build-up using a laser or ultrasonic lithotripsy device deployed down a working channel of the ureteroscope **302**. After lithotripsy is complete, the resulting stone fragments may be removed using baskets deployed down the working channel of the ureteroscope **302**.

FIG. 3B illustrates another embodiment of the system **100** of FIG. 1 arranged for a vascular procedure. In a vascular procedure, the system **100** may be configured such that the cart **102** may deliver a medical instrument **306**, such as a steerable catheter, to an access point in the femoral artery in the patient's leg. The femoral artery presents both a larger diameter for navigation as well as a relatively less circuitous and tortuous path to the patient's heart, which simplifies navigation. As in a ureteroscopic procedure, the cart **102** may be positioned towards the patient's legs and lower abdomen to allow the robotic arms **104** to provide a virtual rail **308** with direct linear access to the femoral artery access point in the patient's thigh/hip region. After insertion into the artery, the medical instrument **306** may be directed and advanced by translating the instrument drivers **108**. Alternatively, the cart **102** may be positioned around the patient's upper abdomen in order to reach alternative vascular access points, such as, for example, the carotid and brachial arteries near the patient's shoulder and wrist.

## B. Robotic System—Table.

Embodiments of the robotically-enabled medical system may also incorporate the patient's table. Incorporation of the table reduces the amount of capital equipment within the operating room by removing the cart, which allows greater access to the patient. FIG. 4 illustrates an embodiment of such a robotically-enabled system 400 arranged for a bronchoscopy procedure. As illustrated, the system 400 includes a support structure or column 402 for supporting platform 404 (shown as a "table" or "bed") over the floor. Much like in the cart-based systems, the end effectors of the robotic arms 406 of the system 400 comprise instrument drivers 408 that are designed to manipulate an elongated medical instrument, such as a bronchoscope 410, through or along a virtual rail 412 formed from the linear alignment of the instrument drivers 408. In practice, a C-arm for providing fluoroscopic imaging may be positioned over the patient's upper abdominal area by placing the emitter and detector around the table 404.

FIG. 5 provides an alternative view of the system 400 without the patient and medical instrument for discussion purposes. As shown, the column 402 may include one or more carriages 502 shown as ring-shaped in the system 400, from which the one or more robotic arms 406 may be based. The carriages 502 may translate along a vertical column interface 504 that runs the length (height) of the column 402 to provide different vantage points from which the robotic arms 406 may be positioned to reach the patient. The carriage(s) 502 may rotate around the column 402 using a mechanical motor positioned within the column 402 to allow the robotic arms 406 to have access to multiples sides of the table 404, such as, for example, both sides of the patient. In embodiments with multiple carriages 502, the carriages 502 may be individually positioned on the column 402 and may translate and/or rotate independent of the other carriages 502. While carriages 502 need not surround the column 402 or even be circular, the ring-shape as shown facilitates rotation of the carriages 502 around the column 402 while maintaining structural balance. Rotation and translation of the carriages 502 allows the system 400 to align medical instruments, such as endoscopes and laparoscopes, into different access points on the patient.

In other embodiments (discussed in greater detail below with respect to FIG. 9A), the system 400 can include a patient table or bed with adjustable arm supports in the form of bars or rails extending alongside it. One or more robotic arms 406 (e.g., via a shoulder with an elbow joint) can be attached to the adjustable arm supports, which can be vertically adjusted. By providing vertical adjustment, the robotic arms 406 are advantageously capable of being stowed compactly beneath the patient table or bed, and subsequently raised during a procedure.

The arms 406 may be mounted on the carriages 502 through a set of arm mounts 506 comprising a series of joints that may individually rotate and/or telescopically extend to provide additional configurability to the robotic arms 406. Additionally, the arm mounts 506 may be positioned on the carriages 502 such that when the carriages 502 are appropriately rotated, the arm mounts 506 may be positioned on either the same side of the table 404 (as shown in FIG. 5), on opposite sides of table 404 (as shown in FIG. 7B), or on adjacent sides of the table 404 (not shown).

The column 402 structurally provides support for the table 404, and a path for vertical translation of the carriages 502. Internally, the column 402 may be equipped with lead screws for guiding vertical translation of the carriages, and motors to mechanize the translation of said carriages based

the lead screws. The column 402 may also convey power and control signals to the carriage 502 and robotic arms 406 mounted thereon.

A table base 508 serves a similar function as the cart base 204 of the cart 102 shown in FIG. 2, housing heavier components to balance the table/bed 404, the column 402, the carriages 502, and the robotic arms 406. The table base 508 may also incorporate rigid casters to provide stability during procedures. Deployed from the bottom of the table base 508, the casters may extend in opposite directions on both sides of the base 508 and retract when the system 400 needs to be moved.

In some embodiments, the system 400 may also include a tower (not shown) that divides the functionality of system 400 between table and tower to reduce the form factor and bulk of the table 404. As in earlier disclosed embodiments, the tower may provide a variety of support functionalities to the table 404, such as processing, computing, and control capabilities, power, fluidics, and/or optical and sensor processing. The tower may also be movable to be positioned away from the patient to improve physician access and de-clutter the operating room. Additionally, placing components in the tower allows for more storage space in the table base 508 for potential stowage of the robotic arms 406. The tower may also include a master controller or console that provides both a user interface for user input, such as keyboard and/or pendant, as well as a display screen (or touchscreen) for pre-operative and intra-operative information, such as real-time imaging, navigation, and tracking information. In some embodiments, the tower may also contain holders for gas tanks to be used for insufflation.

In some embodiments, a table base may stow and store the robotic arms when not in use. FIG. 6 illustrates an embodiment of the system 400 that is configured to stow robotic arms in an embodiment of the table-based system. In the system 400, one or more carriages 602 (one shown) may be vertically translated into a base 604 to stow one or more robotic arms 606, one or more arm mounts 608, and the carriages 602 within the base 604. Base covers 610 may be translated and retracted open to deploy the carriages 602, the arm mounts 608, and the arms 606 around the column 612, and closed to stow and protect them when not in use. The base covers 610 may be sealed with a membrane 614 along the edges of its opening to prevent dirt and fluid ingress when closed.

FIG. 7A illustrates an embodiment of the robotically-enabled table-based system 400 configured for a ureteroscopy procedure. In ureteroscopy, the table 404 may include a swivel portion 702 for positioning a patient off-angle from the column 402 and the table base 508. The swivel portion 702 may rotate or pivot around a pivot point (e.g., located below the patient's head) in order to position the bottom portion of the swivel portion 702 away from the column 402. For example, the pivoting of the swivel portion 702 allows a C-arm (not shown) to be positioned over the patient's lower abdomen without competing for space with the column (not shown) below table 404. By rotating the carriage (not shown) around the column 402, the robotic arms 406 may directly insert a ureteroscope 704 along a virtual rail 706 into the patient's groin area to reach the urethra. In ureteroscopy, stirrups 708 may also be fixed to the swivel portion 702 of the table 404 to support the position of the patient's legs during the procedure and allow clear access to the patient's groin area.

FIG. 7B illustrates an embodiment of the system 400 configured for a laparoscopic procedure. In a laparoscopic procedure, through small incision(s) in the patient's abdomi-

nal wall, minimally invasive instruments may be inserted into the patient's anatomy. In some embodiments, the minimally invasive instruments comprise an elongated rigid member, such as a shaft, which is used to access anatomy within the patient. After inflation of the patient's abdominal cavity, the instruments may be directed to perform surgical or medical tasks, such as grasping, cutting, ablating, suturing, etc. In some embodiments, the instruments can comprise a scope, such as a laparoscope. As shown in FIG. 7B, the carriages 502 of the system 400 may be rotated and vertically adjusted to position pairs of the robotic arms 406 on opposite sides of the table 404, such that an instrument 710 may be positioned using the arm mounts 506 to be passed through minimal incisions on both sides of the patient to reach his/her abdominal cavity.

To accommodate laparoscopic procedures, the system 400 may also tilt the platform to a desired angle. FIG. 7C illustrates an embodiment of the system 400 with pitch or tilt adjustment. As shown in FIG. 7C, the system 400 may accommodate tilt of the table 404 to position one portion of the table 404 at a greater distance from the floor than the other. Additionally, the arm mounts 506 may rotate to match the tilt such that the arms 406 maintain the same planar relationship with table 404. To accommodate steeper angles, the column 402 may also include telescoping portions 712 that allow vertical extension of the column 402 to keep the table 404 from touching the floor or colliding with the base 508.

FIG. 8 provides a detailed illustration of the interface between the table 404 and the column 402. Pitch rotation mechanism 802 may be configured to alter the pitch angle of the table 404 relative to the column 402 in multiple degrees of freedom. The pitch rotation mechanism 802 may be enabled by the positioning of orthogonal axes A and B at the column-table interface, each axis actuated by a separate motor 804a and 804b responsive to an electrical pitch angle command. Rotation along one screw 806a would enable tilt adjustments in one axis A, while rotation along another screw 806b would enable tilt adjustments along the other axis B. In some embodiments, a ball joint can be used to alter the pitch angle of the table 404 relative to the column 402 in multiple degrees of freedom.

For example, pitch adjustments are particularly useful when trying to position the table in a Trendelenburg position, i.e., position the patient's lower abdomen at a higher position from the floor than the patient's lower abdomen, for lower abdominal surgery. The Trendelenburg position causes the patient's internal organs to slide towards his/her upper abdomen through the force of gravity, clearing out the abdominal cavity for minimally invasive tools to enter and perform lower abdominal surgical or medical procedures, such as laparoscopic prostatectomy.

FIGS. 9A and 9B illustrate isometric and end views, respectively, of an alternative embodiment of a table-based surgical robotics system 900. The surgical robotics system 900 includes one or more adjustable arm supports 902 that can be configured to support one or more robotic arms (see, for example, FIG. 9C) relative to a table 904. In the illustrated embodiment, a single adjustable arm support 902 is shown, though an additional arm support can be provided on an opposite side of the table 904. The adjustable arm support 902 can be configured so that it can move relative to the table 904 to adjust and/or vary the position of the adjustable arm support 902 and/or any robotic arms mounted thereto relative to the table 904. For example, the adjustable arm support 902 may be adjusted in one or more degrees of freedom relative to the table 904. The adjustable arm support

902 provides high versatility to the system 900, including the ability to easily stow the one or more adjustable arm supports 902 and any robotics arms attached thereto beneath the table 904. The adjustable arm support 902 can be elevated from the stowed position to a position below an upper surface of the table 904. In other embodiments, the adjustable arm support 902 can be elevated from the stowed position to a position above an upper surface of the table 904.

The adjustable arm support 902 can provide several degrees of freedom, including lift, lateral translation, tilt, etc. In the illustrated embodiment of FIGS. 9A and 9B, the arm support 902 is configured with four degrees of freedom, which are illustrated with arrows in FIG. 9A. A first degree of freedom allows for adjustment of the adjustable arm support 902 in the z-direction ("Z-lift"). For example, the adjustable arm support 902 can include a carriage 906 configured to move up or down along or relative to a column 908 supporting the table 904. A second degree of freedom can allow the adjustable arm support 902 to tilt. For example, the adjustable arm support 902 can include a rotary joint, which can allow the adjustable arm support 902 to be aligned with the bed in a Trendelenburg position. A third degree of freedom can allow the adjustable arm support 902 to "pivot up," which can be used to adjust a distance between a side of the table 904 and the adjustable arm support 902. A fourth degree of freedom can permit translation of the adjustable arm support 902 along a longitudinal length of the table.

The surgical robotics system 900 in FIGS. 9A and 9B can comprise a table 904 supported by a column 908 that is mounted to a base 910. The base 910 and the column 908 support the table 904 relative to a support surface. A floor axis 912 and a support axis 914 are shown in FIG. 9B.

The adjustable arm support 902 can be mounted to the column 908. In other embodiments, the arm support 902 can be mounted to the table 904 or the base 910. The adjustable arm support 902 can include a carriage 906, a bar or rail connector 916 and a bar or rail 918. In some embodiments, one or more robotic arms mounted to the rail 918 can translate and move relative to one another.

The carriage 906 can be attached to the column 908 by a first joint 920, which allows the carriage 906 to move relative to the column 908 (e.g., such as up and down a first or vertical axis 922). The first joint 920 can provide the first degree of freedom ("Z-lift") to the adjustable arm support 902. The adjustable arm support 902 can include a second joint 924, which provides the second degree of freedom (tilt) for the adjustable arm support 902. The adjustable arm support 902 can include a third joint 926, which can provide the third degree of freedom ("pivot up") for the adjustable arm support 902. An additional joint 928 (shown in FIG. 9B) can be provided that mechanically constrains the third joint 926 to maintain an orientation of the rail 918 as the rail connector 916 is rotated about a third axis 930. The adjustable arm support 902 can include a fourth joint 932, which can provide a fourth degree of freedom (translation) for the adjustable arm support 902 along a fourth axis 934.

FIG. 9C illustrates an end view of the surgical robotics system 900 with two adjustable arm supports 902a and 902b mounted on opposite sides of the table 904. A first robotic arm 936a is attached to the first bar or rail 918a of the first adjustable arm support 902a. The first robotic arm 936a includes a base 938a attached to the first rail 918a. The distal end of the first robotic arm 936a includes an instrument drive mechanism or input 940a that can attach to one or more robotic medical instruments or tools. Similarly, the

second robotic arm **936b** includes a base **938a** attached to the second rail **918b**. The distal end of the second robotic arm **936b** includes an instrument drive mechanism or input **940b** configured to attach to one or more robotic medical instruments or tools.

In some embodiments, one or more of the robotic arms **936a,b** comprises an arm with seven or more degrees of freedom. In some embodiments, one or more of the robotic arms **936a,b** can include eight degrees of freedom, including an insertion axis (1-degree of freedom including insertion), a wrist (3-degrees of freedom including wrist pitch, yaw and roll), an elbow (1-degree of freedom including elbow pitch), a shoulder (2-degrees of freedom including shoulder pitch and yaw), and base **938a,b** (1-degree of freedom including translation). In some embodiments, the insertion degree of freedom can be provided by the robotic arm **936a,b**, while in other embodiments, the instrument itself provides insertion via an instrument-based insertion architecture.

#### C. Instrument Driver & Interface.

The end effectors of a system's robotic arms comprise (i) an instrument driver (alternatively referred to as "tool driver," "instrument drive mechanism," "instrument device manipulator," and "drive input") that incorporate electro-mechanical means for actuating the medical instrument, and (ii) a removable or detachable medical instrument, which may be devoid of any electro-mechanical components, such as motors. This dichotomy may be driven by the need to sterilize medical instruments used in medical procedures, and the inability to adequately sterilize expensive capital equipment due to their intricate mechanical assemblies and sensitive electronics. Accordingly, the medical instruments may be designed to be detached, removed, and interchanged from the instrument driver (and thus the system) for individual sterilization or disposal by the physician or the physician's staff. In contrast, the instrument drivers need not be changed or sterilized, and may be draped for protection.

FIG. **10** illustrates an example instrument driver **1000**, according to one or more embodiments. Positioned at the distal end of a robotic arm, the instrument driver **1000** includes one or more drive outputs **1002** arranged with parallel axes to provide controlled torque to a medical instrument via corresponding drive shafts **1004**. Each drive output **1002** comprises an individual drive shaft **1004** for interacting with the instrument, a gear head **1006** for converting the motor shaft rotation to a desired torque, a motor **1008** for generating the drive torque, and an encoder **1010** to measure the speed of the motor shaft and provide feedback to control circuitry **1012**, which can also be used for receiving control signals and actuating the drive output **1002**. Each drive output **1002** being independently controlled and motorized, the instrument driver **1000** may provide multiple (at least two shown in FIG. **10**) independent drive outputs to the medical instrument. In operation, the control circuitry **1012** receives a control signal, transmits a motor signal to the motor **1008**, compares the resulting motor speed as measured by the encoder **1010** with the desired speed, and modulates the motor signal to generate the desired torque.

For procedures that require a sterile environment, the robotic system may incorporate a drive interface, such as a sterile adapter connected to a sterile drape that sits between the instrument driver and the medical instrument. The chief purpose of the sterile adapter is to transfer angular motion from the drive shafts of the instrument driver to the drive inputs of the instrument while maintaining physical separation, and thus sterility, between the drive shafts and drive inputs. Accordingly, an example sterile adapter may com-

prise a series of rotational inputs and outputs intended to be mated with the drive shafts of the instrument driver and drive inputs on the instrument. Connected to the sterile adapter, the sterile drape, comprised of a thin, flexible material such as transparent or translucent plastic, is designed to cover the capital equipment, such as the instrument driver, robotic arm, and cart (in a cart-based system) or table (in a table-based system). Use of the drape would allow the capital equipment to be positioned proximate to the patient while still being located in an area not requiring sterilization (i.e., non-sterile field). On the other side of the sterile drape, the medical instrument may interface with the patient in an area requiring sterilization (i.e., sterile field). D. Medical Instrument.

FIG. **11** illustrates an example medical instrument **1100** with a paired instrument driver **1102**. Like other instruments designed for use with a robotic system, the medical instrument **1100** (alternately referred to as a "surgical tool") comprises an elongated shaft **1104** (or elongate body) and an instrument base **1106**. The instrument base **1106**, also referred to as an "instrument handle" due to its intended design for manual interaction by the physician, may generally comprise rotatable drive inputs **1108**, e.g., receptacles, pulleys or spools, that are designed to be mated with drive outputs **1110** that extend through a drive interface on the instrument driver **1102** at the distal end of a robotic arm **1112**. When physically connected, latched, and/or coupled, the mated drive inputs **1108** of the instrument base **1106** may share axes of rotation with the drive outputs **1110** in the instrument driver **1102** to allow the transfer of torque from the drive outputs **1110** to the drive inputs **1108**. In some embodiments, the drive outputs **1110** may comprise splines that are designed to mate with receptacles on the drive inputs **1108**.

The elongated shaft **1104** is designed to be delivered through either an anatomical opening or lumen, e.g., as in endoscopy, or a minimally invasive incision, e.g., as in laparoscopy. The elongated shaft **1104** may be either flexible (e.g., having properties similar to an endoscope) or rigid (e.g., having properties similar to a laparoscope) or contain a customized combination of both flexible and rigid portions. When designed for laparoscopy, the distal end of the shaft **1104** may be connected to an end effector extending from a jointed wrist formed from a clevis with at least one degree of freedom and a surgical tool or medical instrument, such as, for example, a grasper or scissors, that may be actuated based on force from the tendons as the drive inputs **1008** rotate in response to torque received from the drive outputs **1110** of the instrument driver **1102**. When designed for endoscopy, the distal end of the flexible elongated shaft **1104** may include a steerable or controllable bending section that may be articulated and bent based on torque received from the drive outputs **1110** of the instrument driver **1102**.

In some embodiments, torque from the instrument driver **1102** is transmitted down the elongated shaft **1104** using tendons along the shaft **1104**. These individual tendons, such as pull wires, may be individually anchored to individual drive inputs **1108** within the instrument handle **1106**. From the handle **1106**, the tendons are directed down one or more pull lumens along the elongated shaft **1104** and anchored at the distal portion of the elongated shaft **1104**, or in the wrist at the distal portion of the elongated shaft. During a surgical procedure, such as a laparoscopic, endoscopic, or a hybrid procedure, these tendons may be coupled to a distally mounted end effector, such as a wrist, a grasper, or scissors. Under such an arrangement, torque exerted on the drive inputs **1108** would transfer tension to the tendon, thereby

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causing the end effector to actuate in some way. In some embodiments, during a surgical procedure, the tendon may cause a joint to rotate about an axis, thereby causing the end effector to move in one direction or another. Alternatively, the tendon may be connected to one or more jaws of a grasper at distal end of the elongated shaft **1104**, where tension from the tendon cause the grasper to close.

In endoscopy, the tendons may be coupled to a bending or articulating section positioned along the elongated shaft **1104** (e.g., at the distal end) via adhesive, control ring, or other mechanical fixation. When fixedly attached to the distal end of a bending section, torque exerted on drive inputs **1108** would be transmitted down the tendons, causing the softer, bending section (sometimes referred to as the articulable section or region) to bend or articulate. Along the non-bending sections, it may be advantageous to spiral or helix the individual pull lumens that direct the individual tendons along (or inside) the walls of the endoscope shaft to balance the radial forces that result from tension in the pull wires. The angle of the spiraling and/or spacing there between may be altered or engineered for specific purposes, wherein tighter spiraling exhibits lesser shaft compression under load forces, while lower amounts of spiraling results in greater shaft compression under load forces, but also exhibits limits bending. On the other end of the spectrum, the pull lumens may be directed parallel to the longitudinal axis of the elongated shaft **1104** to allow for controlled articulation in the desired bending or articulable sections.

In endoscopy, the elongated shaft **1104** houses a number of components to assist with the robotic procedure. The shaft may comprise a working channel for deploying surgical tools (or medical instruments), irrigation, and/or aspiration to the operative region at the distal end of the shaft **1104**. The shaft **1104** may also accommodate wires and/or optical fibers to transfer signals to/from an optical assembly at the distal tip, which may include of an optical camera. The shaft **1104** may also accommodate optical fibers to carry light from proximally-located light sources, such as light emitting diodes, to the distal end of the shaft.

At the distal end of the instrument **1100**, the distal tip may also comprise the opening of a working channel for delivering tools for diagnostic and/or therapy, irrigation, and aspiration to an operative site. The distal tip may also include a port for a camera, such as a fiberscope or a digital camera, to capture images of an internal anatomical space. Relatedly, the distal tip may also include ports for light sources for illuminating the anatomical space when using the camera.

In the example of FIG. **11**, the drive shaft axes, and thus the drive input axes, are orthogonal to the axis of the elongated shaft. This arrangement, however, complicates roll capabilities for the elongated shaft **1104**. Rolling the elongated shaft **1104** along its axis while keeping the drive inputs **1108** static results in undesirable tangling of the tendons as they extend off the drive inputs **1108** and enter pull lumens within the elongated shaft **1104**. The resulting entanglement of such tendons may disrupt any control algorithms intended to predict movement of the flexible elongated shaft during an endoscopic procedure.

FIG. **12** illustrates an alternative design for a circular instrument driver **1200** and corresponding instrument **1202** (alternately referred to as a "surgical tool") where the axes of the drive units are parallel to the axis of the elongated shaft **1206** of the instrument **1202**. As shown, the instrument driver **1200** comprises four drive units with corresponding drive outputs **1208** aligned in parallel at the end of a robotic arm **1210**. The drive units and their respective drive outputs

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**1208** are housed in a rotational assembly **1212** of the instrument driver **1200** that is driven by one of the drive units within the assembly **1212**. In response to torque provided by the rotational drive unit, the rotational assembly **1212** rotates along a circular bearing that connects the rotational assembly **1212** to a non-rotational portion **1214** of the instrument driver **1200**. Power and control signals may be communicated from the non-rotational portion **1214** of the instrument driver **1200** to the rotational assembly **1212** through electrical contacts maintained through rotation by a brushed slip ring connection (not shown). In other embodiments, the rotational assembly **1212** may be responsive to a separate drive unit that is integrated into the non-rotatable portion **1214**, and thus not in parallel with the other drive units. The rotational assembly **1212** allows the instrument driver **1200** to rotate the drive units and their respective drive outputs **1208** as a single unit around an instrument driver axis **1216**.

Like earlier disclosed embodiments, the instrument **1202** may include an elongated shaft **1206** and an instrument base **1218** (shown in phantom) including a plurality of drive inputs **1220** (such as receptacles, pulleys, and spools) that are configured to mate with the drive outputs **1208** of the instrument driver **1200**. Unlike prior disclosed embodiments, the instrument shaft **1206** extends from the center of the instrument base **1218** with an axis substantially parallel to the axes of the drive inputs **1220**, rather than orthogonal as in the design of FIG. **11**.

When coupled to the rotational assembly **1212** of the instrument driver **1200**, the medical instrument **1202**, comprising instrument base **1218** and instrument shaft **1206**, rotates in combination with the rotational assembly **1212** about the instrument driver axis **1216**. Since the instrument shaft **1206** is positioned at the center of the instrument base **1218**, the instrument shaft **1206** is coaxial with the instrument driver axis **1216** when attached. Thus, rotation of the rotational assembly **1212** causes the instrument shaft **1206** to rotate about its own longitudinal axis. Moreover, as the instrument base **1218** rotates with the instrument shaft **1206**, any tendons connected to the drive inputs **1220** in the instrument base **1218** are not tangled during rotation. Accordingly, the parallelism of the axes of the drive outputs **1208**, the drive inputs **1220**, and the instrument shaft **1206** allows for the shaft rotation without tangling any control tendons.

FIG. **13** illustrates a medical instrument **1300** having an instrument based insertion architecture in accordance with some embodiments. The instrument **1300** (alternately referred to as a "surgical tool") can be coupled to any of the instrument drivers discussed herein above and, as illustrated, can include an elongated shaft **1302**, an end effector **1304** connected to the shaft **1302**, and a handle **1306** coupled to the shaft **1302**. The elongated shaft **1302** comprises a tubular member having a proximal portion **1308a** and a distal portion **1308b**. The elongated shaft **1302** comprises one or more channels or grooves **1310** along its outer surface and configured to receive one or more wires or cables **1312** therethrough. One or more cables **1312** thus run along an outer surface of the elongated shaft **1302**. In other embodiments, the cables **1312** can also run through the elongated shaft **1302**. Manipulation of the cables **1312** (e.g., via an instrument driver) results in actuation of the end effector **1304**.

The instrument handle **1306**, which may also be referred to as an instrument base, may generally comprise an attachment interface **1314** having one or more mechanical inputs **1316**, e.g., receptacles, pulleys or spools, that are designed

to be reciprocally mated with one or more drive outputs on an attachment surface of an instrument driver.

In some embodiments, the instrument **1300** comprises a series of pulleys or cables that enable the elongated shaft **1302** to translate relative to the handle **1306**. In other words, the instrument **1300** itself comprises an instrument-based insertion architecture that accommodates insertion of the instrument **1300**, thereby minimizing the reliance on a robot arm to provide insertion of the instrument **1300**. In other embodiments, a robotic arm can be largely responsible for instrument insertion.

#### E. Controller.

Any of the robotic systems described herein can include an input device or controller for manipulating an instrument attached to a robotic arm. In some embodiments, the controller can be coupled (e.g., communicatively, electronically, electrically, wirelessly and/or mechanically) with an instrument such that manipulation of the controller causes a corresponding manipulation of the instrument e.g., via master slave control.

FIG. **14** is a perspective view of an embodiment of a controller **1400**. In the present embodiment, the controller **1400** comprises a hybrid controller that can have both impedance and admittance control. In other embodiments, the controller **1400** can utilize just impedance or passive control. In other embodiments, the controller **1400** can utilize just admittance control. By being a hybrid controller, the controller **1400** advantageously can have a lower perceived inertia while in use.

In the illustrated embodiment, the controller **1400** is configured to allow manipulation of two medical instruments, and includes two handles **1402**. Each of the handles **1402** is connected to a gimbal **1404**, and each gimbal **1404** is connected to a positioning platform **1406**.

As shown in FIG. **14**, each positioning platform **1406** includes a selective compliance assembly robot arm (SCARA) **1408** coupled to a column **1410** by a prismatic joint **1412**. The prismatic joints **1412** are configured to translate along the column **1410** (e.g., along rails **1414**) to allow each of the handles **1402** to be translated in the z-direction, providing a first degree of freedom. The SCARA arm **1408** is configured to allow motion of the handle **1402** in an x-y plane, providing two additional degrees of freedom.

In some embodiments, one or more load cells are positioned in the controller **1400**. For example, in some embodiments, a load cell (not shown) is positioned in the body of each of the gimbals **1404**. By providing a load cell, portions of the controller **1400** are capable of operating under admittance control, thereby advantageously reducing the perceived inertia of the controller **1400** while in use. In some embodiments, the positioning platform **1406** is configured for admittance control, while the gimbal **1404** is configured for impedance control. In other embodiments, the gimbal **1404** is configured for admittance control, while the positioning platform **1406** is configured for impedance control. Accordingly, for some embodiments, the translational or positional degrees of freedom of the positioning platform **1406** can rely on admittance control, while the rotational degrees of freedom of the gimbal **1404** rely on impedance control.

#### F. Navigation and Control.

Traditional endoscopy may involve the use of fluoroscopy (e.g., as may be delivered through a C-arm) and other forms of radiation-based imaging modalities to provide endoluminal guidance to an operator physician. In contrast, the robotic systems contemplated by this disclosure can provide

for non-radiation-based navigational and localization means to reduce physician exposure to radiation and reduce the amount of equipment within the operating room. As used herein, the term “localization” may refer to determining and/or monitoring the position of objects in a reference coordinate system. Technologies such as pre-operative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to achieve a radiation-free operating environment. In other cases, where radiation-based imaging modalities are still used, the pre-operative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to improve upon the information obtained solely through radiation-based imaging modalities.

FIG. **15** is a block diagram illustrating a localization system **1500** that estimates a location of one or more elements of the robotic system, such as the location of the instrument, in accordance to an example embodiment. The localization system **1500** may be a set of one or more computer devices configured to execute one or more instructions. The computer devices may be embodied by a processor (or processors) and computer-readable memory in one or more components discussed above. By way of example and not limitation, the computer devices may be in the tower **112** shown in FIG. **1**, the cart **102** shown in FIGS. **1-3B**, the beds shown in FIGS. **4-9**, etc.

As shown in FIG. **15**, the localization system **1500** may include a localization module **1502** that processes input data **1504a**, **1504b**, **1504c**, and **1504d** to generate location data **1506** for the distal tip of a medical instrument. The location data **1506** may be data or logic that represents a location and/or orientation of the distal end of the instrument relative to a frame of reference. The frame of reference can be a frame of reference relative to the anatomy of the patient or to a known object, such as an EM field generator (see discussion below for the EM field generator).

The various input data **1504a-d** are now described in greater detail. Pre-operative mapping may be accomplished through the use of the collection of low dose CT scans. Pre-operative CT scans are reconstructed into three-dimensional images, which are visualized, e.g. as “slices” of a cutaway view of the patient’s internal anatomy. When analyzed in the aggregate, image-based models for anatomical cavities, spaces and structures of the patient’s anatomy, such as a patient lung network, may be generated. Techniques such as center-line geometry may be determined and approximated from the CT images to develop a three-dimensional volume of the patient’s anatomy, referred to as model data **1504a** (also referred to as “preoperative model data” when generated using only preoperative CT scans). The use of center-line geometry is discussed in U.S. patent application Ser. No. 14/523,760, the contents of which are herein incorporated in its entirety. Network topological models may also be derived from the CT-images, and are particularly appropriate for bronchoscopy.

In some embodiments, the instrument may be equipped with a camera to provide vision data **1504b**. The localization module **1502** may process the vision data **1504b** to enable one or more vision-based location tracking. For example, the preoperative model data may be used in conjunction with the vision data **1504b** to enable computer vision-based tracking of the medical instrument (e.g., an endoscope or an instrument advance through a working channel of the endoscope). For example, using the preoperative model data **1504a**, the robotic system may generate a library of expected endoscopic images from the model based on the expected path of travel of the endoscope, each image linked to a location

within the model. Intra-operatively, this library may be referenced by the robotic system in order to compare real-time images captured at the camera (e.g., a camera at a distal end of the endoscope) to those in the image library to assist localization.

Other computer vision-based tracking techniques use feature tracking to determine motion of the camera, and thus the endoscope. Some features of the localization module **1502** may identify circular geometries in the preoperative model data **1504a** that correspond to anatomical lumens and track the change of those geometries to determine which anatomical lumen was selected, as well as the relative rotational and/or translational motion of the camera. Use of a topological map may further enhance vision-based algorithms or techniques.

Optical flow, another computer vision-based technique, may analyze the displacement and translation of image pixels in a video sequence in the vision data **1504b** to infer camera movement. Examples of optical flow techniques may include motion detection, object segmentation calculations, luminance, motion compensated encoding, stereo disparity measurement, etc. Through the comparison of multiple frames over multiple iterations, movement and location of the camera (and thus the endoscope) may be determined.

The localization module **1502** may use real-time EM tracking to generate a real-time location of the endoscope in a global coordinate system that may be registered to the patient's anatomy, represented by the preoperative model. In EM tracking, an EM sensor (or tracker) comprising of one or more sensor coils embedded in one or more locations and orientations in a medical instrument (e.g., an endoscopic tool) measures the variation in the EM field created by one or more static EM field generators positioned at a known location. The location information detected by the EM sensors is stored as EM data **1504c**. The EM field generator (or transmitter), may be placed close to the patient to create a low intensity magnetic field that the embedded sensor may detect. The magnetic field induces small currents in the sensor coils of the EM sensor, which may be analyzed to determine the distance and angle between the EM sensor and the EM field generator. These distances and orientations may be intra-operatively "registered" to the patient anatomy (e.g., the preoperative model) in order to determine the geometric transformation that aligns a single location in the coordinate system with a position in the pre-operative model of the patient's anatomy. Once registered, an embedded EM tracker in one or more positions of the medical instrument (e.g., the distal tip of an endoscope) may provide real-time indications of the progression of the medical instrument through the patient's anatomy.

Robotic command and kinematics data **1504d** may also be used by the localization module **1502** to provide localization data **1506** for the robotic system. Device pitch and yaw resulting from articulation commands may be determined during pre-operative calibration. Intra-operatively, these calibration measurements may be used in combination with known insertion depth information to estimate the position of the instrument. Alternatively, these calculations may be analyzed in combination with EM, vision, and/or topological modeling to estimate the position of the medical instrument within the network.

As FIG. **15** shows, a number of other input data can be used by the localization module **1502**. For example, although not shown in FIG. **15**, an instrument utilizing shape-sensing fiber can provide shape data that the localization module **1502** can use to determine the location and shape of the instrument.

The localization module **1502** may use the input data **1504a-d** in combination(s). In some cases, such a combination may use a probabilistic approach where the localization module **1502** assigns a confidence weight to the location determined from each of the input data **1504a-d**. Thus, where the EM data **1504c** may not be reliable (as may be the case where there is EM interference) the confidence of the location determined by the EM data **1504c** can be decrease and the localization module **1502** may rely more heavily on the vision data **1504b** and/or the robotic command and kinematics data **1504d**.

As discussed above, the robotic systems discussed herein may be designed to incorporate a combination of one or more of the technologies above. The robotic system's computer-based control system, based in the tower, bed and/or cart, may store computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, or the like, that, upon execution, cause the system to receive and analyze sensor data and user commands, generate control signals throughout the system, and display the navigational and localization data, such as the position of the instrument within the global coordinate system, anatomical map, etc.

## 2. Description.

### Insertion Coupled Inserting Surgical Instrument

FIG. **16** is a side view of an example surgical tool **1600** that may incorporate some or all of the principles of the present disclosure. The surgical tool **1600** may be similar in some respects to any of the surgical tools and medical instruments described above with reference to FIGS. **11-13** and, therefore, may be used in conjunction with a robotic surgical system, such as the robotically-enabled systems **100**, **400**, and **900** of FIGS. **1-9C**. As illustrated, the surgical tool **1600** includes an elongated shaft **1602**, an end effector **1604** arranged at the distal end of the shaft **1602**, and an articulable wrist **1606** (alternately referred to as a "wrist joint") that interposes and couples the end effector **1604** to the distal end of the shaft **1602**. In some embodiments, the wrist **1606** may be omitted, without departing from the scope of the disclosure.

The terms "proximal" and "distal" are defined herein relative to a robotic surgical system having an interface configured to mechanically and electrically couple the surgical tool **1600** to a robotic manipulator. The term "proximal" refers to the position of an element closer to the robotic manipulator and the term "distal" refers to the position of an element closer to the end effector **1604** and thus closer to the patient during operation. Moreover, the use of directional terms such as above, below, upper, lower, upward, downward, left, right, and the like are used in relation to the illustrative embodiments as they are depicted in the figures, the upward or upper direction being toward the top of the corresponding figure and the downward or lower direction being toward the bottom of the corresponding figure.

The surgical tool **1600** can have any of a variety of configurations capable of performing one or more surgical functions. In the illustrated embodiment, the end effector **1604** comprises a tissue grasper capable of grasping onto tissue or vessels. The end effector **1604** includes opposing jaws **1608**, **1610** configured to move (articulate) between open and closed positions. Alternatively, the end effector **1604** may comprise other types of instruments with opposing jaws such as, but not limited to, a surgical stapler, a vessel sealer, surgical scissors, clip applicators, needle drivers,

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a babcock including a pair of opposed grasping jaws, bipolar jaws (e.g., bipolar Maryland grasper, forceps, a fenestrated grasper, etc.), etc.

One or both of the jaws **1608**, **1610** may be configured to pivot to actuate the end effector **1604** between open and closed positions. In the illustrated example, both jaws **1608**, **1610** simultaneously move to pivot the jaws **1608**, **1610** between an open, unclamped position and a closed, clamped position and are thus referred to as “bifurcating” jaws. In other embodiments, however, only one of the jaws **1608**, **1610** may be rotatable (pivotable) relative to the opposing jaw to actuate the end effector **1604** between the open and closed positions.

The wrist **1606** enables the end effector **1604** to articulate (pivot) relative to the shaft **1602** and thereby position the end effector **1604** at various desired orientations and locations relative to a surgical site. In the illustrated embodiment, the wrist **1606** is designed to allow the end effector **1604** to pivot (swivel) left and right relative to a longitudinal axis  $A_1$  of the shaft **1602**. In other embodiments, however, the wrist **1606** may be designed to provide multiple degrees of freedom, including one or more translational variables (i.e., surge, heave, and sway) and/or one or more rotational variables (i.e., Euler angles or roll, pitch, and yaw). The translational and rotational variables describe the position and orientation of a component of a surgical system (e.g., the end effector **1604**) with respect to a given reference Cartesian frame. “Surge” refers to forward and backward translational movement, “heave” refers to translational movement up and down, and “sway” refers to translational movement left and right. With regard to the rotational terms, “roll” refers to tilting side to side, “pitch” refers to tilting forward and backward, and “yaw” refers to turning left and right.

The end effector **1604** is depicted in FIG. **16** in the unarticulated position where the longitudinal axis of the end effector **1604** is substantially aligned with the longitudinal axis  $A_1$  of the shaft **1602**, such that the end effector **1604** is at a substantially zero angle relative to the shaft **1602**. In the articulated position, the longitudinal axis of the end effector **1604** would be angularly offset from the longitudinal axis  $A_1$  such that the end effector **1604** would be oriented at a non-zero angle relative to the shaft **1602**.

Still referring to FIG. **16**, the surgical tool **1600** may include a drive housing or “handle” **1612**, and the shaft **1602** extends longitudinally through the handle **1612**. The handle **1612** houses an actuation system designed to facilitate articulation of the wrist **1606** and actuation (operation) of the end effector **1604** (e.g., clamping, firing, rotation, articulation, energy delivery, etc.). As discussed in more detail below, the handle **1612** may include and otherwise house a plurality of drive gears that are actuatable to axially move (translate) a plurality of sliding rack gears nested within corresponding longitudinal channels defined along all or a portion of the shaft **1602**. In some embodiments, the distal end of each rack gear is attached to distal cables that extend to the end effector **1604** or the wrist **1606** at the distal end of the shaft **1602**, and the proximal end of the each rack gear is attached to proximal cables that extend to a proximal end of the shaft **1602**. In other embodiments, however, the distal end of one or more of the rack gears may be directly attached to portions of the end effector **1604** or the wrist **1606** to enable a push-pull translation action (e.g., in the case of push/pull rods).

Selective actuation of one or more of the sliding rack gears, for example, may cause the end effector **1604** to articulate (pivot) relative to the shaft **1602** at the wrist **1606**. Selective actuation of one or more additional sliding rack

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gears may cause the end effector **1604** to actuate (operate). Actuating the end effector **1604** depicted in FIG. **16** may entail closing and/or opening the jaws, **1608**, **1610** and thereby enabling the end effector **1604** to grasp (clamp) onto tissue. In embodiments where the end effector **1604** comprises a vessel sealer, once tissue is grasped or clamped between the opposing jaws **1608**, **1610**, actuating the end effector **1604** may further include “firing” the end effector **1604**, which may refer to causing a cutting element or knife (not visible) to advance distally within a slot or “guide track” defined in the first jaw **1610**. As it moves distally within the guide track, the knife transects tissue grasped between the opposing jaws **1608**, **1610**.

The actuation system housed within the handle **1612** may further be designed to move the shaft **1602** relative to (through) the handle **1612** and along the longitudinal axis  $A_1$ . More particularly, the actuation system may also include a drive gear actuatable to engage a rack gear defined on the shaft **1602** itself; i.e., a “shaft rack gear”. When the drive gear drives against the shaft rack gear, the shaft **1602** along with the nested sliding rack gears are moved (translated) axially relative to the handle **1612**, as indicated by the arrows B. Moreover, as the shaft **1602** moves, the end effector **1604** and the wrist **1606** are simultaneously advanced or retracted, depending on the driving direction.

The handle **1612** may be operatively coupled to an instrument driver **1614** of a robotic surgical system. The instrument driver **1614** may be similar in some respects to the instrument drivers **1102**, **1200** of FIGS. **11** and **12**, respectively, and therefore may be best understood with reference thereto. Similar to the instrument drivers **1102**, **1200**, for example, the instrument driver **1614** may be mounted to or otherwise positioned at the end of a robotic arm (not shown) and is designed to provide the motive forces required to operate the surgical tool **1600**. Unlike the instrument drivers **1102**, **1200**, however, the shaft **1602** of the surgical tool **1600** extends through and penetrates the instrument driver **1614**.

The handle **1612** includes a plurality of rotatable drive inputs (not visible) that can be driven by a corresponding plurality of drive outputs (not visible) of the instrument driver **1614**. Each drive input is actuatable to independently drive (actuate) various portions of the actuation system housed within the handle **1612** and thereby operate the surgical tool **1600**, as generally described above. The number of drive outputs will generally be the same as the number of drive inputs, but the instrument driver **1614** can have additional drive outputs, without departing from the scope of the disclosure. Movement (rotation) of a given drive output correspondingly moves (rotates) an associated drive input and thereby operates the surgical tool **1600**. Actuation of the drive inputs drives the various drive gears mentioned above, which may be arranged to drive the corresponding rack gears, and moving the rack gears causes the end effector **1604** to articulate and/or actuate (operate) and the shaft **1602** to axially move (translate) relative to the handle **1612**.

In the illustrated embodiment, a decoupler subassembly or “decoupler” **1616** is arranged between and otherwise interposes the handle **1612** and the instrument driver **1614**. Among other functions described herein, the decoupler **1616** transfers torque from the drive outputs of the instrument driver **1614** to the drive inputs of the handle **1612**, and thus operates as a type of torque transfer apparatus. Once the drive outputs are operatively and indirectly coupled to corresponding drive inputs via the decoupler **1616**, rotational torque may be transferred from the drive outputs to the corresponding drive inputs through (via) the decoupler

1616, thus being able to operate the handle 1612. As discussed in more detail below, the decoupler 1616 may also be advantageous in transferring insertion motion (e.g., movement of the shaft 1602) to all the drive inputs, thus allowing one robot motor to control insertion of the surgical tool 1600, while allowing the other motors of the instrument driver 1614 to drive the sliding rack gears independent of insertion.

For procedures that require a sterile environment, an instrument sterile adaptor 1618 (shown in dashed lines) connected to a sterile drape may be incorporated into the surgical tool 1600 upon mounting to the instrument driver 1614. Depending on design or application, the decoupler 1616 can form part of the instrument driver 1614, the sterile adaptor 1618, the handle 1612, or a combination of the foregoing. Since it has many complex parts, it may be advantageous for the decoupler 1616 to not be integrated with the handle 1612, which is commonly the disposable portion of the surgical tool 1600. This reduces the cost of the surgical tool 1600 and the quantity of decouplers 1616 needed per procedure. In some embodiments, the decoupler 1616 could be integrated into the sterile adaptor 1618, as shown in FIG. 16. The sterile adaptor 1618 can be designed to have a longer lifetime than the surgical tool 1600 and fewer are needed per procedure, thus, moving the decoupler 1616 can reduce the cost of the device. Moreover, this has the benefit that the instrument driver 1614 remains unchanged, which is important for driving non-inserting tools, such as endoscopy tools.

In other embodiments, however, the decoupler 1616 may be integrated into and otherwise form part of the instrument driver 1614. In such embodiments, the decoupler 1616 is never disposed of and thus entails a one-time cost for the user. However, this may create an issue when trying to drive non-inserting instruments. In such scenarios, the decoupler 1616 may be configured such that each motor of the instrument driver 1614 has two outputs; one output would be the insertion-decoupled motion while the other output is not. The instrument driver 1614 would then be designed to selectively engage the appropriate outputs. This may be helpful for instruments where both coupled and uncoupled motions are desired. For example, a suction irrigator with valves driven by the inputs could operate with coupled motions for the wrist 1606 and uncoupled motion for the valves that are not inserting with the tool shaft 1602.

In yet other embodiments, the decoupler 1616 could be a stand-alone separable assembly that can attach to the instrument driver 1614 as desired. In such embodiments, the decoupler 1616 could be designed to be mounted below the sterile barrier 1618, or outside of it. The stand-alone version of the decoupler 1616 may be designed and otherwise capable of being sterilized for multiple procedures (e.g., 100 uses), and multiple different tools of single or multiple use design could be attached on top of it.

Lastly, in some embodiments, the surgical tool 1600 may include a tailpiece 1620 arranged at the proximal end of the shaft 1602. The tailpiece 1620, its operation, and various embodiments will be described in more detail below. Briefly, however, the tailpiece 1620 may comprise a mechanical device that provides a means for manually controlling the end effector 1604 and/or the wrist 1606. The tailpiece 1620 can be designed to hold a mechanism referred to herein as a “pantograph” or a “pantograph button”. The pantograph essentially acts as a mirror to the wrist 1606 ensuring that the system maintains tension in the surgical tool 1600 when disconnected from the instrument driver 1614 (e.g., the robot). As described herein, however, the tailpiece 1620 can

be provided in various forms and still perform essentially the same function, without departing from the scope of the present disclosure. In at least one embodiment, as described herein, the pantograph can receive a separate input for manual activation, such as a slider that allows the user to manually actuate one of the degrees of freedom of the wrist 1606, such as jaw opening.

FIGS. 17A and 17B are isometric and bottom views, respectively, of an example actuation system 1700, according to one or more embodiments. The actuation system 1700 may be housed within the handle 1612 (FIG. 16) and operable to actuate (operate) the end effector 1604 (FIG. 16), articulate the wrist 1606 (FIG. 16), and axially move (translate) the shaft 1602 relative to (i.e., through) the handle 1612. Several parts and structural elements of the handle 1612, such as the outer housing, are omitted from FIGS. 17A-17B to enable ease of viewing of the actuation system 1700 for purposes of discussion.

As illustrated, the actuation system 1700 may include a plurality of rotatable drive inputs 1702. As discussed above, the drive inputs 1702 can be driven by a corresponding plurality of drive outputs of the instrument driver 1614. In embodiments that include the decoupler 1616, however, a transmission differential or differential assembly provided by the decoupler 1616 will axially interpose corresponding drive inputs and drive outputs, such that any rotational torque provided by a given drive output will be transferred to the corresponding drive input via the corresponding transmission differential.

Rotating a given drive input 1702 will cause a corresponding drive gear 1704 to rotate (operate), and each drive gear 1704 may be arranged to drive against an adjacent sliding rack gear 1706. Each sliding rack gear 1706 is movably nested within a corresponding longitudinal channel 1708 defined along all or a portion of the shaft 1602. In the illustrated embodiment, the drive gears 1704 are depicted as spur gears with teeth matable with corresponding teeth defined on the opposing sliding rack gear 1706. Accordingly, the drive gears 1704 may alternatively be referred to herein as “spur” gears, and operation of the spur gears 1704 and corresponding rack gears 1706 may be similar to rack-and-pinion operation. Driving a given sliding rack gear 1706 will urge the sliding rack gear 1706 to move (slide) within the corresponding longitudinal channel 1708, and moving the rack gears 1706 within the channels 1708 may actuate (operate) the end effector 1604 (FIG. 16) and/or articulate the wrist 1606 (FIG. 16).

The rack gears 1706 may prove advantageous over cable-based systems since they have a much larger cross-sectional area than cables, so their stiffness is much higher than cables. Consequently, the performance of the tool can be more robust and predictable. Moreover, rack gears 1706 enable the driving of instrument end effectors that do not use cables, such as push/pull rods that are often used in certain designs, such as vessel sealers or staplers.

One of the drive inputs, shown as drive input 1702a, may be configured to cause a corresponding shaft drive gear 1704a to rotate (operate), and the shaft drive gear 1704a may be arranged to drive against an adjacent shaft rack gear 1710. Accordingly, the drive input 1702a may be referred to herein as the “shaft” drive input 1702a. As illustrated, the shaft rack gear 1710 may form part of and may otherwise be defined along all or a portion of the outer surface of the shaft 1602. Driving against the shaft rack gear 1710 will cause the shaft 1602 to move (translate) axially relative to (i.e., through) the handle 1612 (FIG. 16).

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Movement of the shaft **1602** via actuation of the shaft drive input **1702a** is one reason why the decoupler **1616** (FIG. 16) may be advantageous or even needed in the surgical tool **1600** (FIG. 16). As the shaft drive gear **1704a** is actuated (driven) to move the shaft **1602**, the remaining drive gears **1704** must also be driven or the sliding rack gears **1706** will be inadvertently moved, thus affecting articulation, end effector operation, etc. This would require all motors in the instrument driver **1614** (FIG. 16) to operate concurrently whenever the shaft **1602** is axially translated. Moreover, if the shaft drive gear **1704a** is actuated (driven) without driving any of the remaining drive gears **1704**, the shaft **1602** would bind against opposing forces or would inadvertently move the wrist **1606** (FIG. 16) or actuate the end effector **1604** (FIG. 16). As described in more detail below, the differential assemblies included in the decoupler **1616** may be configured to transfer insertion motion (e.g., movement of the shaft **1602**) to all the drive inputs **1702**, thus allowing one robot motor to control insertion of the surgical tool **1600** (FIG. 16), while allowing the other motors of the instrument driver **1614** to drive the sliding rack gears **1706** independent of insertion. In addition to applying insertion motion to all inputs, the decoupler **1616** may be designed to cancel out reaction loads, such that tensions applied to wrist actuators are not realized as reaction torques on the insertion driver.

In some embodiments, a gear train including one or more intermediate gears may interpose each drive input **1702** and the corresponding drive gear **1704**, such that rotating the drive input **1702** will correspondingly rotate the corresponding drive gear **1704** via the interposing gear train. A similar type of gear train may interpose the shaft drive input **1702a** and the shaft drive gear **1704a**. Those skilled in the art will readily appreciate that this gear train can assume a variety of configurations. In the illustrated embodiment, for example, the gear train includes mating helical gears that transition the rotational axis of the drive inputs **1702**, **1702a** 90° to enable the drive gears **1704** and the shaft drive gear **1704a** to properly engage the rack gears **1706** and the shaft rack gear **1710**, respectively. In at least one embodiment, however, the gear train may be omitted and rotating the drive inputs **1702**, **1702a** may directly drive the adjacent rack gear **1706**, **1710**, without departing from the scope of the disclosure.

FIG. 17C is another example of a gear train that may be used to transfer rotational torque from the drive inputs **1702**, **1702a** to the rack gears **1706** or the shaft rack gear **1710**, respectively. In the illustrated embodiment, the drive input **1702**, **1702a** may include a bevel drive gear **1712** arranged to drive a bevel driven gear **1714**, and the bevel driven gear **1714** may further include a spur gear **1716** arranged to drive the drive gear **1704**, **1704a**. The intermeshed bevel gears **1712**, **1714** facilitates the directional change required in the handle **1612** (FIG. 16). Through the depicted gear train arrangement, rotation of the drive input **1702**, **1702a** will correspondingly move the rack gears **1706** or the shaft rack gear **1710**. In other embodiments, other known gearing mechanisms may be utilized or combined in any number of configurations and dimensioned for optimal torque and/or speed outputs. In at least one embodiment, for example, the gear train may incorporate worm gears or a combination of bevel and spur gears.

FIGS. 18A-18D are schematic diagrams of example operation of a portion of the actuation system **1700**, according to one or more embodiments. As mentioned above, the actuation system **1700** is housed within the handle **1612** and includes the drive gears **1704** rotatable (operable) to drive against adjacent sliding rack gears **1706**. As also mentioned

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above, the rack gears **1706** are nested within corresponding longitudinal channels **1708** defined along all or a portion of the shaft **1602**. As depicted, the drive and rack gears **1704**, **1706** provide or define meshing gear teeth that allows the drive gears **1704** to drive the rack gears **1706** longitudinally along the length of the shaft **1602**. A distal structure **1802a** (shown in dashed lines) is provided at the distal end of the shaft **1602** and is representative of the end effector **1604**, the wrist **1606** (FIG. 16), or another mechanism that may be driven or operated through operation of the actuation system **1700**. Similarly, a proximal structure **1802b** (shown in dashed lines) is positioned at the proximal end of the shaft **1602** and is representative of the tailpiece **1620** (FIG. 16) or another mechanism that may be driven or operated through operation of the actuation system **1700**.

In some embodiments, a distal cable **1804a** may be attached to the distal end of each rack gear **1706**, and a proximal cable **1804b** may be attached to the proximal end of each rack gear **1706**. In the illustrated embodiment, the distal cable **1804a** extends from one of the rack gears **1706**, wraps around a distal pulley **1806a**, and is attached to an opposing rack gear **1706**. Similarly, the proximal cable **1804b** extends from one of the rack gears **1706**, wraps around a proximal pulley **1806b**, and is attached to the opposing rack gear **1706**. The cables **1804a,b** can be made from a variety of materials including, but not limited to, a metal (e.g., tungsten, stainless steel, nitinol, etc.) a polymer (e.g., ultra-high molecular weight polyethylene), a synthetic fiber (e.g., KEVLAR®, VECTRAN®, etc.), an elastomer, or any combination thereof.

The coupled assembly of the rack gears **1706**, the distal and proximal cables **1804a,b**, and the distal and proximal pulleys **1806a,b** completes or provides a cable loop, which allows the rack gears **1706** to operate in an antagonistic relationship where movement of one rack gear **1706** in one longitudinal direction can cause the other rack gear to move in the opposite direction and thereby manipulate the angular position of the distal structure **1802a** (e.g., the end effector **1604** or the wrist **1806**). In this example, selective actuation of the drive gears **1704** can cause the end effector **1604** (FIG. 16) to open or close, or may alternatively cause the wrist **1606** (FIG. 16) to articulate.

In FIG. 18B, for example, actuation (articulation) of the wrist **1606** may be accomplished by rotating the drive gears **1704** in the same direction, as indicated by the arrows C. This allows the rack gears **1706** to antagonistically operate and cause the distal structure **1802a** to move (rotate) in the direction D. Alternatively, and depending on how the distal pulley **1806a** is configured, rotating the drive gears **1704** in the same direction C may also cause one of the jaws **1608**, **1610** (FIG. 16) of the end effector **1604** to open or close.

In FIG. 18C, axial movement of the shaft **1602** (e.g., insertion of the instrument) may be accomplished by rotating the drive gears **1704** in opposite directions, as shown by the arrows C and E. In this scenario, the shaft **1602** and the rack gears **1706** will be moved (translated) along its axis through the handle **1612** in the direction F. As will be appreciated, the configuration of the actuation system **1700** may be altered and otherwise modified to facilitate pitch, yaw, and jaw actuation of the distal structure **1802a**.

This concept extends to more complicated wrists (e.g., the wrist **1606** of FIG. 16) that have more than one degree of freedom and follows N+1 rules where there are N+1 cables **1804a,b** for N degrees of freedom in the wrist **1606**. The handle **1612** would contain N+1 drive gears **1704** that each drive a rack gear **1706**. The proximal structure **1802b** at the proximal end of the shaft **1602** would also change to

accommodate a wrist with more degrees of freedom. For instance, an example needle driver wrist may be driven by four cables **1804a,b**, so the instrument would have four drive gears **1704** and four rack gears **1706**. Additionally, this design could support more wrist concepts, such as a 2N wrist, and the proximal structure **1802b** would have to change to appropriately mirror the wrist configuration.

In FIG. **18D**, the actuation system **1700** is shown with the shaft drive gear **1704a** arranged to drive against the adjacent shaft rack gear **1710**, where the teeth of the shaft rack gear **1710** are defined along all or a portion of the shaft **1602**. In some embodiments, as discussed above, driving against the shaft rack gear **1710** will cause the shaft **1602** to move (translate) axially relative to (i.e., through) the handle **1612**. In other embodiments, however, driving against the shaft rack gear **1710** may help adjust tension in the cables (e.g., the distal cable **1804a**). More specifically, in order to facilitate on-robot tensioning, each of the drive gears **1704** (only one shown) may be rotated in the direction G, which places the distal cables **1804a** (only one shown) in tension. The generated tension may then be resolved by actuating the shaft drive gear **1704a** in the direction E, which urges the shaft **1602** in the direction F. Consequently, the insertion degree of freedom may also be used to place tension on the robot, and, theoretically, all backlash in the system is now taken up when the robot tensions the tool.

In this scenario, portions of the shaft **1602** and the shaft rack gear **1710** located distal to the handle **1612** (i.e., to the left of the handle **1612** in FIG. **18D**) will be under load when the robot puts tension in the tool. More specifically, portions of the shaft **1602** distal to the handle **1612** will be in compression and portions of the rack gear **1706** distal to the handle **1612** will be in tension. The remaining portions of the shaft **1602** and the rack gear **1706** proximal to the handle **1612**, however, will not be loaded. This is in contrast to cable-based surgical tool designs, where the loaded regions extend the entire length of the tool shaft with the cable in tension and the tool shaft in compression. Here, the actuation system **1700** results in a shorter loading path, which may be advantageous in generating smaller amounts of deflection at the wrist **1606** (FIG. **16**) because the potential “spring” length is shorter. This may also result in the deflection length being dependent on the insertion length, but the input motion may be software adjusted based on the insertion length.

This architecture may further prove advantageous in removing backlash from insertion of the surgical tool **1600** (FIG. **16**) and the sterile adaptor **1618** (FIG. **16**). In this architecture, tension in the cables extending through the wrist **1606** (FIG. **16**) may be able to react against the compressive load through the shaft **1602**. This means that the gear trains located in the handle **1612** and the decoupler **1616** (FIG. **16**) are all preloaded and all backlash in the decoupler **1616** and the surgical tool **1600** are removed. If the decoupler **1616** is located in the instrument driver **1614** (FIG. **16**), however, the driver motors can be attached directly to the drive inputs provided on the decoupler **1616**, all components between the wrist **1606** and the motors of the instrument driver **1614** are preloaded, and all backlash is theoretically eliminated.

Additionally, this architecture allows for more complicated components to be moved or designed into the instrument driver **1614** (FIG. **16**). If the decoupler **1616** (FIG. **16**) is integrated into the instrument driver **1614** or the sterile adaptor **1618** (FIG. **16**), the surgical tool **1600** (FIG. **16**) would contain fewer parts than a cable-differential design. As will be appreciated, this will help drive down the cost of

the disposable component of the system, reduce manufacturing time, and lower the number of failure points in the instrument.

FIG. **19** is an isometric view of one example of the tailpiece **1620**, according to one or more embodiments. As mentioned above, the tailpiece **1620** may be alternately referred to as a “pantograph” or “pantograph button,” and may provide a means for maintaining tension in the surgical tool **1600** (FIG. **16**) when the instrument is detached from the instrument driver **1614** (FIG. **16**). The pantograph **1620** may additionally provide a way to manually control the end effector **1604** (FIG. **16**) and/or the wrist **1606** (FIG. **16**).

In the illustrated embodiment, the tailpiece **1620** includes a housing **1902** arranged at the distal end of the shaft **1602** (shown in dashed lines). A manual actuation device may be mounted to the housing **1902**. In this embodiment, the manual actuation device includes first and second buttons **1904a** and **1904b** pivotably coupled to the housing **1902** and laterally offset from each other. Each button **1904a,b** may be coupled to a corresponding one of the proximal cables **1804b**. In the illustrated configuration, manipulation of the buttons **1904a,b** may cause the jaws **1608**, **1610** (FIG. **16**) to open and close. More specifically, when the first button **1904a** is depressed, the jaws **1608**, **1610** open, and when the second button **1904b** is depressed, the jaws **1608**, **1610** close. The architecture of the pantograph **1620** could be altered, however, such that manipulation of the buttons **1904a,b** may cause the wrist **1606** (FIG. **16**) to articulate, without departing from the scope of the disclosure.

FIGS. **20A** and **20B** depict example operation of the tailpiece (pantograph) **1620** of FIG. **19**, according to one or more embodiments. Each of FIGS. **20A-20B** include two images of the pantograph **1620**, the left image being a partial, cross-sectional view, and the right image including portions in phantom (dashed lines) to enable viewing of various internal components. As illustrated, the first and second buttons **1904a,b** may be coupled to a toggle or “rocker” **2002** pivotably coupled to the housing **1902** such that pressing down on one button **1904a,b** simultaneously causes the other button **1904a,b** to pivot upward.

First and second proximal pulleys **2004a** and **2004b** may be rotatably mounted to the first and second buttons **1904a,b** and are axially movable therewith during actuation of the pantograph **1620** and within the housing **1902**. The proximal pulleys **2004a,b** may be similar to the proximal pulley **1806b** of FIGS. **18A-18C** and thus have corresponding proximal cables **2006a** and **2006b** extending (wrapped) thereabout, where the proximal cables **2006a,b** are similar to the proximal cable **1804b** of FIGS. **18A-18C**.

As mentioned above, the purpose of the pantograph **1620** is to maintain tension in the surgical tool **1600** (FIG. **16**) when disconnected from the instrument driver **1614** (FIG. **16**), but additionally, to provide a way to manually control the end effector **1604** (FIG. **16**) and/or the wrist **1606** (FIG. **16**). The pantograph **1620** ensures that cable length is conserved within the surgical tool **1600**, which prevents the cables **2006a,b** from derailing off the pulleys **2004a,b** (also prevents the distal cables **1804a** of FIGS. **18A-18C** from derailing off the distal pulleys **1806a** of FIGS. **18A-18C**). The pantograph **1620** maintains tension in the surgical tool **1600** by completing the cable loops and conserving cable length.

In the illustrated embodiment, manually depressing the first button **1904a** may cause the jaws **1608**, **1610** (FIG. **16**) to open, and manually depressing the second button **1904b** may cause the jaws **1608**, **1610** to close. The buttons **1904a,b** are pivotably constrained together by the rocker

**2002** such that if one button **1904a,b** is pressed down, the opposite button **1904a,b** correspondingly moves up with the same axial displacement. Moreover, since the pulleys **2004a,b** are coupled to the buttons **1904a,b**, as one button **1904a,b** is pressed down, the corresponding pulley **2004a,b** also moves down, and the opposing button **1904a,b** and its corresponding pulley **2004a,b** moves up with the same displacement. Consequently, the net cable length of the cables **2006a,b** remains the same because the two pulleys **2004a,b** have the same displacement, but the change in length of each pair changes, which causes movement of the jaws **1608, 1610**.

FIG. **21** is an enlarged side view of the tailpiece (pantograph) **1620** of FIG. **19**, according to one or more embodiments. The pantograph **1620** can also simplify the process of tensioning the surgical tool **1600** (FIG. **16**). In the illustrated embodiment, for example, the housing **1902** is movably mounted to the shaft **1602** adjacent a tensioning nut **2102** threaded to the proximal end of the shaft **1602**. Rotating the tensioning nut **2102** may apply an axial load on the bottom of the housing **1902**, which may correspondingly extend the cable paths and thus tension the cables **2006a,b** (FIGS. **20A-20B**) in the system. Accordingly, pushing against the housing **1902** with the tensioning nut **2102** applies tension equally to all cables **1804a,b** (FIGS. **18A-18C**) in the system.

FIG. **22** is a side view of the surgical tool **1600**, according to one or more additional embodiments. The tailpiece (pantograph) **1620** arranged at the proximal end of the shaft **1602** may further prove advantageous in converting the surgical tool **1600** into a type of manual laparoscopic surgical tool. More specifically, a manual accessory **2202** may be mounted to or otherwise secured to the tailpiece **1620**. In the illustrated embodiment, for instance, the manual accessory may include ergonomic features, such as opposing handles **2204** and finger holes **2206**, that allow a user to operate the manual accessory **2202** and thereby manually operate at least one of the buttons **1904a,b** to control the end effector **1604** (FIG. **16**) or the wrist **1606** (FIG. **16**). In at least one embodiment, the manual accessory **2202** may be used to lock the wrist **1606** so manual actuation of the pantograph **1620** only articulates the jaws **1608, 1610** (FIG. **16**) open and closed.

FIGS. **23A** and **23B** are isometric and bottom views, respectively, of one example of the decoupler **1616** of FIG. **16**, according to one or more embodiments. As discussed above, the decoupler **1616** may interpose the handle **1612** (FIG. **16**) and the instrument driver **1614** (FIG. **16**), and may otherwise operate to transfer torque from the drive outputs of the instrument driver **1614** to the drive inputs of the handle **1612**. The architecture of the decoupler **1616** further allows for the transfer of insertion motion (e.g., movement of the shaft **1602**) to all the drive inputs of the handle **1612**, thus allowing one robot motor from the instrument driver **1614** to control insertion of the surgical tool **1600**, while allowing the other driver motors to drive the sliding rack gears independent of insertion.

As best seen in FIG. **23A**, the decoupler **1616** may include a first or "upper" flange **2302a** and a second or "lower" flange **2302b** axially offset from the upper flange **2302a**. Concentrically aligned central apertures **2303** are defined in each flange **2302a,b** and through which the shaft **1602** (FIG. **16**) can extend. The flanges **2302a,b** may be operatively coupled with one or more columns **2304** that extend between the flanges **2302a,b**. In FIG. **23B**, the lower flange **2302b**

and the columns **2304** are omitted to enable viewing of the internal components of the decoupler **1616** from a bottom perspective.

The decoupler **1616** includes a plurality of differential assemblies **2306** rotatably mounted between the upper and lower flanges **2302a,b**. While five differential assemblies **2306** are depicted in FIGS. **23A-23B**, more or less than five may be included in the decoupler **1616**, without departing from the scope of the disclosure. As best seen in FIG. **23B**, each differential assembly **2306** includes a differential input **2308**, which is matable with a corresponding one of the drive outputs of the instrument driver **1614** (FIG. **16**); each motor in the instrument driver **1614** drives one of the differential assemblies **2306**. As illustrated, each differential input **2308** may define a receptacle that defines splines, protrusions, or other structural features designed to mate with corresponding features of the drive outputs of the instrument driver **1614**, or vice versa. Once properly mated, the differential inputs **2308** will share axes of rotation with the corresponding drive outputs of the instrument driver **1614** to allow the transfer of rotational torque from the drive output to the corresponding differential input **2308**.

As best seen in FIG. **23A**, each differential assembly **2306** may also include a differential output **2310**, which is matable with a corresponding one of the drive inputs **1702** (FIGS. **17A-17B**) of the actuation system **1700** (FIGS. **17A-17B**). As illustrated, each differential output **2310** may define splines, protrusions, or other structural features designed to mate with corresponding receptacles of the drive inputs **1702**, or vice versa. Once properly mated, the drive inputs **1702** will share axes of rotation with a corresponding differential output **2310** to allow rotational torque from the differential output **2310** to be transferred to the corresponding drive inputs **1702**. Accordingly, once the decoupler **1616** is properly installed between the handle **1612** (FIG. **16**) and the instrument driver **1614** (FIG. **16**), the drive inputs **1702** will share axes of rotation with the corresponding drive outputs of the instrument driver **1614** (FIG. **16**) via the decoupler **1616**, which allows the transfer of rotational torque from the drive outputs to the corresponding drive inputs **1720**.

The decoupler **1616** further includes an insertion assembly **2312** also rotatably mounted between the upper and lower flanges **2302a,b**. As best seen in FIG. **23B**, the insertion assembly **2312** includes an insertion input **2314**, which is matable with a corresponding one of the drive outputs of the instrument driver **1614** (FIG. **16**); referred to herein as the "shaft drive output". The insertion input **2314** may define a receptacle that defines splines, protrusions, or other structural features designed to mate with corresponding features of the corresponding drive output of the instrument driver **1614**, or vice versa. Once properly mated, the insertion input **2314** will share an axis of rotation with the corresponding drive output of the instrument driver **1614** to allow the transfer of rotational torque from the drive output to the insertion input **2314**.

As seen in FIG. **23A**, the insertion assembly **2312** further includes an insertion output **2316**, which is matable with a corresponding one of the drive inputs **1702** (FIGS. **17A-17B**) of the actuation system **1700** (FIGS. **17A-17B**); referred to herein as the "shaft drive input". As illustrated, the insertion output **2316** may define splines, protrusions, or other structural features designed to mate with a receptacle of a corresponding drive input **1702**, or vice versa. Once properly mated, the insertion output **2316** will share an axis of rotation with the corresponding drive input **1702** to allow the transfer of rotational torque from the insertion output

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2316 to the corresponding drive input 1702. Accordingly, once the decoupler 1616 is properly installed between the handle 1612 (FIG. 16) and the instrument driver 1614 (FIG. 16), the drive input 1702 mated to the insertion output 2316 will share an axis of rotation with the drive output of the instrument driver 1614 (FIG. 16) mated with the insertion input 2314 via the decoupler 1616, which allows the transfer of rotational torque from the drive output to the corresponding drive input 1720.

In the case of an insertion-coupled instrument, such as the surgical tool 1600 (FIG. 16), when the shaft 1602 (FIG. 16) is inserted or retracted, all the sliding rack gears 1706 (FIGS. 17A-17B and 18A-18D) need to move at the same rate to maintain the position of the wrist 1606 (FIG. 16). One solution to accomplish this may be to have the motors of the instrument driver 1614 (FIG. 16) jointly perform the coupled motion. In this solution, however, the power requirements for the motors increases because the performance of the surgical tool 1600 is based on maintaining a grip force and a tip velocity. If the motors were to drive in a coupled fashion, then the motors would have to move faster and still maintain the same force output. This is particularly difficult because the required insertion velocity is much higher than the velocity to drive the sliding rack gears 1706.

In contrast, the decoupler 1616 allows the surgical tool 1600 (FIG. 1) with coupled motions to be driven by uncoupled inputs. More specifically, the decoupler 1616 allows one motor from the instrument driver 1614 (FIG. 16) to control insertion of the surgical tool 1600 (i.e., axial movement of the shaft 1602 of FIG. 16) while simultaneously allowing the other motors of the instrument driver 1614 to drive the sliding rack gears 1706 (FIGS. 17A-17B and 18A-18D) independent of insertion. To accomplish this, the decoupler 1616 includes a differential gear train 2318, which can include a system of mechanically-linked differential sub-assemblies that allow the insertion input 2314 to simultaneously drive each differential output 2310 of the decoupler 1616, while also allowing each differential input 2308 to independently drive the corresponding differential output 2310.

As best seen in FIG. 23B, the differential gear train 2318 includes an insertion transmission gear 2320 that may be driven by an insertion input gear 2322 mounted to and otherwise forming part of the insertion assembly 2312. As the insertion assembly 2312 is driven (rotated), the insertion input gear 2322 drives against and causes the insertion transmission gear 2320 to rotate, and the insertion transmission gear 2320 is arranged to interface with a differential insertion input gear 2324 mounted to and otherwise forming part of each differential assembly 2306. As the insertion transmission gear 2320 drives against the differential insertion input gears 2324 simultaneously, each differential assembly 2306 correspondingly rotates. Accordingly, the insertion assembly 2312 is coupled to all the differential assemblies 2306 in the decoupler 1616 by a single gear; i.e., the insertion transmission gear 2320. Thus, when insertion is driven by operation (rotation) of the insertion assembly 2312, each of the differential assemblies 2306 is simultaneously rotated, which means that by driving the insertion input 2314, all the differential outputs 2310 simultaneously rotate. Additionally, if a motor input to any of the differential assemblies 2306 is rotated, then the corresponding differential output 2310 is simultaneously rotated. As will be appreciated, the result of this is the separation of the insertion motion from the motions (articulation) of the wrist 1606 (FIG. 16).

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FIG. 24 is an isometric view of one example of the insertion assembly 2312, according to one or more embodiments. As illustrated, the insertion assembly 2312 includes a drive shaft 2402, and the insertion input 2314 is located at a first end of the drive shaft 2402 while the insertion output 2316 is located at a second (opposite) end of the drive shaft 2402. The insertion input gear 2322 is mounted to the drive shaft 2402 at a location between the insertion input and output 2314, 2316. In some embodiments, as illustrated, the insertion input and output 2314, 2316 and the insertion input gear 2322 may form an integral part of the drive shaft 2402, thus making the insertion assembly 2312 a monolithic part. In other embodiments, however, one or more of the insertion input and output 2314, 2316 and the insertion input gear 2322 may be coupled (fixed) to the drive shaft 2402, without departing from the scope of the disclosure.

FIG. 25 is an isometric view of an example differential assembly 2306, according to one or more embodiments. As illustrated, the differential assembly 2306 includes a drive shaft 2502, and the differential input 2308 is located at a first end of the drive shaft 2502 while the differential output 2310 is located at a second (opposite) end of the drive shaft 2502. The differential insertion input gear 2324 is mounted to or otherwise provided on the drive shaft 2502 at a location between the differential input 2308 and the differential output 2310.

The differential assembly 2306 may further include a differential 2504 also mounted to the drive shaft 2502 at a location between the differential input 2308 and the differential output 2310. The differential 2504 allows the differential output 2310 to be driven by the difference of two inputs; i.e., one from the differential input 2308 and another from the differential insertion input gear 2324 as driven by operation of the insertion assembly 2312 (FIGS. 23A-23B and 24).

FIGS. 26A-26C are side views of the differential assembly 2306 of FIG. 25 in various stages of deconstruction, according to one or more embodiments. In FIG. 26A, various parts of the differential assembly 2306 are shown in phantom (dashed lines) to enable viewing of various internal parts of the differential 2504. As illustrated, the differential 2504 may include a differential gear box or "carrier" 2602, a first or "upper" bevel gear 2604a, a second or "lower" bevel gear 2604b, and a pair of opposing side bevel gears or "carrier gears" 2606 extending between the upper and lower bevel gears 2604a,b.

The lower bevel gear 2604b operates as an input gear and may be secured to the drive shaft 2502 such that rotation of the drive shaft 2502 rotates the lower bevel gear 2604b and the differential output 2310. In some embodiments, the lower bevel gear 2604b may form part of the differential insertion input gear 2324, but may alternatively be nested within a portion of the differential insertion input gear 2324. The differential insertion input gear 2324, however, may nonetheless be able to drive rotation of the lower bevel gear 2604b, which drives rotation of the drive shaft 2502 and the differential output 2310 via interaction with the carrier gears 2606 and the upper bevel gear 2604a. The upper bevel gear 2604a operates as an output gear and may rotatably mounted to the drive shaft 2502 (i.e., free floating). While rotation of the lower bevel gear 2604b creates rotation input to the carrier 2602, the upper bevel gear 2604a rotates as a combination of rotation of the lower bevel gear 2604b and the carrier 2602. As the upper bevel gear 2604a rotates, the differential output 2310 correspondingly rotates.

An axle 2608 may be mounted to the carrier 2602, and the carrier gears 2606 may be rotatably mounted to the axle

2608. When the differential input 2308 is actuated, as discussed above, the drive shaft 2502 will rotate and correspondingly rotate the lower bevel gear 2604b and the differential output 2310. Rotation of the lower bevel gear 2604b will also drive against and cause the carrier gears 2606 to rotate about (on) the upper bevel gear 2604a, which results in rotation of the carrier 2602 while the upper bevel gear 2604a remains stationary. In contrast, however, when the differential insertion input gear 2324 is rotated via actuation of the insertion assembly 2312, the upper bevel gear 2604a will rotate and thereby cause the carrier gears 2606 to drive against the lower bevel gear 2604b, which correspondingly rotates the drive shaft 2502 and the differential output 2310. In this scenario, the carrier 2602 remains stationary. Accordingly, the differential 2504 may allow two inputs to create rotation of the differential output 2310. As will be appreciated, the insertion motion generated by the insertion assembly 2312 is transferred to all differential assemblies 2306 of the decoupler 1616.

While this design uses bevel gear differentials, those skilled in the art will readily appreciate that this same concept may be realized with any type of differential, such as a planetary or spur differential, which are described in more detail herein. Moreover, as mentioned above, the decoupler 1616 (FIGS. 23A-23B) could be integrated into the instrument driver 1614 (FIG. 16). In such embodiments, the decoupler 1616 may be integrated into the gearbox of the motors of the instrument driver 1614 because the differentials can provide a gear reduction. This may require a redesign of the decoupler 1616 such that each motor of the instrument driver 1614 has two outputs. One would be the insertion-decoupled motion while the other is not, and the instrument driver 1614 would then be designed to selectively engage the appropriate outputs. One way to accomplish this is graphically depicted in FIGS. 27A and 27B.

FIGS. 27A and 27B are isometric and cross-sectional side views, respectively, of another example differential assembly 2306, according to one or more additional embodiments. The differential assembly 2306 of FIGS. 27A-27B may be similar in some respects to the differential assembly 2306 of FIGS. 25 and 26A-26B, and therefore may be best understood with reference thereto, where like numerals will represent like components not described again in detail. As illustrated, for example, the differential assembly 2306 of FIGS. 27A-27B includes the drive shaft 2502, the differential input 2308 located at the first end of the drive shaft 2502, the differential insertion input gear 2324, and the differential 2504.

Unlike the differential assembly 2306 of FIGS. 25 and 26A-26C, however, the differential assembly 2306 of FIGS. 27A-27B may include first and second differential outputs 2702a and 2702b located at the second (opposite) end of the drive shaft 2502. The first and second differential outputs 2702a,b are coaxially aligned with each other and able to rotate independent of the other. More specifically, the first differential output 2702a is coupled to the drive shaft 2502 such that actuating the differential input 2308 drives the drive shaft 2502 and rotates the first differential output 2702a. In contrast, when the differential insertion input gear 2324 is rotated via actuation of the insertion assembly 2312 (FIGS. 23A-23B and 24), the differential 2504 operates to drive against the upper bevel gear 2604a, which correspondingly rotates the second differential output 2702b. In such embodiments, a corresponding drive input of the handle 1612 (FIG. 16) will be configured to engage with one of the differential outputs 2702a,b, while the other differential

output 2702a,b is able to freely spin. This allows the instrument input design to determine which differential output 2702a,b is used.

Another solution could be a mechanism that uncouples the insertion assembly 2312 (FIGS. 23A-23B and 24) from the differential insertion input gears 2324 on each differential assembly 2306 then keeps the differential insertion input gears 2324 on all the differential assemblies 2306 stationary. With this setup, the differential assembly 2306 only has one input driving the differential output 2310, so they drive the differential output 2310 as normal. One design to execute this concept may be to have the insertion transmission gear 2320 (FIG. 23) on a shifter such that it could be selectively engaged or disengaged with the differential insertion input gears 2324 on each differential assembly 2306, and then lock the insertion transmission gear 2320. This design could be actuated by a single solenoid in the instrument driver 1614 (FIG. 16), for example.

FIG. 28 is a top, schematic view of another example decoupler 2802, according to one or more additional embodiments. The decoupler 2802 may be similar in some respects to the decoupler 1616 of FIGS. 23A-23B, and therefore may be best understood with reference thereto. Similar to the decoupler 1616, for example, the decoupler 2802 may be incorporated into the surgical tool 1600 (FIG. 16) and may be configured to interpose the instrument driver 1614 (FIG. 16) and the handle 1612 (FIG. 16). The decoupler 2802 may be operable to transfer torque from the drive outputs of the instrument driver 1614 to the drive inputs of the handle 1612. Moreover, the architecture of the decoupler 2802 further allows for the transfer of insertion motion (e.g., movement of the shaft 1602) to all the drive inputs of the handle 1612, thus allowing one robot motor from the instrument driver 1614 to control insertion of the surgical tool 1600, while allowing the other driver motors to drive the sliding rack gears independent of insertion. The decoupler 2802 defines a central aperture 2803 through which the shaft 1602 is able to extend and translate.

Unlike the decoupler 1616 of FIGS. 23A-23B, however, which incorporated a bevel gear differential design, the decoupler 2802 incorporates a planetary or epicyclic gearbox as the differential and primary axle driver. Those skilled in the art will appreciate that the specific design of the decoupler 2802 shown herein can allow for additional options to vary gear ratios and thereby augment load output for the cable drivers that may require higher loads. The decoupler 2802 may also prove beneficial in facilitating smaller packaging, which allows for a reduced overall height for the tool assembly and thereby improves internal reach for a common shaft.

As illustrated, the decoupler 2802 includes a housing 2804 and a plurality of differential assemblies 2806 are mounted to the housing 2804. While five differential assemblies 2806 are depicted in FIG. 28, more or less than five may be included in the decoupler 2802, without departing from the scope of the disclosure. On the bottom of the decoupler 2802, each differential assembly 2806 includes a differential input 2808 (see FIGS. 29, 31, and 32) matable with a corresponding one of the drive outputs of the instrument driver 1614 (FIG. 16); each motor in the instrument driver 1614 drives one of the differential assemblies 2806 via a corresponding one of the differential inputs 2808. Once properly mated, the differential input 2808 will share an axis of rotation with the corresponding drive output of the instrument driver 1614 to allow the transfer of rotational torque from the drive output to the corresponding differential input 2808.

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Each differential assembly **2806** may also include a differential output **2810** (see FIGS. **29**, **31**, and **32**), which is matable with a corresponding one of the drive inputs **1702** (FIGS. **17A-17B**) of the actuation system **1700** (FIGS. **17A-17B**). Once properly mated, the drive inputs **1702** will share axes of rotation with corresponding differential outputs **2810** to allow rotational torque from the differential output **2810** to be transferred to the corresponding drive input **1702**. Accordingly, once the decoupler **2802** is properly installed between the handle **1612** (FIG. **16**) and the instrument driver **1614** (FIG. **16**), the drive inputs **1702** will share axes of rotation with the corresponding drive outputs of the instrument driver **1614** (FIG. **16**) via the decoupler **2802**, which allows the transfer of rotational torque from the drive outputs to the corresponding drive inputs **1720**.

The decoupler **2802** further includes an insertion assembly **2812** also mounted to the housing **2804**. The insertion assembly **2812** includes an insertion input **2814** (see FIGS. **33-34**), which is matable with a corresponding one of the drive outputs of the instrument driver **1614** (FIG. **16**); referred to herein as a “shaft drive output”. Once properly mated, the insertion input **2814** will share an axis of rotation with the shaft drive output of the instrument driver **1614** to allow the transfer of rotational torque from the drive output to the insertion input **2814**.

In the illustrated embodiment, the insertion assembly **2812** is provided in the form of a planetary gear box, which includes a sun gear **2816** (see FIGS. **33-34**), a plurality of planetary gears **2818** surrounding and intermeshed with the sun gear **2816**, and a planet carrier **2820** (see FIGS. **33-34**) coupled to each planetary gear **2818**. As best seen in FIGS. **33-34**, and in some embodiments, the planet carrier **2820** may form part of an insertion output **2824** matable with a corresponding one of the drive inputs **1702** (FIGS. **17A-17B**) of the actuation system **1700** (FIGS. **17A-17B**); referred to herein as the “shaft drive input”. Once properly mated, the insertion output **2824** will share an axis of rotation with the corresponding shaft drive input **1702** to allow the transfer of rotational torque from the insertion output **2824** to the corresponding drive input **1702**. Accordingly, once the decoupler **2802** is properly installed between the handle **1612** (FIG. **16**) and the instrument driver **1614** (FIG. **16**), the shaft drive input **1702** mated to the insertion output **2824** will share an axis of rotation with the shaft drive output of the instrument driver **1614** (FIG. **16**) mated with the insertion input **2814** via the decoupler **2802**. This allows the transfer of rotational torque from the shaft drive output of the instrument driver **1614** to the corresponding shaft drive input **1720**.

Referring again to FIG. **28**, each differential assembly **2806** may further include a differential **2826** that allows the differential output **2810** (FIGS. **29**, **31**, and **32**) to be driven by the difference of two inputs; i.e., one from the differential input **2808** (FIGS. **29**, **31**, and **32**) and another from the insertion assembly **2812**. Similar to the insertion assembly **2812**, each differential **2826** is provided in the form of a planetary gearbox, which includes a sun gear **2828**, a plurality of planetary gears **2830** surrounding the sun gear **2828**, and a planet carrier **2832** (see FIGS. **29-31**) coupled to each planetary gear **2816**. In some embodiments, as seen in FIGS. **31** and **32**, the planet carrier **2832** may form part of the differential output **2810**. Each differential **2826** may further include a differential ring gear **2834** surrounding and intermeshed with the planetary gears **2830**. As described below, the differential ring gear **2834** may be driven by operation of the insertion assembly **2812**, and thereby take into account insertion movement during operation.

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Similar to the decoupler **1616** of FIGS. **23A-23B**, the decoupler **2802** allows the surgical tool **1600** (FIG. **1**) with coupled motions to be driven by uncoupled inputs. More specifically, the decoupler **2802** allows one motor from the instrument driver **1614** (FIG. **16**) to control insertion of the surgical tool **1600** (i.e., axial movement of the shaft **1602** of FIG. **16**) while simultaneously allowing the other motors of the instrument driver **1614** to drive the sliding rack gears **1706** (FIGS. **17A-17B** and **18A-18D**) independent of insertion. To accomplish this, the decoupler **2802** includes a differential gear train **2836**, which can include a system of mechanically-linked differential sub-assemblies that allows the insertion input **2814** (see FIGS. **33-34**) to simultaneously drive each differential output **2810** (see FIGS. **29**, **31**, and **32**) of each differential assembly **2806**, while also allowing each differential input **2808** (see FIGS. **29**, **31**, and **32**) of each differential assembly **2806** to independently drive the corresponding differential output **2810**.

In the illustrated embodiment, the differential gear train **2836** includes an insertion transmission gear **2838** and an insertion input gear **2840** arranged to drive the insertion transmission gear **2838**. As illustrated, the insertion input gear **2840** surrounds and is intermeshed with the planetary gears **2818**. As the insertion assembly **2812** is driven (rotated) to move the shaft **1602** (FIG. **16**), the insertion input gear **2840** is simultaneously driven against and causes the insertion transmission gear **2838** to rotate. The insertion transmission gear **2838** is arranged to interface with (drive) the differential ring gear **2834** of each differential assembly **2806** such that each differential assembly **2806** correspondingly rotates.

Accordingly, the insertion assembly **2812** is coupled to the all the differential assemblies **2806** in the decoupler **2802** by a single gear; i.e., the insertion transmission gear **2838**. Thus, when insertion is driven by operation (rotation) of the insertion assembly **2812**, each of the differential assemblies **2806** is simultaneously rotated, which means that by driving the insertion input **2814** (FIGS. **33-34**), all the differential outputs **2810** (FIGS. **29**, **31**, and **32**) simultaneously rotate. Additionally, if a motor input to any of the differential assemblies **2806** is rotated, then the corresponding differential output **2810** is simultaneously rotated. As will be appreciated, the result of this is the separation of the insertion motion from the motions (articulation) of the wrist **1606** (FIG. **16**).

In some embodiments, the differential gear train **2836** may further include a second insertion transmission gear **2842**. In such embodiments, the first insertion transmission gear **2838** may be characterized as a central spur gear and the second insertion transmission gear **2842** may be characterized as an outer ring gear. The second insertion transmission gear **2842** may operate similar to the first differential gear **2838**, in that the second insertion transmission gear **2842** may intermesh with and be driven via rotation of the insertion input gear **2840**. Moreover, the second insertion transmission gear **2842** may be arranged to interface with the differential ring gear **2834** of each differential assembly **2806**, and rotation of the second insertion transmission gear **2842** simultaneously drives against each differential ring gear **2834**. In at least one embodiment, the first insertion transmission gear **2838** may be omitted and the differential gear train **2836** may only include the second insertion transmission gear **2842** to drive the differential ring gears **2834**.

FIG. **29** is an isometric view of an example differential assembly **2806**, FIG. **30** is a cross-sectional top view of the differential assembly **2806** taken along the lines shown in

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FIG. 29, and FIG. 31 is a schematic cross-sectional side view of the differential assembly 2806 of FIG. 29, according to one or more embodiments. As best seen in FIGS. 29 and 31, the differential assembly 2806 includes opposing first and second ends 2902a and 2902b. The differential input 2808 is located at the first end 2902a and is matable with a corresponding one of the drive outputs of the instrument driver 1614 (FIG. 16). As best seen in FIG. 29, the differential input 2808 may define a receptacle that defines splines, protrusions, or other structural features designed to mate with corresponding features of the drive outputs of the instrument driver 1614, or vice versa. The differential output 2810 is located at the second end 2902b and is matable with a corresponding one of the drive inputs 1702 (FIGS. 17A-17B) of the actuation system 1700 (FIGS. 17A-17B). As best seen in FIG. 29, the differential output 2810 may define splines, protrusions, or other structural features designed to mate with a corresponding receptacle of the drive input 1702, or vice versa.

The differential assembly 2806 also includes the differential 2826, which allows the differential output 2810 to be driven by both the differential input 2808 and the insertion assembly 2812 (FIG. 28). As best seen in FIGS. 30-31, the differential 2826 includes the sun gear 2828, the plurality of planetary gears 2830 surrounding the sun gear 2828, and the planet carrier 2832 coupled to each planetary gear 2830. In some embodiments, as shown in FIG. 31, at least a portion of the planet carrier 2832 may form part of the differential output 2810. The differential 2826 further includes the differential ring gear 2834, which surrounds and intermeshes with the planetary gears 2830 and is driven by the insertion transmission gear 2838 or 2842 (FIG. 28).

FIG. 32 is a schematic, cross-sectional side view of another example of the differential assembly 2806 of FIG. 28, according to one or more additional embodiments. In the illustrated embodiment, the differential assembly 2812 includes a drive shaft 3202 having opposing first and second ends 3204a and 3204b. The differential input 2808 is located at the first end 3404a and the differential output 2810 is located at the second end 3404b. The sun gear 2828 either forms part of the drive shaft 3202 or is fixed thereto such that rotation of the drive shaft 3202 correspondingly rotates the sun gear 2828. Moreover, the planet carrier 2832 may be coupled to the drive shaft 3202 such that rotation of the drive shaft 3202 correspondingly rotates the planet carrier 2832. Accordingly, the illustrated embodiment provides a non-coupled direct drive motion through the center of the differential assembly 2806. Those skilled in the art will readily appreciate that this may be done as an opportunity to drive a non-coupled actuator on a different instrument architecture.

FIG. 33 is a cross-sectional side view of one example of the insertion assembly 2812 of FIG. 28, according to one or more embodiments. As illustrated, the insertion assembly 2812 has opposing first and second ends 3202a and 3202b. The insertion input 2814 is provided at the first end 3202a, and the insertion output 2824 is provided at the second end 3202b. In some embodiments, as illustrated, the insertion input 2814 may define a receptacle that defines splines, protrusions, or other structural features designed to mate with corresponding features of a corresponding drive output of the instrument driver 1614 (FIG. 16), or vice versa. Similarly, in some embodiments, the insertion output 2824 define splines, protrusions, or other structural features designed to mate with a corresponding receptacle of a drive input 1702 (FIG. 17), or vice versa.

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The insertion assembly 2812 further includes the sun gear 2816, the planetary gears 2818 surrounding and intermeshed with the sun gear 2816, and the planet carrier 2820 coupled to each planetary gear 2818. In at least one embodiment, the planet carrier 2820 may extend into and otherwise form part of the insertion output 2824. The insertion assembly 2812 also includes the insertion input gear 2840 operable to drive the insertion transmission gear 2838 (FIG. 28).

FIG. 34 is a schematic, cross-sectional side view of another example of the insertion assembly 2812 of FIG. 28, according to one or more additional embodiments. In the illustrated embodiment, the sun gear 2816 may include a distally extending extension 3402 that is keyed (fixed) to the housing 2804 of the decoupler 2802 (FIG. 28) at or near the insertion input 2814. Moreover, the insertion drive for the insertion assembly 2812 may be linked directly to the bottom of the planet carrier 2820. More specifically, the planet carrier 2820 may be mounted to a drive shaft 3404 that extends along the axial length of the insertion assembly 2812. The insertion input 2814 may be provided at one end of the drive shaft 3404, and the insertion output 2824 may be provided at the opposing end of the drive shaft 3404. Accordingly, instead of the insertion axis just having a gear on it, in the illustrated embodiment, the insertion axis directly drives the insertion output 2824 via rotation of the drive shaft 3404 and the planet carrier 2820; i.e., insertion directly drives the output.

Actuation of the insertion input 2814 simultaneously drives the planet carrier 2820, which drives the planetary gears 2818 against the insertion input gear 2840. This ensures that the rotation of the insertion axis is scaled to the correct output that occurs at each individual axis of the differential assemblies 2806 (FIGS. 28-31), without having to pair uneven gear ratios. Moreover, each rotation of the planet carrier 2820 drives the shaft rack gear 1710 (FIG. 17A) on the shaft 1602 (FIGS. 16 and 17A), while simultaneously rotating the insertion input gear 2840. As the insertion input gear 2840 rotates, it drives rotation of the insertion transmission gear 2838 (FIG. 28) and/or the second insertion transmission gear 2842 (FIG. 28). This motion then acts on the differential 2826 (FIG. 28) of each differential assembly 2806 (FIG. 28), which rotation is summed with the sun drive motion of each differential assembly 2806 to obtain a total output planet carrier motion at the individual differential assemblies 2806.

FIGS. 35A and 35B are isometric and cross-sectional side views of another example of the differential assembly 2806 of FIG. 28, according to one or more additional embodiments. Similar to the differential assembly 2806 shown in FIGS. 29-31, the differential assembly 2806 in FIGS. 35A-35B includes opposing first and second ends 2902a and 2902b, where the differential input 2808 is located at the first end 2902a and the differential output 2810 is located at the second end 2902b. Moreover, the differential assembly 2806 further includes the differential ring gear 2834, which is driven by the insertion transmission gear 2838 or 2842 (FIG. 28).

Unlike the differential assembly 2806 of FIGS. 29-31, however, where the differential 2826 was provided in the form of a planetary gearbox, the differential assembly 2806 of FIGS. 35A-35B includes a differential 3502 in the form of an offset axle gearbox. At least one advantage to the offset axle gearbox is that it can achieve a more ideal gear ratio (closer to 1:1) while limiting the height of the overall differential mechanism.

As illustrated, the differential 3502 includes an input shaft 3504a and an output shaft 3504b, where the input and output

shafts **3504a,b** are axially aligned and extend between the first and second ends **2902a,b**. The differential input **2808** either forms part of or is coupled to the input shaft **3504a**, and the differential output **2810** either forms part of or is coupled to the output shaft **3504b**.

The differential **3502** further includes a housing **3506**, and an offset axle **3508** (FIG. 35B) extends between the housing **3506** and the differential ring gear **2834**. The offset axle **3508** extends parallel to but eccentric from the input and output shafts **3504a,b**. A first offset gear **3510a** is coupled to the offset axle **3508** and arranged to engage an input shaft gear **3512a** mounted to the input shaft **3504a**, and a second offset gear **3510b** is coupled to the offset axle **3508** and arranged to engage an output shaft gear **3512b** mounted to the output shaft **3504a**.

The differential **3502** allows the differential output **2810** to be driven by both the differential input **2808** and the insertion assembly **2812** (FIG. 28). More specifically, rotating the differential input **2808** will cause the input shaft **3504a** and the input shaft gear **3512a** to rotate, which will drive against the first offset gear **3510a** and thereby cause the offset axle **3508** and the second offset gear **3510b** to drive against the output shaft gear **3512b**, thus resulting in rotation of the output shaft **3504b** and the differential output **2810**. In contrast, rotating the differential ring gear **2834**, as driven by the insertion transmission gear **2838** or **2842** (FIG. 28), will cause the offset axle **3508** and the housing **3506** to rotate about the input and output shafts **3504a,b**, and thus driving the second offset gear **3510b** against the output shaft gear **3512b** and resulting in rotation of the output shaft **3504b** and the differential output **2810**. In such a scenario, the motor driving the differential input **2808** and the input shaft **3504a** may be static or operating.

The differential **3502** shown in FIGS. 35A-35B is closer to 1.2:1 ratio. Utilizing a true 1:1 ratio through this gearbox would have the first and second offset gears **3510a,b** of equal size, which for this configuration of differential **3502** would create a unique singularity wherein the motion of the differential **3502** would translate only to rotation of the offset axle **3508**, and not the intended motion of the output shaft **3504b** and the differential output **2810**.

FIGS. 36A and 36B are isometric and cross-sectional side views, respectively, of another example of the insertion assembly **2812** of FIG. 28, according to one or more additional embodiments. The insertion assembly **2812** of FIGS. 36A-36B may be similar in some respects to the insertion assembly **2812** of FIGS. 33-34, and thus may be best understood with reference thereto, where like numerals will refer to similar components. As illustrated, the insertion assembly **2812** includes a drive shaft **3602** that has opposing first and second ends **3604a** and **3604b**. The insertion input **2814** is provided at the first end **3604a**, and the insertion output **2824** is provided at the second end **3604b**. Moreover, the insertion assembly also includes the insertion input gear **2840**, which is arranged to drive the insertion transmission gear **2838** (FIG. 28) to enable simultaneous driving of each differential ring gear **2834** (FIG. 28) of each differential assembly **2806** (FIG. 28), as discussed above.

Unlike the insertion assembly **2812** of FIGS. 23-34, however, which included a planetary gearbox, the insertion assembly **2812** of FIGS. 36A-36B includes an offset axle gearbox. More specifically, the insertion assembly **2812** includes a housing **3606**, and an offset axle **3608** (FIG. 36B) extends between the housing **3606** and the insertion input gear **2840**. The offset axle **3608** extends parallel to but eccentric from the drive shaft **3602**. A first offset gear **3610a** is coupled to the offset axle **3608** and arranged to engage a

stationary gear **3612a** forming part of a stationary structure **3614** through which the drive shaft **3602** extends. The stationary structure **3614** may be keyed into the housing **3606**, such as through a hex head mating engagement or the like. A second offset gear **3610b** is coupled to the offset axle **3608** and arranged to engage a drive shaft gear **3612b** mounted to the drive shaft **3602**.

In example operation, rotating the insertion input **2814** will cause the drive shaft **3602** to rotate, which will cause insertion of the tool. Rotating the drive shaft **3602**, however, will also cause the offset axle **3608** and the housing **3606** to rotate about the stationary structure **3614**, which drives the insertion input gear **2840** in rotation. As discussed herein, rotating the insertion input gear **2840** drives the insertion transmission gear **2838** (FIG. 28) and thereby enables simultaneous driving of each differential ring gear **2834** (FIG. 28) of each differential assembly **2806** (FIG. 28). Accordingly, the insertion assembly **2812** of FIGS. 36A-36B ensures that all rotation of the insertion axis output (i.e., the drive shaft **3602**) is simultaneously transferred to the differential assemblies **2806** at the inverted rate of the other gearboxes. Thus, when the motion is converted from a given differential assembly **2806** to its output **2810**, the output rotation is equal to that of insertion. In some embodiments, to accomplish this, the same size offset gears **3510a,b** (FIGS. 35A-35B) and **3610A,B** may be used.

FIG. 37 is an isometric view of another example surgical tool **3700** that may incorporate some or all of the principles of the present disclosure. The surgical tool **3700** may be similar in some respects to the surgical tool **1600** of FIG. 16 and therefore may be best understood with reference thereto, where like numerals correspond to similar components or structures not described again in detail. Similar to the surgical tool **1600**, for example, the surgical tool **3700** includes the elongated shaft **1602**, the end effector **1604** arranged at the distal end of the shaft **1602**, and the articulating wrist **1606** that interposes and couples the end effector **1604** to the distal end of the shaft **1602**. Moreover, a tailpiece **3702** may be arranged at a proximal end of the shaft **1602**.

The surgical tool **3700** may include a drive housing or "handle" **3704** similar in some respects to the handle **1612** of FIG. 16. The shaft **1602** extends longitudinally through the handle **3704**, and the handle **3704** houses an actuation system designed to facilitate articulation of the wrist **1606**, actuation (operation) of the end effector **1604** (e.g., clamping, firing, rotation, articulation, energy delivery, etc.), and movement of the shaft **1602** relative to (through) the handle **3704** and along the longitudinal axis  $A_1$ . The actuation system may be the same as or similar to the actuation system **1700** of FIGS. 17A-17B, but could alternatively be another type of actuation system.

The handle **3704** may be operatively coupled to an instrument driver **3706** (shown in dashed lines) of a robotic surgical system. The instrument driver **3706** may be the same as or similar to the instrument driver **1614** of FIG. 16, and therefore may be best understood with reference thereto. Similar to the instrument driver **1614**, for example, the instrument driver **3706** may be mounted to or otherwise positioned at the end of a robotic arm (not shown) and is designed to provide the motive forces required to operate the surgical tool **3700**.

The handle **3704** includes a plurality of rotatable drive inputs (not visible) that can be driven by a corresponding plurality of drive outputs (not visible) of the instrument driver **3706**. Each drive input is actuatable to independently drive (actuate) various portions of the actuation system

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housed within the handle **3704** and thereby operate the surgical tool **3700**. Movement (rotation) of a given drive output correspondingly moves (rotates) the associated drive input and thereby operates the surgical tool **3700**. Actuation of the drive inputs operates the actuation system, which can cause the end effector **1604** to articulate and/or actuate (operate) and the shaft **1602** to axially move (translate) relative to the handle **3704**.

In the illustrated embodiment, a decoupler subassembly or “decoupler” **3708** is arranged between and otherwise interposes the handle **3704** and the instrument driver **3706**. The decoupler **3708** may be the same as or similar in some respects to the decouplers **1616**, **2802** of FIGS. **16** and **28**, respectively. Similar to the decouplers **1616**, **2802**, for example, the decoupler **3708** transfers torque from the drive outputs of the instrument driver **3706** to the drive inputs of the handle **3704**. Once the drive outputs are operatively and indirectly coupled to corresponding drive inputs via the decoupler **3708**, the drive inputs will share axes of rotation with the corresponding drive outputs to allow the transfer of rotational torque from the drive outputs to the corresponding drive inputs, thus being able to operate the handle **3704**. The decoupler **3708** may also be advantageous in transferring insertion motion (e.g., movement of the shaft **1602**) to all the drive inputs, thus allowing one robot motor to control insertion of the surgical tool **3700**, while allowing the other motors of the instrument driver **3706** to drive the sliding rack gears independent of insertion.

FIG. **38** depicts separated isometric end views of the instrument driver **3706** and the surgical tool **3700** of FIG. **37**. With the jaws **1608**, **1610** closed, the shaft **1602** and the end effector **1604** can penetrate the instrument driver **3706** by extending through a central aperture **3802** defined longitudinally through the instrument driver **3706** between first and second ends **3804a, b**. A drive interface **3806** is provided at the first end **3804a** of the instrument driver **3706** and is matable with a driven interface **3808** provided on the distal end of the handle **3704** and, more particularly, the distal end (bottom) of the decoupler **3708**. The drive and driven interfaces **3806**, **3808** may be configured to mechanically, magnetically, and/or electrically couple the handle **3704** (e.g., the decoupler **3708**) to the instrument driver **3706**.

The instrument driver **3706** includes a plurality of drive outputs **3810** that extend through the drive interface **3806**. In embodiments where the surgical tool **3700** includes the decoupler **3708**, the drive outputs **3810** are configured to mate with corresponding differential inputs **3812** provided at the distal end of the decoupler **3708**. At least one of the differential inputs **3812** may be an insertion input **3814** operable to facilitate axial translation of the shaft **1602**. The number of drive outputs **3810** will generally be the same as the number of differential and insertion inputs **3812**, **3814**, but it is contemplated herein that the instrument driver **3706** can have additional drive outputs, without departing from the scope of the disclosure.

The drive outputs **3810** may define splines, protrusions, or other mechanical features designed to mate with corresponding receptacles of the differential and insertion inputs **3812**, **3814**, or vice versa. One of the drive outputs **3810** may be configured to mate with the insertion input **3814**, this drive output **3810** is referred to herein as a “shaft drive output.” Once properly mated, the differential and insertion inputs **3812**, **3814** will share axes of rotation with the corresponding drive outputs **3810** to allow the transfer of rotational torque from the drive outputs **3810** to the corresponding differential inputs **3812**, **3814**. In some embodiments, each drive output **3810** may be spring loaded and otherwise

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biased to spring outwards away from the drive interface **3806**. Each drive output **3810** may be capable of partially or fully retracting into the drive interface **3806**.

FIGS. **39A** and **39B** are isometric open (disengaged) and closed (engaged) views, respectively, of another example actuation system **3900**, according to one or more embodiments. The actuation system **3900**, also referred to as an axle redirect assembly, may be similar in some respects to the actuation system **1700** of FIGS. **17A-17B**, and may thus be best understood with reference thereto. The actuation **3900** may be housed within the handle **3704** and operable to actuate (operate) the end effector **1604** (FIG. **16**), articulate the wrist **1606** (FIG. **16**), and axially move (translate) the shaft **1602** relative to (i.e., through) the handle **3704**. Several parts and structural elements of the handle **3704**, such as the outer housing, are omitted from FIGS. **39A-39B** to enable ease of viewing of the actuation system **3900** for purposes of discussion.

As illustrated, the actuation system **3900** may include a first or “upper” mounting assembly **3901a** and a second or “bottom” mounting assembly **3901b**. Briefly, the upper mounting assembly **3901a** is a translating ring that moves to allow and enable the assembly of the instrument shaft **1602** to the actuation system **3900** and the decoupler **3708**, and the bottom mounting assembly **3901b** is the base of the axle redirect assembly or actuation system **3900**. The mounting assemblies **3901a, b** may be vertically offset from each other and concentrically arranged about the shaft **1602**. A plurality of spur linkage subassemblies **3902** may be pivotably mounted to the mounting assemblies **3901a, b** and configured to pivot between a first or “disengaged” position, as shown in FIG. **39A**, and a second or “engaged” position, as shown in FIG. **39B**.

Each spur linkage subassembly **3902** includes a corresponding drive gear **3904** arranged to drive against an adjacent sliding rack gear **3906** (two visible in FIGS. **39A-39B**). Each sliding rack gear **3906** is movably nested within a corresponding longitudinal channel **3908** defined along all or a portion of the shaft **1602**. In the illustrated embodiment, the drive gears **3904** are depicted as spur gears with teeth matable with corresponding teeth defined on the opposing sliding rack gear **3906**. Accordingly, the drive gears **3904** may alternatively be referred to herein as “spur” gears, and operation of the spur gears **3904** and corresponding rack gears **3906** may conform to known rack-and-pinion operation. Driving a given sliding rack gear **3906** will urge the sliding rack gear **3906** to move (slide) within its corresponding longitudinal channel **3908**, and moving the sliding rack gear **3906** within the channel **3908** may actuate (operate) the end effector **1604** (FIG. **16**) and/or articulate the wrist **1606** (FIG. **16**).

One of the drive gears is indicated in FIGS. **39A-39B** as a shaft drive gear **3904a**, which may be arranged to drive against an adjacent shaft rack gear **3910**. The shaft rack gear **3910** may form part of and may otherwise be defined along all or a portion of the outer surface of the shaft **1602**. Driving against the shaft rack gear **3910** will cause the shaft **1602** to move (translate) axially relative to (i.e., through) the handle **3704** (FIG. **16**).

Referring briefly to FIG. **40**, illustrated is an isometric view of an example spur linkage subassembly **3902**, according to one or more embodiments. As illustrated, the spur linkage subassembly **3902** includes a rotatable drive input **4002**, which, in some embodiments, can be driven by one of the drive outputs **3810** (FIG. **38**) of the instrument driver **3706** (FIGS. **37-38**). In embodiments that include the decoupler **1616** (FIG. **37**), however, a transmission differential or

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differential assembly provided by the decoupler 1616 will axially interpose the drive input 4002 and a corresponding drive output 3810, as generally described herein. In such embodiments, any rotational torque provided by a given drive output 3810 will be transferred to the corresponding drive input 4002 via a corresponding transmission differential of the decoupler 1616.

A gear train including one or more intermediate gears may interpose the drive input 4002 and the corresponding drive gear 3904, such that rotating the drive input 4002 will cause the corresponding drive gear 3904 to rotate. Those skilled in the art will readily appreciate that this gear train can assume a variety of configurations. In the illustrated embodiment, for example, the gear train includes mating bevel gears, shown as a bevel drive gear 4004a arranged to drive a bevel driven gear 4004b. The bevel drive gear 4004a may be coupled to or otherwise form part of the drive input 4002 such that rotation of the drive input 4002 correspondingly rotates the bevel drive gear 4004a. In some embodiments, the bevel driven gear 4004b may be mounted to an axle 4006 configured to be rotatably mounted to the lower mounting assembly 3901b (FIGS. 39A-39B). A spur gear 4008 may also be mounted to the axle 4006 and axially offset from the bevel driven gear 4004b. In at least one embodiment, however, the spur gear 4008 may be coupled to or otherwise form part of the bevel driven gear 4004b. In either scenario, rotation of the bevel driven gear 4004b will cause the spur gear 4008 to correspondingly rotate, and the spur gear 4008 may be arranged to drive the drive gear 3904.

The intermeshed bevel gears 4004a,b facilitate the directional change required in the handle 3704 (FIGS. 39A-39B). More specifically, the bevel gears 4004a,b redirect motor rotation of the drive input 4002 to be perpendicular to the shaft 1602 (FIGS. 39A-39B). Through the depicted gear train arrangement, rotation of the drive input 4002 will correspondingly move the rack gears 3906 (FIGS. 39A-39B). In other embodiments, other known gearing mechanisms may be utilized or combined in any number of configurations and dimensioned for optimal torque and/or speed outputs. In at least one embodiment, for example, the gear train may incorporate worm gears or a combination of bevel and spur gears. Moreover, in at least one embodiment, the gear train may be omitted and rotating the drive input 4002 may directly drive an adjacent rack gear 3906 without departing from the scope of the disclosure.

While the spur linkage assembly 3902 shown in FIG. 40 is described with reference to interfacing with a sliding rack gear 3906 (FIGS. 39A-39B), the foregoing description is equally applicable to a spur linkage assembly 3902 configured to interface with the shaft rack gear 3910. In such embodiments, the drive input 4002 may be referred to as a “shaft drive input,” which may be configured to cause the shaft drive gear 3904a (FIGS. 39A-39B) to rotate (operate) and drive against the shaft rack gear 3910. Driving against the shaft rack gear 3910, as mentioned above, will cause the shaft 1602 (FIGS. 39A-39B) to move (translate) axially relative to (i.e., through) the handle 3704 (FIGS. 39A-39B).

Still referring to FIG. 40, the spur linkage subassembly 3902 may further include a first or “upper” linkage 4010a and a second or “lower” linkage 4010b. As illustrated, one end of the upper linkage 4010a may provide or otherwise define a pair of pins 4012 configured to help pivotably couple the upper linkage 4010a to the upper mounting assembly 3901a (FIGS. 39A-39B). The pins 4012, however, could be replaced with any other type of pivotable coupling engagement mechanism or structure, without departing from the scope of the disclosure. The opposing end of the upper

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linkage 4010a may be pivotably coupled to one end of the lower linkage 4010b, and the opposing end of the lower linkage 4010b may be rotatably mounted to the axle 4006, which, as mentioned above, can be rotatably mounted to the lower mounting assembly 3901b (FIGS. 39A-39B).

Referring again to FIGS. 39A-39B, with continued reference to FIG. 40, in some embodiments, the spur linkage subassemblies 3902 may be spring biased to the disengaged position, shown in FIG. 39A. In the disengaged position, the drive gears 3904 are disengaged from the sliding rack gears 3906 and the shaft drive gear 3904a is disengaged from the shaft rack gear 3910. The spur linkage subassemblies 3902 are transitioned to the engaged position by collapsing the axial distance between the upper and lower mounting assemblies 3901a,b. As the upper mounting assembly 3901a is lowered toward the lower mounting assembly 3901b, the spur linkage subassemblies 3902 will pivot in tandem at the pivotably coupled linkages 4010a,b toward the engaged (closed) position, as shown in FIG. 39B. In the engaged position, each drive gear 3904, 3904a engages and mates with the adjacent sliding rack gear 3906 or shaft rack gear 3910 simultaneously. Accordingly, the upper linkage 4010a may be configured to move linearly up and down to move the drive gears 3904, 3904a away from and toward the axis of the shaft 1602 as the actuation system 3900 transitions between the open and closed positions.

Referring now to FIG. 41, in some embodiments, the actuation system 3900 may further include a clocking wheel linkage subassembly 4102 configured to help properly align (e.g., angularly, rotationally, etc.) the actuation system 3900 with the shaft 1602. FIG. 42 is an enlarged, isometric view of one example of the clocking wheel linkage subassembly 4102, according to one or more embodiments. As illustrated, the clocking wheel linkage subassembly 4102 may include a first or “upper” linkage 4202a and a second or “lower” linkage 4202b. As illustrated, one end of the upper linkage 4202a may provide or otherwise define a pair of pins 4204 configured to help pivotably couple the upper linkage 4202a to the upper mounting assembly 3901a (FIG. 41). Similarly, one end of the lower linkage 4202b may also provide or otherwise define a pair of pins 4206 configured to help pivotably couple the lower linkage 4202b to the lower mounting assembly 3901b. The pins 4204, 4206, however, could be replaced with any other type of pivotable coupling engagement mechanism or structure, without departing from the scope of the disclosure. The opposing ends of the linkages 4202a,b may be pivotably attached to each other at a pivot axis 4208.

The clocking wheel linkage subassembly 4102 may also include a clocking wheel 4210 rotatably mounted at the pivot axis 4208. In some embodiments, one or more thrust bearings and/or washers 4212 (e.g., Belleville washers) may help maintain the pivotably mounted linkages 4202a,b and clocking wheel 4210 axially tight and rotationally free at the pivot axis 4208.

Referring again to FIG. 41, with continued reference to FIG. 42, similar to the spur linkage subassemblies 3902, the clocking wheel linkage subassembly 4102 may be pivotably mounted to the mounting assemblies 3901a,b and configured to pivot between a first or “disengaged” position, as shown in FIG. 39A, and a second or “engaged” position, as shown in FIGS. 39B and 41. Upon transitioning to the engaged position, the clocking wheel 4210 may be configured to be received within and otherwise mate with a longitudinal groove 4104 defined along all or a portion of the shaft 1602. Receiving the clocking wheel 4210 within the groove 4104 will help properly align all the spur linkage

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subassemblies **3902** rotationally (angularly) with the proper sliding rack gears **3906** or shaft rack gear **3910** (FIGS. 39A-39B).

FIG. 43 is an enlarged isometric view of the tailpiece **3702**, according to one or more embodiments. The tailpiece **3702** may be similar in respects to the tailpiece **1620** of FIG. 16, and may thus be best understood with reference thereto. For example, similar to the tailpiece **1620**, the tailpiece **3702** may be arranged at the proximal end of the shaft **1602**. Moreover, the tailpiece **3702** may comprise a mechanical device that provides a means for manually controlling the end effector **1604** (FIG. 37) and/or the wrist **1606** (FIG. 37). The tailpiece **3702** may also be used to maintain tension in the surgical tool **3700** (FIG. 37) when disconnected from the instrument driver **3706** (FIG. 37). As described herein, however, the tailpiece **3702** can be provided in various forms and still perform essentially the same function, without departing from the scope of the present disclosure.

In the illustrated embodiment, the tailpiece **3702** includes a housing **4302** that may be attached to the proximal end of the shaft **1602**. The housing **4302** may include first and second housing portions **4304a** and **4304b** that are matable and coupled together using one or more mechanical fasteners **4306** or the like. The tailpiece **3702** may further include a manual actuation device, which, in this embodiment, comprises a slider **4308** that extends out of the housing **4302** and provides a means for manually actuating the tailpiece **3702**. More specifically, the slider **4308** extends past an outer circumferential periphery of the housing **4302**, and as the slider **4308** is manually moved (rotated) about the outer circumference, the end effector **1604** (FIG. 37) may be actuated or the wrist **1606** (FIG. 37) may be articulated.

FIG. 44 is an isometric view of the interior of the tailpiece **3702**, according to one or more embodiments. The second or "upper" housing portion **4304b** is omitted to enable viewing of the internal components and mechanics of the tailpiece **3702**. As illustrated, the tailpiece **3702** includes first and second pulley arms **4402a** and **4402b** that can be secured to the slider **4308** such that manually rotating the slider **4308** about the axis of the shaft **1602** correspondingly rotates the pulley arms **4402a,b** in the same angular direction. First and second proximal pulleys **4404a** and **4404b** may also be rotatably mounted to the first and second pulley arms **4402a,b** and move with the pulley arms **4402a,b** within the housing **4302** as the slider **4308** is manually moved. The proximal pulleys **4404a,b** may be similar to the proximal pulley **1806b** discussed above with reference to FIGS. 18A-18C, and thus have corresponding proximal cables **4406a** and **4406b** extending (wrapped) thereabout, where the proximal cables **4406a,b** are similar to the proximal cable **1804b** of FIGS. 18A-18C.

As illustrated, the proximal cables **4406a,b** may be routed to and from the shaft **1602** using one or more rerouting pulleys **4408**. Some of the rerouting pulleys **4408** may be located within the shaft **1602** and may exhibit horizontal axes (e.g., perpendicular to shaft **1602**), while other rerouting pulleys **4408** may be located along the outer diameter of the housing **4302** and exhibit vertical axes (e.g., parallel to the shaft **1602**). Each proximal cable **4406a,b** is distributed through corresponding rerouting pulleys **4408** and wrapped around the corresponding proximal pulley **4404a,b**, respectively.

When the slider **4308** is manually moved in either angular direction relative to the outer circumference of the housing **4302**, the pulley arms **4402a,b** move in the same angular direction, which causes one of the proximal pulleys **4404a,b** to draw in its proximal cable **4406a,b** from the shaft **1602**

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while simultaneously causing the other proximal pulley **4404a,b** to pay out its proximal cable **4406a,b** to the shaft **1602**. More specifically, as the slider **4308** moves counter-clockwise, for example, as indicated by the arrow C, the first proximal pulley **4404a** moves away from its corresponding rerouting pulleys **4408** and thereby draws a length of the first proximal cable **4406a** into the tailpiece **3702** from the shaft **1602**. At the same time, the second proximal pulley **4404b** simultaneously moves toward its corresponding rerouting pulleys **4408** and thereby pays out a length of the second proximal cable **4406b** into the shaft **1602**. Consequently, the net cable length of the cables **4406a,b** remains the same because the two pulleys **4404a,b** have the same displacement, but the change in length of each pair changes, which causes movement (actuation) of the jaws **1608**, **1610** (FIG. 37). Moving the slider **4308** clockwise, as indicated by arrow D, will act on the first and second proximal cables **4406a,b** in the opposite manner, and actuate the jaws **1608**, **1610** in the opposite direction. In alternative embodiments, this cable configuration may alternatively allow pitch and yaw articulations at the wrist **1606** (FIG. 37) to create motion where each cable loop rolls around its respective pulley system.

The tailpiece **3702** may also be useful in facilitating assembly pretension in the cable system. More specifically, during assembly of the tool, the pulley arms **4402a,b** may be able to move relative to the slider **4308**. When the full system is assembled with cable routing, the jaws **1608**, **1610** (FIG. 37) are held still and the pulley arms **4402a,b** are rotated (drawn) towards each other and the center of the slider **4308**. This rotation may be done with a set torque load to create a desired pretension in the system, and once the predetermined tension is achieved, one or more mechanical fasteners **4410** may be used to rigidly secure the pulley arms **4402a,b** to the slider **4308**. Consequently, by setting the relative position of the two pulley arms **4402a,b** to the slider **4308**, the pretension of the cable system may be set and held.

FIG. 45 is an isometric view of another example tailpiece **4502**, according to one or more additional embodiments. The tailpiece **4502** may be similar in some respects to the tailpiece **1620** of FIG. 16 or the tailpiece **3702** of FIG. 37, and thus may comprise a mechanical device that provides a means for manually controlling the end effector **1604** (FIG. 37) and/or the wrist **1606** (FIG. 37). The tailpiece **4502** may also be used to maintain tension in the surgical tool **3700** (FIG. 37) when disconnected from the instrument driver **3706** (FIG. 37).

As illustrated, the tailpiece **4502** includes a housing **4504** that may be attached to the proximal end of the shaft **1602**. The tailpiece **4502** may further include a manual actuation device, which, in this embodiment, comprises a slider **4506** that extends out of the housing **4504** and provides a means for manually actuating the tailpiece **4502**. In some embodiments, as illustrated, the slider **4506** may be manually moved side to side relative to the housing **4504** to actuate the tailpiece **4502**. In other embodiments, however, the slider **4506** may be configured to slide (move) in other directions, without departing from the scope of the disclosure. As the slider **4506** is manually moved, the end effector **1604** (FIG. 37) may be actuated or the wrist **1606** (FIG. 37) may be articulated.

FIGS. 46A and 46B are isometric views of the interior of the tailpiece **4502**, according to one or more embodiments. A top portion of the housing **4504** is omitted in FIGS. 46A-46B to enable viewing of the internal components and mechanisms of the tailpiece **4502**. As illustrated, the tailpiece **4502** includes first and second camming plates **4602a**

and **4602b** that are vertically offset from each other within the housing **4504**. In FIG. **46B**, only the second or “lower” camming plate **4602b** is shown to facilitate better viewing of various component parts of the tailpiece **4502**. The camming plates **4602a,b** may be secured together and mounted within the housing **4504** such that they are able to move (slide) in tandem relative to housing **4504**. As illustrated, the first camming plate **4602a** may define or otherwise provide the slider **4506**.

The tailpiece **4502** may further include first and second proximal pulleys **4604a** and **4604b** rotatably mounted between the first and second camming plates **4602a,b**.

Each pulley **4604a,b** may be rotatably mounted to a corresponding pin **4606** extending between the camming plates **4602a,b**, and the pins **4606** may allow the pulleys **4604a,b** to move laterally within the housing **4504** as the slider **4506** is manually moved. More specifically, each camming plate **4602a,b** defines and otherwise provides distal and proximal slots **4608a** and **4608b** configured to receive opposing ends of each pin **4606**. The pin **4606** of the first pulley **4604a** may be arranged to extend between the camming plates **4602a,b** and into the distal slots **4608a**, and the pin **4606** of the second pulley **4604b** may be arranged to extend between the camming plates **4602a,b** and into the proximal slots **4608b**. As illustrated, the distal and proximal slots **4608a,b** of each camming plate **4602a,b** are angled and extend in the same direction such that the distal slots **4608a** align and the proximal slots **4608b** align. Moreover, the slots **4608a,b** converge toward or away from each other in a lateral direction, which allows the pulleys **4604a,b** to progressively converge or diverge as the pins **4606** traverse the slots **4608a,b**.

The proximal pulleys **4604a,b** may be similar to the proximal pulley **1806b** discussed above with reference to FIGS. **18A-18C**, and may thus have a corresponding proximal cable **4610a** and **4610b** extending (wrapped) thereabout, where the proximal cables **4610a,b** are similar to the proximal cable **1804b** of FIGS. **18A-18C**. As the slider **4506** is moved laterally, the camming plates **4602a,b** correspondingly move laterally and force the pulleys **4604a,b** (i.e., the pins **4606**) to traverse the slots **4608a,b** defined in the camming plates **4602a,b**. As they traverse the slots **4608a,b**, the pulleys **4604a,b** converge toward each other or diverge away from each other, depending on the direction of the slider **4506**, which correspondingly changes the length of the cables **4610a,b** within the system. For example, moving the slider **4506** to the left in FIGS. **46A-46B** will tend make the pulleys converge toward each other, and moving the slider **4506** to the right in FIGS. **46A-46B** will tend make the pulleys diverge away from each other. The net cable length of the cables **4610a,b**, however, remains the same because the two pulleys **4604a,b** have the same displacement, but the change in length of each pair changes, which causes movement (actuation) of the jaws **1608**, **1610** (FIG. **37**). In alternative embodiments, this cable configuration may be configured to allow pitch and yaw articulations of the wrist **1606** (FIG. **37**) to create motion where each cable loop rolls around its respective pulley system.

As best seen in FIG. **46B**, the pulleys **4604a,b** may be configured as spools rotatably mounted to their corresponding pins **4606**, and each spool has two parts that are matable at a corresponding engagement feature **4612**. When the two parts are properly mated, the engagement feature **4612** prevents the two parts of each pulley **4604a,b** from rotating relative to the other. When separated, however, the two parts may be able to rotate relative to one another. In the illustrated embodiment, the engagement features **4612** comprise

castellated features, but could alternatively comprise other types of structural features that prevent relative rotation of the opposing parts when engaged, but allow relative rotation when separated. Assembly pretension in the cable system can be achieved by rotating the two parts of each pulley **4604a,b** in opposite directions in order to apply a set torque load on the associated cables **4610a,b**, and thus creating the desired pretension in the system. Once the desired (predetermined) tension is reached, the two parts may be mated once again at the engagement feature **4612** to hold the pretension of the system.

FIG. **47** is an enlarged isometric view of another example tailpiece **4702**, according to one or more additional embodiments. The tailpiece **4702** may be similar in respects to the tailpiece **3702** of FIGS. **43-44**, and may thus be best understood with reference thereto. For example, similar to the tailpiece **3702**, the tailpiece **4702** may be arranged at the proximal end of the shaft **1602** and may comprise a mechanical device that provides a means for manually controlling the end effector **1604** (FIG. **37**) and/or the wrist **1606** (FIG. **37**). The tailpiece **4702** may also be used to maintain tension in the surgical tool **3700** (FIG. **37**) when disconnected from the instrument driver **3706** (FIG. **37**).

The tailpiece **4702** may include a housing **4704** that may be attached to the proximal end of the shaft **1602**. In FIG. **47**, an upper portion of the housing **4704** is omitted to enable viewing of the internal components and mechanics of the tailpiece **4702**. The tailpiece **4702** may further include a manual actuation device, which, in this embodiment, comprises a slider **4706** that extends out of the housing **4704** and provides a means for manually actuating the tailpiece **4702**. Similar to the slider **4308** of FIGS. **43-44**, the slider **4706** extends past an outer circumferential periphery of the housing **4704**. As the slider **4706** is manually moved (rotated) about a portion of the outer circumference, the end effector **1604** (FIG. **37**) may be actuated or the wrist **1606** (FIG. **37**) may be articulated.

Unlike the tailpiece **3702** of FIGS. **43-44**, however, the tailpiece **4702** includes an internal linear slider **4708** movably mounted within the housing **4704**. A first end **4710a** of the slider **4706** extends out of the housing **4704** to be manually manipulated by the user, but the opposing second end **4710b** of the slider **4706** is configured to interact with the linear slider **4708** such that manually sliding (rotating) the slider **4706** causes the linear slider **4708** to move laterally within the housing **4704**. In some embodiments, for example, the linear slider **4708** may define or otherwise provide a rack gear **4712**, and the second end **4710b** of the slider **4706** may define or otherwise provide a pinion gear **4714** (mostly occluded) arranged to engage and mate with the rack gear **4712**. As the slider **4706** is moved (rotated), the pinion gear **4714** drives against the rack gear **4712** and thereby causes the linear slider **4708** to move laterally within the housing **4704**, depending on the driving direction of the pinion gear **4714**.

The tailpiece **4702** may further include first and second proximal pulleys **4714a** and **4714b** rotatably mounted to opposing ends of the linear slider **4708**. The proximal pulleys **4714a,b** may be similar to the proximal pulley **1806b** discussed above with reference to FIGS. **18A-18C**, and thus have corresponding proximal cables **4716a** and **4716b** extending (wrapped) thereabout, where the proximal cables **4716a,b** are similar to the proximal cable **1804b** of FIGS. **18A-18C**. The proximal cables **4716a,b** may be routed to and from the shaft **1602** using one or more rerouting pulleys (not shown).

When the slider **4706** is manually moved in either angular direction relative to the outer circumference of the housing **4704**, the linear slider **4708** moves linearly within the housing **4704**. As the linear slider **4708** moves, the axis of each proximal pulley **4714a,b** is moved relative to the axis of the shaft **1602**. As a result, and depending on the sliding direction of the linear slider **4708**, one of the proximal pulleys **4714a,b** will draw in its proximal cable **4716a,b** from the shaft **1602**, and the other proximal pulley **4714a,b** will simultaneously pay out its proximal cable **4716a,b** to the shaft **1602**. Consequently, the net cable length of the cables **4716a,b** remains the same because the two pulleys **4714a,b** will have the same displacement, but the change in length of each pair changes, which causes movement (actuation) of the jaws **1608**, **1610** (FIG. 37). In alternative embodiments, this cable configuration may alternatively allow pitch and yaw articulations at the wrist **1606** (FIG. 37) to create motion where each cable loop rolls around its respective pulley system.

The tailpiece **4702** may also be useful in facilitating assembly pretension in the cable system. More specifically, in some embodiments, the pulleys **4714a,b** may be rotatably mounted to the linear slider **4708** using corresponding pulley brackets **4718a,b**. During assembly of the tool, the pulley brackets **4718a,b** can be adjusted relative to the linear slider **4708** so that a set torque load (tension) on the cables **4716a,b** is created; e.g., the axis of the pulleys **4714a,b** may be pulled outward from the device or shaft **1602** centerline. Once the predetermined tension is achieved, the pulley brackets **4718a,b** may be secured to the linear slider **4708** in that position using one or more mechanical fasteners (not shown) accessible to the pulley brackets **4718a,b** via one or more apertures **4720** (one shown) defined in the linear slider **4708**.

Embodiments disclosed herein include:

- A. A robotic surgical tool that includes a handle providing a plurality of drive inputs, an elongate shaft extendable through the handle and including a plurality of sliding rack gears movably nested within a corresponding plurality of longitudinal channels defined along the shaft, an end effector arranged at a distal end of the shaft and an articulable wrist interposing the end effector and the distal end, an actuation system housed within the handle and operatively coupling the plurality of drive inputs to the plurality of sliding rack gears such that actuation of the plurality of the drive inputs moves the plurality of sliding rack gears and causes operation of at least one of the end effector and the wrist, and a tailpiece arranged at a proximal end of the shaft and operably coupled to the plurality of sliding rack gears, the tailpiece including a housing and a manual actuation device mounted to the housing and manually actuatable to act on the plurality of sliding rack gears and thereby operate at least one of the end effector and the wrist.
- B. A method of operating a robotic surgical tool including arranging a robotic surgical tool adjacent a patient, the robotic surgical tool including a handle providing a plurality of drive inputs, an elongate shaft extendable through the handle and including a plurality of sliding rack gears movably nested within a corresponding plurality of longitudinal channels defined along the shaft, an end effector arranged at a distal end of the shaft and an articulable wrist interposing the end effector and the distal end, an actuation system housed within the handle and operatively coupling the plurality of drive inputs to the plurality of sliding rack gears such that actuation of the plurality of the drive inputs moves

the plurality of sliding rack gears and causes operation of at least one of the end effector and the wrist, and a tailpiece arranged at a proximal end of the shaft and operably coupled to the plurality of sliding rack gears, the tailpiece including a housing and a manual actuation device mounted to the housing. The method further including manually actuating the manual actuation device and thereby acting on the plurality of sliding rack gears to operate at least one of the end effector and the wrist.

Each of embodiments A and B may have one or more of the following additional elements in any combination: Element 1: wherein the manual actuation device comprises first and second buttons pivotably coupled to the housing, a first pulley rotatably mounted to the first button, a second pulley rotatably mounted to the second button, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, wherein manually depressing one of the first and second buttons simultaneously causes the other of the first and second buttons to pivot in an opposite direction and thereby move the first and second pairs of sliding rack gears to operate at least one of the end effector and the wrist. Element 2: further comprising a tensioning nut threadably attached to the shaft at the proximal end and adjacent the housing, wherein rotating the tensioning nut applies an axial load on the housing and thereby places the first and second cables in tension. Element 3: further comprising a manual accessory attachable to the tailpiece and including opposing handles, wherein at least one of the opposing handles is manually manipulatable to act on at least one of the first and second buttons. Element 4: wherein the manual actuation device comprises a slider that extends out of the housing and past an outer circumference of the housing, a first pulley arm mounted to the slider within the housing and having a first pulley rotatably mounted thereto, a second pulley arm mounted to the slider within the housing and having a second pulley rotatably mounted thereto, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, wherein manually moving the slider about the outer circumference of the housing correspondingly moves the first and second pulley arms within the housing and thereby draws a length of the first cable into the tailpiece from the shaft while simultaneously paying out a length of the second cable into the shaft to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist. Element 5: wherein the manual actuation device comprises first and second camming plates movably mounted within the housing and vertically offset from each other, each camming plate defining distal and proximal slots that are angled to converge or diverge from one another, a slider defined on the first camming plate and extending out of the housing, a first pulley rotatably mounted to a first pin that extends between the first and second camming plates and into the distal slot of each camming plate, a second pulley rotatably mounted to a second pin that extends between the first and second camming plates and into the proximal slot of each camming plate, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped

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around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, wherein manually moving the slider correspondingly moves the first and second camming plates laterally within the housing and thereby urges the first and second pins to traverse the distal and proximal slots, respectively, which moves the first and second pulleys toward or away from each other, and wherein moving the first and second pulleys toward or away from each other acts on the first and second cables to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist. Element 6: wherein each pulley comprises a spool having two parts matable at an engagement feature. Element 7: wherein the manual actuation device comprises a linear slider movably mounted within the housing and defining a rack gear, a slider having a first end that extends out of the housing and past an outer circumference of the housing and a second end that defines a pinion gear engageable with the rack gear, first and second pulleys rotatably mounted to opposing ends of the linear slider, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, wherein manually moving the slider about the outer circumference of the housing correspondingly shifts the linear slider within the housing and thereby draws a length of the first cable into the tailpiece from the shaft while simultaneously paying out a length of the second cable into the shaft to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist. Element 8: wherein the actuation system further includes a plurality of drive gears, each drive gear being engageable with a corresponding one of the plurality of sliding rack gears, and a gear train extending between each drive input and each drive gear such that actuation of a given drive input drives the corresponding one of the plurality of sliding rack gears within a corresponding one of the plurality of longitudinal channels. Element 9: wherein the gear train includes one or more intermediate gears that transition a rotational axis of the given drive input by 90° to enable the plurality of drive gears to engage the plurality of sliding rack gears. Element 10: wherein the actuation system further includes first and second mounting assemblies vertically offset from each other and concentrically arranged about the shaft; a plurality of spur linkage subassemblies pivotably mounted to the first and second mounting assemblies, each spur linkage subassembly including a drive gear engageable with a corresponding one of the plurality of sliding rack gears, wherein moving the first mounting assembly toward the second mounting assembly pivots the plurality of spur linkage assemblies between a disengaged position, where the drive gear of each spur linkage subassembly is disengaged from the corresponding one of the plurality of sliding rack gears, and an engaged position, where the drive gear of each spur linkage subassembly is engaged with the corresponding one of the plurality of sliding rack gears. Element 11: wherein each spur linkage subassembly includes a drive gear engageable with a corresponding one of the plurality of sliding rack gears, and a gear train extending between a corresponding one of the plurality of drive inputs and the drive gear such that actuation of the corresponding one of the plurality of drive inputs drives the corresponding one of the plurality of sliding rack gears within a corresponding one of the plurality of longitudinal channels. Element 12: wherein the gear train includes one or more intermediate gears that transition a

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rotational axis of the corresponding one of the plurality of drive inputs by 90° to enable the drive gear to engage the corresponding one of the plurality of sliding rack gears. Element 13: wherein each spur linkage subassembly further includes a first linkage pivotably coupled to the first mounting assembly, and a second linkage having a first end pivotably coupled to the first linkage and a second end pivotably coupled to the second mounting assembly. Element 14: wherein the actuation system further includes a clocking wheel linkage subassembly comprising a first linkage pivotably coupled to the first mounting assembly, a second linkage having a first end pivotably coupled to the first linkage at a pivot axis and a second end pivotably coupled to the second mounting assembly, and a clocking wheel rotatably mounted to the first and second linkages at the pivot axis, wherein moving the first mounting assembly toward the second mounting assembly pivots the clocking wheel linkage subassembly between the disengaged position, where the clocking wheel is disengaged from a longitudinal groove defined in the shaft, and the engaged position, where the clocking wheel is engaged with the longitudinal groove and thereby angularly aligns each spur linkage subassembly with the corresponding one of the plurality of sliding rack gears.

Element 15: wherein the manual actuation device comprises first and second buttons pivotably coupled to the housing, a first pulley rotatably mounted to the first button, a second pulley rotatably mounted to the second button, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, the method further comprising manually depressing one of the first and second buttons and thereby simultaneously causing the other of the first and second buttons to pivot in an opposite direction to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist. Element 16: wherein the tailpiece further includes a tensioning nut threadably attached to the shaft at the proximal end and adjacent the housing, the method further comprising rotating the tensioning nut and thereby applying an axial load on the housing and placing the first and second cables in tension. Element 17: wherein the manual actuation device comprises a slider that extends out of the housing and past an outer circumference of the housing, a first pulley arm mounted to the slider within the housing and having a first pulley rotatably mounted thereto, a second pulley arm mounted to the slider within the housing and having a second pulley rotatably mounted thereto, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, the method further comprising manually moving the slider about the outer circumference of the housing and thereby correspondingly moving the first and second pulley arms within the housing, and as the first and second pulley arms move within the housing, drawing a length of the first cable into the tailpiece from the shaft while simultaneously paying out a length of the second cable into the shaft to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist. Element 18: wherein the manual actuation device comprises first and second camming plates movably mounted within the housing and vertically offset from each other, each camming

plate defining distal and proximal slots that are angled to converge or diverge from one another, a slider defined on the first camming plate and extending out of the housing, a first pulley rotatably mounted to a first pin that extends between the first and second camming plates and into the distal slot of each camming plate, a second pulley rotatably mounted to a second pin that extends between the first and second camming plates and into the proximal slot of each camming plate, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, the method further comprising manually moving the slider and thereby correspondingly moving the first and second camming plates laterally within the housing, as the first and second camming plates laterally within the housing, urging the first and second pins to traverse the distal and proximal slots, respectively, which moves the first and second pulleys toward or away from each other, and moving the first and second pulleys toward or away from each and thereby acting on the first and second cables to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist.

By way of non-limiting example, exemplary combinations applicable to A and B include: Element 1 with Element 2; Element 1 with Element 3; Element 8 with Element 9; Element 10 with Element 11; Element 11 with Element 12; Element 11 with Element 13; Element 11 with Element 14; and Element 15 with Element 16.

### 3. Implementing Systems and Terminology.

Implementations disclosed herein provide systems, methods and apparatus for instruments for use with robotic systems. It should be noted that the terms “couple,” “coupling,” “coupled” or other variations of the word couple as used herein may indicate either an indirect connection or a direct connection. For example, if a first component is “coupled” to a second component, the first component may be either indirectly connected to the second component via another component or directly connected to the second component.

The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

As used herein, the term “plurality” denotes two or more. For example, a plurality of components indicates two or more components. The term “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and the like.

The phrase “based on” does not mean “based only on,” unless expressly specified otherwise. In other words, the phrase “based on” describes both “based only on” and “based at least on.”

As used herein, the terms “generally” and “substantially” are intended to encompass structural or numeral modifica-

tion, which do not significantly affect the purpose of the element or number modified by such term.

To aid the Patent Office and any readers of this application and any resulting patent in interpreting the claims appended herein, applicants do not intend any of the appended claims or claim elements to invoke 35 U.S.C. 112(f) unless the words “means for” or “step for” are explicitly used in the particular claim.

The foregoing previous description of the disclosed implementations is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these implementations will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the scope of the invention. For example, it will be appreciated that one of ordinary skill in the art will be able to employ a number corresponding alternative and equivalent structural details, such as equivalent ways of fastening, mounting, coupling, or engaging tool components, equivalent mechanisms for producing particular actuation motions, and equivalent mechanisms for delivering electrical energy. Thus, the present invention is not intended to be limited to the implementations shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

What is claimed is:

#### 1. A robotic surgical tool, comprising:

a handle providing a plurality of drive inputs including first and second drive inputs, wherein the handle is couplable to an instrument driver of a robotic surgical system, the instrument driver comprising first and second drive outputs for driving the first and second drive input, respectively;

an elongate shaft extendable through the handle and including a plurality of sliding rack gears movably nested within a corresponding plurality of longitudinal channels defined along the shaft, the plurality of sliding rack gears including first and second sliding rack gears independently moveable relative to one another;

an end effector arranged at a distal end of the shaft and an articulable wrist interposing the end effector and the distal end;

an actuation system housed within the handle and operatively coupling the plurality of drive inputs to the plurality of sliding rack gears, wherein:

actuation of the first drive output drives the first drive input, thereby moving the first sliding rack gear to perform a first function; and

actuation of the second drive output drives the second drive input, thereby moving the second sliding rack gear to perform a second function different than the first function; and

a tailpiece arranged at a proximal end of the shaft, the tailpiece including a housing and a manual actuation device mounted to the housing and manually and physically engageable by a user, wherein manual engagement of the manual actuation device by the user causes movement of the first sliding rack gear to perform the first function.

#### 2. The robotic surgical tool of claim 1, wherein the manual actuation device comprises:

first and second buttons pivotably coupled to the housing; a first pulley rotatably mounted to the first button;

a second pulley rotatably mounted to the second button;

a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears; and

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a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, wherein manually depressing one of the first and second buttons simultaneously causes the other of the first and second buttons to pivot in an opposite direction and thereby move the first and second pairs of sliding rack gears to operate at least one of the end effector and the wrist.

3. The robotic surgical tool of claim 2, further comprising a tensioning nut threadably attached to the shaft at the proximal end and adjacent the housing, wherein rotating the tensioning nut applies an axial load on the housing and thereby places the first and second cables in tension.

4. The robotic surgical tool of claim 2, further comprising a manual accessory attachable to the tailpiece and including opposing handles, wherein at least one of the opposing handles is manually manipulatable to act on at least one of the first and second buttons.

5. The robotic surgical tool of claim 1, wherein the manual actuation device comprises:

a slider that extends out of the housing and past an outer circumference of the housing;

a first pulley arm mounted to the slider within the housing and having a first pulley rotatably mounted thereto;

a second pulley arm mounted to the slider within the housing and having a second pulley rotatably mounted thereto;

a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears; and

a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears,

wherein manually moving the slider about the outer circumference of the housing correspondingly moves the first and second pulley arms within the housing and thereby draws a length of the first cable into the tailpiece from the shaft while simultaneously paying out a length of the second cable into the shaft to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist.

6. The robotic surgical tool of claim 1, wherein the manual actuation device comprises:

first and second camming plates movably mounted within the housing and vertically offset from each other, each camming plate defining distal and proximal slots that are angled to converge or diverge from one another;

a slider defined on the first camming plate and extending out of the housing;

a first pulley rotatably mounted to a first pin that extends between the first and second camming plates and into the distal slot of each camming plate;

a second pulley rotatably mounted to a second pin that extends between the first and second camming plates and into the proximal slot of each camming plate;

a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears; and

a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears,

wherein manually moving the slider correspondingly moves the first and second camming plates laterally within the housing and thereby urges the first and second pins to traverse the distal and proximal slots,

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respectively, which moves the first and second pulleys toward or away from each other, and

wherein moving the first and second pulleys toward or away from each other acts on the first and second cables to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist.

7. The robotic surgical tool of claim 6, wherein each pulley comprises a spool having two parts matable at an engagement feature.

8. The robotic surgical tool of claim 1, wherein the manual actuation device comprises:

a linear slider movably mounted within the housing and defining a rack gear;

a slider having a first end that extends out of the housing and past an outer circumference of the housing and a second end that defines a pinion gear engageable with the rack gear;

first and second pulleys rotatably mounted to opposing ends of the linear slider;

a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears; and

a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears,

wherein manually moving the slider about the outer circumference of the housing correspondingly shifts the linear slider within the housing and thereby draws a length of the first cable into the tailpiece from the shaft while simultaneously paying out a length of the second cable into the shaft to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist.

9. The robotic surgical tool of claim 1, wherein the actuation system further includes:

a plurality of drive gears, each drive gear being engageable with a corresponding one of the plurality of sliding rack gears; and

a gear train extending between each drive input and each drive gear such that actuation of a given drive input drives the corresponding one of the plurality of sliding rack gears within a corresponding one of the plurality of longitudinal channels.

10. The robotic surgical tool of claim 9, wherein the gear train includes one or more intermediate gears that transition a rotational axis of the given drive input by 90° to enable the plurality of drive gears to engage the plurality of sliding rack gears.

11. The robotic surgical tool of claim 1, wherein the actuation system further includes:

first and second mounting assemblies vertically offset from each other and concentrically arranged about the shaft; and

a plurality of spur linkage subassemblies pivotably mounted to the first and second mounting assemblies, each spur linkage subassembly including a drive gear engageable with a corresponding one of the plurality of sliding rack gears,

wherein moving the first mounting assembly toward the second mounting assembly pivots the plurality of spur linkage assemblies between a disengaged position, where the drive gear of each spur linkage subassembly is disengaged from the corresponding one of the plurality of sliding rack gears, and an engaged position,

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where the drive gear of each spur linkage subassembly is engaged with the corresponding one of the plurality of sliding rack gears.

12. The robotic surgical tool of claim 11, wherein each spur linkage subassembly includes:

- a drive gear engageable with a corresponding one of the plurality of sliding rack gears; and
- a gear train extending between a corresponding one of the plurality of drive inputs and the drive gear such that actuation of the corresponding one of the plurality of drive inputs drives the corresponding one of the plurality of sliding rack gears within a corresponding one of the plurality of longitudinal channels.

13. The robotic surgical tool of claim 12, wherein each spur linkage subassembly further includes:

- a first linkage pivotably coupled to the first mounting assembly; and
- a second linkage having a first end pivotably coupled to the first linkage and a second end pivotably coupled to the second mounting assembly.

14. The robotic surgical tool of claim 12, wherein the actuation system further includes a clocking wheel linkage subassembly comprising:

- a first linkage pivotably coupled to the first mounting assembly;
- a second linkage having a first end pivotably coupled to the first linkage at a pivot axis and a second end pivotably coupled to the second mounting assembly; and
- a clocking wheel rotatably mounted to the first and second linkages at the pivot axis,

wherein moving the first mounting assembly toward the second mounting assembly pivots the clocking wheel linkage subassembly between the disengaged position, where the clocking wheel is disengaged from a longitudinal groove defined in the shaft, and the engaged position, where the clocking wheel is engaged with the longitudinal groove and thereby angularly aligns each spur linkage subassembly with the corresponding one of the plurality of sliding rack gears.

15. The robotic surgical tool of claim 1, wherein the first function comprises rotating at least a portion of the end effector relative to the elongate shaft about the articulable wrist and the second function comprises translating the elongate shaft and the end effector relative to the handle.

16. A method, comprising:

arranging a robotic surgical tool adjacent a patient, the robotic surgical tool including:

- a handle providing a plurality of drive inputs including first and second drive inputs, the handle being coupleable to an instrument driver of a robotic surgical system, and the instrument driver comprising a first and second drive outputs for driving the first and second drive input, respectively;

an elongate shaft extendable through the handle and including a plurality of sliding rack gears movably nested within a corresponding plurality of longitudinal channels defined along the shaft, the plurality of sliding rack gears including first and second sliding rack gears independently moveable relative to one another;

an end effector arranged at a distal end of the shaft and an articulable wrist interposing the end effector and the distal end;

an actuation system housed within the handle and operatively coupling the plurality of drive inputs to the plurality of sliding rack gears, wherein:

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actuation of the first drive output drives the first drive input, thereby moving the first sliding rack gear to perform a first function;

actuation of the second drive output drives the second drive input, thereby moving the second sliding rack gear to perform a second function different than the first function; and

a tailpiece arranged at a proximal end of the shaft, the tailpiece including a housing and a manual actuation device mounted to the housing and manually and physically engageable by a user;

manually actuating the manual actuation device by the user and thereby causing movement of the first sliding rack gear to perform the first function; and

robotically actuating the actuation system via the plurality of drive outputs and thereby causing movement of the plurality of sliding rack gears to perform at least one of the first and second functions.

17. The method of claim 16, wherein the manual actuation device comprises first and second buttons pivotably coupled to the housing, a first pulley rotatably mounted to the first button, a second pulley rotatably mounted to the second button, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, the method further comprising:

manually depressing one of the first and second buttons and thereby simultaneously causing the other of the first and second buttons to pivot in an opposite direction to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist.

18. The method of claim 17, wherein the tailpiece further includes a tensioning nut threadably attached to the shaft at the proximal end and adjacent the housing, the method further comprising rotating the tensioning nut and thereby applying an axial load on the housing and placing the first and second cables in tension.

19. The method of claim 16, wherein the manual actuation device comprises a slider that extends out of the housing and past an outer circumference of the housing, a first pulley arm mounted to the slider within the housing and having a first pulley rotatably mounted thereto, a second pulley arm mounted to the slider within the housing and having a second pulley rotatably mounted thereto, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears, the method further comprising:

manually moving the slider about the outer circumference of the housing and thereby correspondingly moving the first and second pulley arms within the housing; and as the first and second pulley arms move within the housing, drawing a length of the first cable into the tailpiece from the shaft while simultaneously paying out a length of the second cable into the shaft to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the wrist.

20. The method of claim 16, wherein the manual actuation device comprises first and second camming plates movably mounted within the housing and vertically offset from each other, each camming plate defining distal and proximal slots that are angled to converge or diverge from one another, a

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slider defined on the first camming plate and extending out of the housing, a first pulley rotatably mounted to a first pin that extends between the first and second camming plates and into the distal slot of each camming plate, a second pulley rotatably mounted to a second pin that extends 5 between the first and second camming plates and into the proximal slot of each camming plate, a first cable wrapped around the first pulley and operatively coupled to a first pair of sliding rack gears of the plurality of sliding rack gears, and a second cable wrapped around the second pulley and 10 operatively coupled to a second pair of sliding rack gears of the plurality of sliding rack gears,

the method further comprising:

manually moving the slider and thereby correspondingly moving the first and second camming plates laterally 15 within the housing;

as the first and second camming plates laterally within the housing, urging the first and second pins to traverse the distal and proximal slots, respectively, which moves the first and second pulleys toward or away from each 20 other; and

moving the first and second pulleys toward or away from each and thereby acting on the first and second cables to move the first and second pairs of sliding rack gears and operate at least one of the end effector and the 25 wrist.

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